FinMath Formula Sheet

Plug-and-play. Every card: the formula → when it applies → a past-paper instance and the trick. Part I: foundations, expectation & martingales. Part II: random walks & Brownian motion. Part III: stochastic calculus. Part IV: option pricing & arbitrage. Part V: discrete markets & Black–Scholes. Click **Proof** on a card to open its proof.

0. Routing Table — read the question, jump to the card

THE QUESTION SAYS…	IT IS REALLY ASKING	GO TO
PART I — FOUNDATIONS, EXPECTATION & MARTINGALES		
"Is this collection a σ-field?"	Three axioms; complement fails first	§F1.2
"How many events are in $\sigma(\xi)$?"	Count distinct values k, answer 2^k	§F1.5, §F4.1 ✎ proof
"Show $P(A_n \text{ i.o.}) = 0$" / "prove Borel–Cantelli"	Convergent sum ⇒ BC I; needs $B_n \downarrow$ + union bound	§F2.3, §F2.5 ✎ proof
"Is X $\mathcal{G}$-measurable?"	Constant on every atom of $\mathcal{G}$	§F3.4 ✎ proof
"Show η is a function of ξ"	Doob–Dynkin: ties in ξ must be ties in η	§F4.4 ✎ proof
"Is F a distribution function?"	Non-decreasing, right-continuous, limits 0 /1	§F5.3 ✎ proof
"Does $<$ vs $\leq$ matter here?"	Only at a jump; jump size $= P(\xi = x)$	§F5.4 ✎ proof
"Is ξ integrable?"	Test $E\lvert\xi\rvert$, never $E\xi$	§E1.1
"Construct / define the Lebesgue integral"	Indicators → simple → non-negative (MCT) → general	§E1.5
"Compute $E[h(\xi)]$"	Change of variables: $\int hf\,dx$ or $\sum h(x_i)p_i$	§E2.2 ✎ proof
Two-stage experiment / "given it tested positive"	Total probability, then Bayes	§E3.3 ✎ proof
"Are they mutually independent?"	$2^n - n - 1$ equations, not just the pairs	§E3.4 ✎ proof
"Find $E(X \mid \mathcal{G})$" (discrete)	Block average over each atom	§E4.3 ✎ proof
Any conditional-expectation manipulation	Linearity / take out what is known / independence	§E4.5 ✎ proof
"Is X_t adapted?"	Same atom test as measurability	§M1.2
"Is X_n a martingale? (Yes/No)"	Adapted + integrable + $E[X_{n+1} \mid \mathcal{F}_n] = X_n$	§M2.1
MCQ: "which statement about a MG is TRUE?"	"The mean is constant with respect to time"	§M2.4 ✎ proof

"For which p is S_t a martingale?"	$p = (1-d)/(u-d)$	§M3.3 ✎ proof
Binomial with interest r	$p^* = (1+r-d)/(u-d)$; discounted price is a MG	§M3.4 ✎ proof
"Replicating portfolio has the form (Δ, B) ="	Δ and B formulas, then $V_0 = \Delta S_0 + B$	§M4.3 ✎ proof
"Price the call / put"	Risk-neutral expectation, discounted	§M4.2 ✎ proof
"For which c is there arbitrage / no sure win?"	p from the present price, then $c^* = E^*[$ payoff$]$	§M4.5 ✎ proof
"Verify the strategy is self-financing"	Old portfolio's new value = new portfolio's cost	§M4.7
"Is τ a stopping time?"	$\{\tau = t\} \in \mathcal{F}_t$; first-entry ✓, last-exit ✗	§M5.2–M5.4 ✎ proof
"Use OST to calculate $E(T)$"	$E[X_\tau] = E[X_0]$, substitute the barrier	§M6.3, §M6.5 ✎ proof
Ruin probability *and* duration together	Two martingales: X_n and $X_n^2 - n$	§M6.6 ✎ proof
"Can a betting system beat a fair game?"	No — predictable bounded α keeps Z_n a MG	§M7.2, §M7.5 ✎ proof
"Short note: American vs European call"	Equal for a non-dividend call; differ for a put	§M7.6 ✎ proof
PART II — RANDOM WALKS & BROWNIAN MOTION		
"Compute the distribution of $\int_0^T f(t)\,dW_t$"	It's Gaussian; variance by Itô isometry	§7.5 ✎ proof
"$aW_s + bW_t \sim$?" / "$W_1 - W_2 + 3W_4 \sim$?"	Normal, mean 0, variance from $\min(s,t)$	§6.2 ✎ proof
"Compute $E(W_s^2 W_t^2)$" / "$\mathrm{Cov}(e^{W_s}, e^{W_t})$"	Split $W_t = W_s + (W_t - W_s)$, independence	§6.4, §6.5 ✎ proof
"Is $Y_t = e^{\lambda t}\sin W_t$ a martingale?"	Itô drift $= 0$ test	§7.4 ✎ proof
"Is $\lvert W_t\rvert^2 - t$ a martingale?"	Canonical martingale #2 — yes	§7.2 ✎ proof
"Is $\frac{1}{c}W(c^2 t)$ / $e^{-\lambda t}W(e^{2\lambda t})$ a Wiener / Gaussian process?"	4-point Wiener checklist vs. Gaussian-process test	§5.2, §5.3 ✎ proof
"GBM with drift μ, volatility σ, $Y(0)$. Find $E[Y(t)]$"	Which μ convention? Give both	§11.2 ✎ proof
"Wins/loses 1 rupee, stops at $+W$ or $-L$"	Gambler's ruin: win-prob and duration	§3 ✎ proof
"$P(B(t)$ reaches -2 before $1)$"	Two-barrier exit, drift $0 \Rightarrow$ ratio of distances	§10.4 ✎ proof

"Drift $-\mu$, buy at fixed price K. When to exercise?"	Maximize $(x-K)e^{-2\mu x/\sigma^2}$	§10.6 ✎ proof
"cdf and pdf of τ_a" / "$P(\tau_a<\infty)$" / "$E[\tau_a]$"	Reflection principle ⇒ $2\Phi(-a/\sqrt{t})$	§12.3–12.5 ✎ proof
"$T=\min\{t: W_t=a+bt\}$, find $E[T]$"	OST on W_t: $E[W_T]=0$	§10.7 ✎ proof
"$M(t)=\max_{0\le u\le t}W(u)$. Find $P(\cdot)$" — M alone	$M(t)\overset{d}{=}\lvert W(t)\rvert$	§13 ✎ proof
"$P(\alpha<W(t)<\beta,\ \gamma<M(t)<\delta)$" — both	Master function G + 4-term inclusion–exclusion	§14.3 ✎ proof
"$W(s)$ given $W(t)=b$" / Brownian bridge	Backward interpolation $N(\frac{s}{t}b,\frac{s(t-s)}{t})$	§9.4 ✎ proof
"Random walk $\pm\sqrt{\Delta t}$, $p=\frac{1}{2}(1+\mu\sqrt{\Delta t})$"	Diffusive scaling limit → BM drift μ	§4 ✎ proof
"$X_n=Y_1\cdots Y_n$ (or $2^nU_1\cdots U_n$). Martingale?"	Geometric RW: need $E[Y]=1$	§2.2 ✎ proof
PART III — STOCHASTIC CALCULUS		
"Compute the stochastic differential $d(\cdot)$"	Itô: drift $=\frac{1}{2}f''$, diffusion $=f'$	§I4 ✎ proof
"Derive Itô's formula"	Taylor to 2nd order + the multiplication table	§I3.1, §I2.2 ✎ proof
"Variance / distribution of $\int_0^T f\,dW$"	Itô isometry: $N\big(0,\int_0^T f^2dt\big)$	§I1.4–I1.5 ✎ proof
"$\int_0^t W_s\,dW_s=?$"	$\frac{1}{2}W_t^2-\frac{t}{2}$ — the correction term	§I1.7 ✎ proof
"Is Y_t a martingale? (Yes/No)"	Apply Itô, read the dt coefficient	§I5.1, §I4.4 ✎ proof
"Show $X_t=\ldots$ solves the SDE"	Itô the candidate, match coefficients, check X_0	§I6.3 ✎ proof
"Express X_t as a stochastic integral"	Integrating factor e^{rt}, then integrate	§I7.1, §I6.4
"Find $E(X_t)$ and $\mathrm{Var}(X_t)$" from an SDE	Drift ODE for the mean; Itô on X^2 for the variance	§I5.2–I5.3 ✎ proof
"Is it an OU process? Explain"	Drift $-rX$, constant noise ⇒ mean-reverting, Gaussian	§I7.3 ✎ proof
"Using part (i) or otherwise, evaluate $E[f(W_t)]$"	Constant mean of the martingale from part (i)	§I5.4 ✎ proof
PART IV — OPTION PRICING & ARBITRAGE		
"For which c is there no arbitrage?"	Riskless portfolio, then gain $=0$	§A1.2 ✎ proof

"Price the call / put with interest rate r"	$(1+r)^{-1}E^*[\text{payoff}]$ with p^*	§A1.3, §A1.4 ✎ proof
"Find the replicating portfolio (Δ, B)"	$\Delta =$ payoff spread / price spread	§A1.5 ✎ proof
"Is $E_p[\text{payoff}]/(1+r)$ the price?" (real p given)	No — it is an arbitrage; use p^*	§A1.6 ✎ proof
"State / prove the arbitrage theorem"	Dichotomy: risk-neutral measure XOR sure win	§A2.3 ✎ proof
"Three possible time-1 prices — find c"	Underdetermined: an *interval* of prices	§A2.4 ✎ proof
"Payoff of a forward vs a call vs a put"	$S_T - K$ / $(S_T-K)^+$ / $(K-S_T)^+$	§A2.1
"American call — exercise early?"	Never, if no dividends and $r \geq 0$	§A2.2 ✎ proof
PART V — DISCRETE MARKETS & BLACK–SCHOLES		
"Fair forward price" / "K so the contract costs nothing"	$K = S_0(1+r)^T$ or S_0e^{rT}; no p^*	§D1.2 ✎ proof
"Verify self-financing" (several assets)	Old holdings at new prices = cost of new holdings	§D2.2 ✎ proof
"Define arbitrage formally"	$V_0 = 0$, $V_n \geq 0$, $P(V_n > 0) > 0$	§D2.3
"Find all EMMs" / "is the market arbitrage-free?"	Solve $E^*[S_1/(1+r)] = S_0$, $\sum p = 1$, keep all $p_k^* > 0$	§D3.1, §D3.3 ✎ proof
"State / prove the fundamental theorem"	NA $\iff$ EMM $\iff$ $S_0 \in D^0$, 3 steps	§D3.3 ✎ proof
"Price the claim" when the EMM is not unique	Parameter cancels in $E^*[X/(1+r)]$ ⇒ attainable, one price	§D3.4 ✎ proof
"Is the market complete?"	EMM unique; states = assets	§D3.5 ✎ proof
Two- or three-step tree: price / hedge at each node	Backward induction from the leaves	§D4.2 ✎ proof
"Price a European call over T periods"	Find m, then $S_0P(\hat{U} \geq m) - K(1+r)^{-T}P(U^* \geq m)$	§D4.4 ✎ proof
"Derive Black–Scholes from the binomial model"	$u_n, d_n, r_n \to p_n^* \to \frac{1}{2}$ → CLT	§D5.1 ✎ proof
"Black–Scholes price of the call / put"	$S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2)$; put by parity	§D5.2 ✎ proof
"Solve $dS = \mu S\,dt + \sigma S\,dW$"	Itô on $\ln S$	§D5.3 ✎ proof
"Is the BS market arbitrage-free / complete? Find the EMM"	$e^{-rt}Z_t$ MG $\iff$ $a_t = r - \sigma^2/2$: yes and yes	§D5.4 ✎ proof

Short note: “martingales and claim pricing”	The three facts: EMM, price formula, uniqueness	§D5.5 ✎ proof

Part I — Foundations, Expectation & Martingales

Measure theory → conditional expectation → discrete-time martingales and the binomial market. Markov decision processes are out of syllabus.

F1. Probability Space & σ-Fields

F1.1 THE TRIPLE

$(\Omega, \mathcal{F}, P)$

Ω = outcomes | $\mathcal{F}$ = events we may assign probability to | P = the measure

event $A \subseteq \Omega$ **happened** $\iff \omega \in A$

> **Use when:** every definition question starts here. $\mathcal{F}$ is not decoration — it is *which questions you are allowed to ask*.

> **Q:** “short note on the probability space”.
> **Trick:** one line each: Ω is what can happen, $\mathcal{F}$ is what you can ask, P is how likely. Then give the die example.

F1.2 Σ-FIELD AXIOMS

(i) $\varnothing \in \mathcal{F}$

(ii) $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$

(iii) $A_1, A_2, \ldots \in \mathcal{F} \Rightarrow \bigcup_i A_i \in \mathcal{F}$

> **Use when:** “is this collection a σ-field?” Check in order. **Closure under complement is the usual failure** — test it first, it is fastest.

> **Q:** die, is $\{\varnothing, \{6\}, \Omega\}$ a σ-field? **No** — $\{6\}^c = \{1,2,3,4,5\}$ is missing.
> **Trick:** intersections come free: $A \cap B = (A^c \cup B^c)^c$. You never check them separately.

F1.3 PROBABILITY MEASURE AXIOMS

$P(A) \geq 0; \;\; P(\Omega) = 1;$
$P\left(\bigsqcup_i A_i\right) = \sum_i P(A_i)$ (disjoint)
$P(A^c) = 1 - P(A),$
$A \subseteq B \Rightarrow P(A) \leq P(B)$
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
, $P\left(\bigcup A_i\right) \leq \sum P(A_i)$

> **Use when:** the subadditivity (union bound) on the last line is the one that does real work — it is step 6 of Borel–Cantelli (F2.5).

> **Q:** die, $A = \{1,2\}$, $B = \{2,3\}$:
> $P(A \cup B) = \frac{1}{2} \leq \frac{1}{3} + \frac{1}{3}$.
> **Trick:** the union bound is loose exactly by the double-counted overlap.

▼ **Proof**

> A and A^c are disjoint with union Ω:
> $P(A) + P(A^c) = P(\Omega) = 1$.
> $A \subseteq B$: $B = A \sqcup (B \setminus A)$, so
> $P(B) = P(A) + P(B \setminus A) \geq P(A)$. With
> $A = \varnothing$, $B = \varnothing$: $P(\varnothing) = 2P(\varnothing)$, so
> $P(\varnothing) = 0$.

F1.4 CONTINUITY OF MEASURE

$A_n \uparrow A \;\Rightarrow\; P(A_n) \to P(A)$ (from below)

$A_n \downarrow A \;\Rightarrow\; P(A_n) \to P(A)$ (from above)

> **Use when:** two places on this paper — right-continuity of the d.f. (F5.3) and step 5 of Borel–Cantelli (F2.5). Both times you need the *decreasing* version.

> **Trick:** spot the nesting first. If the sets shrink you may swap $\lim$ and P; if they neither grow nor shrink you may not.

▼ **Proof**

> **From below.** Cut into rings: $C_1 = A_1$, $C_n = A_n \setminus A_{n-1}$. They are disjoint, $A_n = C_1 \sqcup \cdots \sqcup C_n$ and $A = \bigsqcup_n C_n$. So $P(A) = \sum_n P(C_n) = \lim_n \sum_{k \le n} P(C_k) = \lim$.
> **From above.** $A_n \downarrow A$ means $A_n^c \uparrow A^c$; apply the first part and use $P(A^c) = 1 - P(A)$.

F1.5 COUNTING Σ-FIELDS (FINITE Ω)

generated by a partition into k atoms $\Rightarrow$ exactly 2^k events

smallest: $\{\varnothing, \Omega\}$ | largest: 2^Ω, i.e. $2^{|\Omega|}$ events

> **Use when:** "how many events are in $\sigma(\xi)$?" **Count the distinct values of ξ, call it k, answer 2^k.** Never enumerate.

> **Q:** die. $\sigma(\text{even/odd})$: 2 atoms $\to$ **4** events. $\sigma(\omega)$: 6 atoms $\to 2^6 = \mathbf{64}$. Constant r.v.: 1 atom $\to$ **2**.

▼ **Proof**

> Every event of the σ-field is a union of some of the k atoms (an atom is either fully inside or fully outside it). Each atom is in or out: 2^k choices, and different choices give different sets because atoms are disjoint and non-empty.
> Largest: every point is its own atom, $k = |\Omega|$.
> Smallest: one atom Ω, $k = 1$: $\{\varnothing, \Omega\}$.

F2. Almost Surely & Borel–Cantelli

F2.1 ALMOST SURELY

"A holds a.s." $\iff P(A) = 1$

equivalently $P(A^c) = 0$ — exceptions allowed, but only on a null set

> **Use when:** a.s. is *not* "always". A continuous r.v. satisfies $\xi \neq 0.5$ a.s. even though ω with $\xi = 0.5$ exist.

> **Trick:** in a finite Ω with all $P(\omega) > 0$ there are no null sets, so a.s. and always coincide. That is why die examples never show the distinction.

F2.2 THE LIMSUP EVENT ("INFINITELY OFTEN")

$$E = \limsup_n A_n = \bigcap_{n=1}^{\infty} \underbrace{\bigcup_{i=n}^{\infty} A_i}_{B_n}$$

$$= \{\omega : \omega \text{ belongs to } \textbf{infinitely many } A_i\}$$

> **Use when:** the words "infinitely often" or "i.o." appear. Note $B_n \downarrow$, which is what licenses F1.4.

> **Read it aloud:** "for every n, there is some $i \geq n$ with $\omega \in A_i$" — i.e. you never run out of later hits.
> **Q:** coin, $A_i = \{\text{toss } i \text{ is H}\}$.
> $\omega = HTHTHT\ldots \in E$;
> $\omega = HHTTTT\ldots \notin E$.

▼ **Proof**

> $\omega \in \bigcap_n \bigcup_{i\geq n} A_i \iff$ for every n there is some $i \geq n$ with $\omega \in A_i \iff$ however far you go, ω is in a later $A_i \iff \omega$ is in infinitely many A_i.

F2.3 BOREL–CANTELLI I

$$\sum_{i=1}^{\infty} P(A_i) < \infty \implies P(\limsup A_n) = 0$$

i.e. a.s. only **finitely many** A_i occur

No independence needed.

> **Use when:** the sum converges. One direction only — a *divergent* sum tells you nothing on its own.

> **Q:** $\Omega = [0,1]$ uniform, $A_i = [0, 1/i^2]$.
> $\sum P(A_i) = \pi^2/6 < \infty$, so $P(E) = \mathbf{0}$ — and indeed $E = \{0\}$.
> **Trick:** the conclusion is "there is a random cutoff $N(\omega)$ after which A_i never happens again".

▼ **Proof**

> $B_n = \bigcup_{i\geq n} A_i$ decreases to $E = \limsup A_n$, so $P(E) = \lim_n P(B_n)$ (§F1.4).
> Union bound: $P(B_n) \leq \sum_{i\geq n} P(A_i)$, the tail of a convergent series, which $\to 0$. Hence $P(E) = 0$. (This is the chain in §F2.5.)

F2.4 BOREL–CANTELLI II (NEEDS INDEPENDENCE)

A_i **independent** and

$$\sum_{i} P(A_i) = \infty \implies P(\limsup A_n) = 1$$

> **Use when:** divergent sum *and* independence. Drop independence and the conclusion dies.

> **Counterexample:** $A_i = [0, 1/i]$. $\sum 1/i = \infty$, but the A_i are nested (maximally dependent) and $E = \{0\}$, so $P(E) = 0$, not 1.
> **Trick:** state which lemma you are using and verify *its* hypothesis — that is where the mark is.

▼ **Proof**

> $E^c = \bigcup_n \bigcap_{i\geq n} A_i^c$, so $P(E^c) \leq \sum_n P\left(\bigcap_{i\geq n} A_i^c\right)$. Fix n. By independence and $1 - x \leq e^{-x}$:
> $P\left(\bigcap_{i=n}^{m} A_i^c\right) = \prod_{i=n}^{m} \left(1 - P(A_i)\right) \leq \exp\Big(-$
> as $m \to \infty$, since the sum diverges.
> So every term is 0, $P(E^c) = 0$, $P(E) = 1$.

F2.5 THE PROOF CHAIN (BC I)

$0 \leq P(E) = \lim_n P(B_n) \leq \lim_n \sum_{i \geq n} P$

Use when: asked to *prove* it. Exactly three ingredients, in this order:
1. $B_n \downarrow$ 2. continuity from above (F1.4) 3. union bound (F1.3).
The final 0 is because a convergent series has vanishing tails.

Trick: write the sandwich $0 \leq \cdots \leq 0$ explicitly. It is worth full marks on its own and takes four lines.

F3. Random Variables & Measurability

F3.1 DEFINITION

$X : \Omega \to \mathbb{R}$ is $\mathcal{F}$-measurable if
$[X \in B] := X^{-1}(B) = \{\omega : X(\omega) \in B\}$
A r.v. = an $\mathcal{F}$-measurable function on a probability space

Use when: the *reason* for the definition is the marked point: it is exactly what makes $P(X \in B)$ well defined.

Trick: a random variable is not random and is not a variable — it is a deterministic function of ω. All the randomness lives in which ω occurs.

F3.2 YOU ONLY CHECK THE GENERATORS

X is $\mathcal{F}$-measurable $\iff \{X \leq x\} \in \mathcal{F}$ for every $x \in \mathbb{R}$

Because $\{(-\infty, x] : x \in \mathbb{R}\}$ generates $\mathcal{B}(\mathbb{R})$.

Use when: always. Never attempt to test every Borel set — test the half-lines and cite that they generate.

Q: $\Omega = \{H, T\}$, X = #heads, $\mathcal{F} = 2^\Omega$. Pre-images of $(-\infty, x]$: $\varnothing$ ($x < 0$), $\{T\}$ ($0 \leq x < 1$), Ω ($x \geq 1$). All in $\mathcal{F}$ ✓

▼ **Proof**

($\Rightarrow$) $(-\infty, x]$ is a Borel set.
($\Leftarrow$) Let $\mathcal{C} = \{B : X^{-1}(B) \in \mathcal{F}\}$. Preimages respect complements and countable unions, so $\mathcal{C}$ is a σ-field. It contains every $(-\infty, x]$, so it contains the σ-field they generate, which is all Borel sets.

F3.3 MEASURABILITY IS JOINT IN $(X, \mathcal{F})$

the same X can be $\mathcal{F}$-measurable and **not** $\mathcal{G}$-measurable

> **Use when:** the question shrinks the σ-field ("you are only told whether the die was low or high"). Measurability is a statement about *information*, not about the formula for X.

> **Q:** die, $\mathcal{G} = \{\varnothing, \{1,2,3\}, \{4,5,6\}, \Omega\}$.
> $Y = \mathbb{1}\{\omega \geq 4\}$: $Y^{-1}(\{1\}) = \{4,5,6\} \in \mathcal{G}$ ✓ measurable.
> $X = \mathbb{1}\{\omega \text{ even}\}$: $X^{-1}(\{1\}) = \{2,4,6\} \notin \mathcal{G}$ ×. Being told "high" does not tell you "even".

▼ **Proof**

> Example: $\Omega = \{1,2\}$, $X(\omega) = \omega$. With $\mathcal{F} = 2^{\Omega}$, every $\{X \in B\}$ is in $\mathcal{F}$. With $\mathcal{G} = \{\varnothing, \Omega\}$, $\{X \leq 1\} = \{1\} \notin \mathcal{G}$. Same X, different answer.

F3.4 ⭐ THE ATOM TEST

X is $\mathcal{G}$-measurable $\iff$ X is **constant on every atom of $\mathcal{G}$**

Atoms = the finest blocks $\mathcal{G}$ can distinguish.

> **Use when:** every discrete measurability question on the paper. It is also, unchanged, the test for *adapted* (M1.2) and for Doob–Dynkin (F4.4). **Learn this one test and you get three topics.**

> **Trick:** list the atoms, then scan X down each block. Two different values inside one block $\Rightarrow$ not measurable, and you have your counterexample for free.

▼ **Proof**

> ($\Rightarrow$) If X took two values $x \neq y$ on one atom A, the event $\{X = x\} \in \mathcal{G}$ would cut A into two non-empty pieces, but an atom is the smallest non-empty block of $\mathcal{G}$.
> ($\Leftarrow$) Then $X = \sum_k c_k \mathbf{1}_{A_k}$ and $\{X \in B\} = \bigcup\{A_k : c_k \in B\}$, a union of atoms, which is in $\mathcal{G}$.

F4. Generated σ-Fields & the Doob–Dynkin Lemma

F4.1 $\Sigma(\Xi)$

$\sigma(\xi) = \{\xi^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\}$

the smallest σ-field making ξ measurable

atoms of $\sigma(\xi)$ = the level sets $\{\xi = v\}$

> **Use when:** "find $\sigma(\xi)$" means "find the level sets, then take all unions of them". Pre-images automatically satisfy the σ-field axioms, so nothing needs checking.

> **Q:** die, $\xi(\omega) = \omega \bmod 3$. Atoms $\{3,6\},\{1,4\},\{2,5\}$, so $|\sigma(\xi)| = 2^3 = \mathbf{8}$.
> Check: $\xi^{-1}([0.5,5]) = \{1,4\} \cup \{2,5\} = \{1,2,4,5\}$ ✓

▼ **Proof**

> Preimages respect complements and countable unions, so $\{\xi^{-1}(B)\}$ is a σ-field, and ξ is measurable for it by construction. Any $\mathcal{G}$ making ξ measurable must contain every $\xi^{-1}(B)$, so it contains $\sigma(\xi)$: smallest.
> Atoms: two points can be separated by some $\xi^{-1}(B)$ exactly when ξ differs at them, so the atoms are the level sets $\{\xi = x\}$.

F4.2 Σ-FIELD OF A FAMILY & THE SEPARATION PRINCIPLE

$\sigma(\xi_i, i \in I) = \sigma\left(\bigcup_{i\in I} \sigma(\xi_i)\right)$

ω_a, ω_b separable $\iff \exists\, j$ with $\xi_j(\omega_a) \neq \xi_j(\omega_b)$

> **Use when:** comparing how much information different observables carry. Count atoms via the separation principle, then use F1.5.

> **Q:** two tosses, ξ_1, ξ_2 the indicators and $S = \xi_1 + \xi_2$.
> $\sigma(\xi_1)$: 2 atoms → **4** events. $\sigma(\xi_1, \xi_2)$: 4 atoms → **16**. $\sigma(S)$: HT, TH both give $S{=}1$ so they merge → 3 atoms → **8**.

▼ **Proof**

> The joint σ-field must contain every $\sigma(\xi_i)$; the smallest σ-field containing all of them is $\sigma\left(\bigcup_i \sigma(\xi_i)\right)$.
> Separation: if $\xi_j(\omega_a) \neq \xi_j(\omega_b)$, the event $\{\xi_j = \xi_j(\omega_a)\}$ holds at ω_a, not at ω_b. If every ξ_j agrees, the sets containing both points or neither form a σ-field holding all generators, so no event can separate them.

F4.3 BOREL FUNCTIONS COMPOSE

f Borel, ξ a r.v. $\implies f(\xi)$ is $\sigma(\xi)$-measurable

proof: $\{f(\xi) \in B\} = \xi^{-1}\left(f^{-1}(B)\right)$

> **Use when:** two-line proof, always the same two lines. Note it says $\sigma(f(\xi)) \subseteq \sigma(\xi)$ — applying a function can only *lose* information.

> **Q:** die, $\xi(\omega) = \omega$, $f(x) = x^2$.
> $\{f(\xi) = 16\} = \xi^{-1}(\{-4,4\}) = \{4\} \in \sigma(\xi)$ ✓
> **Trick:** f can merge atoms ($f(x) = x^2$ merges $\pm$), never split one.

F4.4 ⭐ DOOB–DYNKIN LEMMA

η is $\sigma(\xi)$-measurable $\iff \exists$ Borel f with $\eta = f(\xi)$

> **Use when:** "show η is a function of ξ" or "is η $\sigma(\xi)$-measurable?"
> **Meaning:** knowing the *value* $\xi(\omega)$ pins down $\eta(\omega)$ — no leftover randomness.

> **The one-line test:** does ξ ever take the *same* value at two ω where η *differs*? If yes → not measurable, and those two ω are your counterexample.
> **Q:** $\xi(\omega_1){=}\xi(\omega_2){=}2$, $\xi(\omega_3){=}5$; $\eta = 9, 9, 25$. Ties in ξ match ties in η ✓, so $\eta = f(\xi)$ with $f(2){=}9$, $f(5){=}25$.

▼ **Proof**

> ($\Leftarrow$) $\{\eta \in B\} = \{\xi \in f^{-1}(B)\}$ and $f^{-1}(B)$ is Borel, so the event is in $\sigma(\xi)$.
> ($\Rightarrow$, discrete ξ) ξ takes values $x_1, x_2, \ldots$; the atoms $A_k = \{\xi = x_k\}$ generate $\sigma(\xi)$. A $\sigma(\xi)$-measurable η is constant on each atom, say $\eta = c_k$ on A_k. Put $f(x_k) = c_k$ (and $f = 0$ elsewhere). Then $\eta = f(\xi)$.
> General ξ: approximate η by simple functions, apply the discrete case to each, pass to the limit.

F5. Distribution Measure & the d.f.

F5.1 PUSH-FORWARD (DISTRIBUTION) MEASURE

$$P_\xi(B) = P(\xi \in B) = P\big(\{\omega : \xi(\omega) \in B\}\big)$$

A probability measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

> **Use when:** P_ξ transports probability from the abstract Ω onto the real line. Once you have P_ξ you never need Ω again (that is E2.1).

> **Q:** two tosses, ξ = #heads, vs. a uniform $\omega \in [0,1]$ cut at 0.25 and 0.75. Completely different Ω, **identical** P_ξ.
> **Trick:** P_ξ records values and their probabilities and forgets everything else.

▼ **Proof**

> $P_\xi(B) = P(\xi^{-1}(B)) \geq 0$;
> $P_\xi(\mathbb{R}) = P(\Omega) = 1$. Disjoint B_i have disjoint preimages and $\xi^{-1}(\bigcup B_i) = \bigcup \xi^{-1}(B_i)$, so $P_\xi(\bigcup B_i) = \sum P_\xi(B_i)$.

F5.2 DISTRIBUTION FUNCTION

$$F_\xi(x) = P(\xi \leq x) = P_\xi\big((-\infty, x]\big)$$
$$P(a < \xi \leq b) = F(b) - F(a)$$

> **Use when:** note the interval formula is for $(a, b]$ — half-open, matching the $\leq$ in the definition. Getting this wrong only matters at a jump (F5.4).

> **Q:** #heads in two tosses: $F = 0,\ \frac{1}{4},\ \frac{3}{4},\ 1$ on $x < 0$, $[0,1)$, $[1,2)$, $x \geq 2$.
> $F(1.99) = \frac{3}{4}$, $F(1) = \frac{3}{4}$, $F(0.5) = \frac{1}{4}$.

▼ **Proof**

> For $a < b$: $\{\xi \leq b\} = \{\xi \leq a\} \sqcup \{a < \xi \leq b\}$, so $F(b) = F(a) + P(a < \xi \leq b)$.

F5.3 THE THREE PROPERTIES

(i) non-decreasing

(ii) right-continuous:

$$\lim_{h \downarrow 0} F(x + h) = F(x)$$

(iii) $F(-\infty) = 0, \quad F(+\infty) = 1$

> **Use when:** "is this function a d.f.?" or "prove the properties". All three, and **right-continuity is the one they mark**.

> **Proof of (ii):** $x_n \downarrow x$ gives $A_n = \{\xi \leq x_n\} \downarrow \{\xi \leq x\}$; apply continuity from above (F1.4). That is the whole argument.
> **Note:** left limits may differ — $F(1^-) = \frac{1}{4} \neq \frac{3}{4} = F(1)$ above.

▼ **Proof**

> (i) $x < y \Rightarrow \{\xi \leq x\} \subseteq \{\xi \leq y\}$. (ii) $\{\xi \leq x + \frac{1}{n}\} \downarrow \{\xi \leq x\}$; continuity from above (§F1.4). (iii) $\{\xi \leq -n\} \downarrow \varnothing$ and $\{\xi \leq n\} \uparrow \Omega$.

F5.4 JUMP SIZE = POINT MASS

$$P(\xi = x) = F(x) - F(x^-)$$
$$F \text{ continuous at } x \iff P(\xi = x) = 0$$

> **Use when:** deciding whether $<$ and $\leq$ can be swapped. **At a jump they cannot.** For an absolutely continuous r.v. there are no jumps, so they always can.

> **Q:** #heads: $P(\xi \leq 1) = \frac{3}{4}$ but $P(\xi < 1) = F(1^-) = \frac{1}{4}$.
> **Trick:** this is why every Brownian-motion formula on this sheet ignores strict vs non-strict — those laws are continuous.

▼ **Proof**

> $\{\xi \leq x - \frac{1}{n}\} \uparrow \{\xi < x\}$, so $F(x^-) = P(\xi < x)$. Then $F(x) - F(x^-) = P(\xi \leq x) - P(\xi < x) = P(\xi$. F is always right-continuous, so it is continuous at x iff this jump is 0.

F6. Densities & Joint Distributions

F6.1 ABSOLUTELY CONTINUOUS

$$P(\xi \in B) = \int_B f_\xi(x)\,dx, \quad f_\xi \geq 0,$$
$\int_{\mathbb{R}} f_\xi = 1$
$f_\xi(x) = F'_\xi(x)$ wherever F is differentiable

Use when: two traps live here. **(a)** a density is not a probability and may exceed 1 (uniform on $[0, 0.5]$ has $f = 2$). **(b)** $P(\xi = x_0) = 0$ for every single point.

Q: $\xi \sim \text{Exp}(1)$: $P(\xi > 1) = e^{-1} \approx 0.368$, $P(1 < \xi \leq 2) = e^{-1} - e^{-2} \approx 0.233$.

▼ **Proof**

$F(x) = P(\xi \leq x) = \int_{-\infty}^{x} f(u)\,du$. By the fundamental theorem of calculus $F'(x) = f(x)$ wherever f is continuous (in general at almost every x). Taking $B = \mathbb{R}$: $\int f = 1$.

F6.2 DISCRETE

$$P(\xi \in B) = \sum_{x_i \in B} P(\xi = x_i)$$

Use when: mass sits at isolated points. Only the listed values inside B count — anything between them contributes nothing.

Q: fair die, $P(\xi \in [2, 4.5]) = P(2) + P(3) + P(4) = \frac{1}{2}$. The endpoint 4.5 carries no mass.

▼ **Proof**

$\{\xi \in B\} = \bigsqcup_{x_i \in B} \{\xi = x_i\}$, a countable disjoint union; add the probabilities.

F6.3 JOINT DENSITY IN $\mathbb{R}^N$

$$P\big((\xi_1, \ldots, \xi_n) \in B\big) = \int \cdots \int_B f(x_1, \ldots$$
marginal: $f_{\xi_1}(x) = \int_{\mathbb{R}} f(x, y)\,dy$
independent $\iff$ $f(x, y) = f_\xi(x)\,f_\eta(y)$ (factorises)

Use when: independence is tested by *factorisation*, not by computing a correlation. Zero correlation does not imply independence — except inside a MVN (§1.5).

Q: unit square, $f \equiv 1$: $P(\xi_1 + \xi_2 \leq 1)$ = area of the triangle = $\frac{1}{2}$. $P(\xi_1 = \xi_2) = 0$ (a line has zero area).

▼ **Proof**

Marginal:
$P(\xi_1 \leq x) = P\big((\xi_1, \xi_2) \in (-\infty, x] \times \mathbb{R}\big) = \int_{-}^{a}$
, so the inner integral is the density of ξ_1.
Independence: if $f = f_\xi f_\eta$, then
$P(\xi \in A, \eta \in B) = \int_A f_\xi \int_B f_\eta = P(\xi \in A)P($
. Conversely $P(\xi \leq x, \eta \leq y) = F_\xi(x)F_\eta(y)$; differentiate in x and y.

F6.4 EVENTS AS REGIONS

$$\{\xi_1 \leq a,\ \xi_2 > b\} \longleftrightarrow B = (-\infty, a] \times$$

Use when: translate the worded event into a region first, *then* integrate. For a uniform density the integral is just the area.

Q: unit square, $P(\xi_1 \leq 0.3,\ \xi_2 > 0.5) = 0.3 \times 0.5 = \mathbf{0.15}$. **Trick:** this is exactly the skill §14.3 needs — a joint probability is a region, and a rectangle is four corner evaluations.

F7. Multivariate Normal & Joint MGF

F7.1 MVN CONSTRUCTION

$\mathbf{X} = A\mathbf{Z} + \boldsymbol{\mu}$, $\mathbf{Z} \sim N(\mathbf{0}, I_n)$ i.i.d. standard normals

$E[\mathbf{X}] = \boldsymbol{\mu}$, $\Sigma = \mathrm{Cov}(\mathbf{X}) = AA^\top$

> **Use when:** the full toolkit — linear combinations, joint MGF, conditional law — is in **§1.5–§1.9**. This card is the construction those rest on.

> **Q:** show Brownian motion is MVN. $W_1 = Z_1$ and $W_3 = W_1 + (W_3 - W_1) = Z_1 + \sqrt{2}Z_2$, so $A = \begin{pmatrix} 1 & 0 \\ 1 & \sqrt{2} \end{pmatrix}$, $\Sigma = AA^\top = \begin{pmatrix} 1 & 1 \\ 1 & 3 \end{pmatrix}$ — exactly $\min(s,t)$ ✓

▼ **Proof**

> $E\mathbf{X} = AE\mathbf{Z} + \boldsymbol{\mu} = \boldsymbol{\mu}$.
> $\mathrm{Cov}(\mathbf{X}) = E[(A\mathbf{Z})(A\mathbf{Z})^\top] = A\,E[\mathbf{Z}\mathbf{Z}^\top]A^\top = $.

F7.2 JOINT MGF DETERMINES EVERYTHING

$\phi_{\mathbf{X}}(\mathbf{t}) = E\left[e^{\mathbf{t}^\top \mathbf{X}}\right] = \exp\left(\mathbf{t}^\top \boldsymbol{\mu} + \tfrac{1}{2}\mathbf{t}^\top \Sigma\, \mathbf{t}\right)$

> **Use when:** an MVN is completely specified by $(\boldsymbol{\mu}, \Sigma)$ — so to identify a distribution you need only match mean and covariance. That is the whole of the §5.2 Wiener checklist.

> **Derivation:** $Y = \mathbf{t}^\top \mathbf{X}$ is scalar normal with mean $\mathbf{t}^\top \boldsymbol{\mu}$ and variance $\mathbf{t}^\top \Sigma \mathbf{t}$; apply $E[e^Y] = e^{\mu_Y + \sigma_Y^2/2}$ (§1.2). Two lines.

▼ **Proof**

> $\mathbf{t}^\top \mathbf{X} = \mathbf{c}^\top \mathbf{Z} + \mathbf{t}^\top \boldsymbol{\mu}$ with $\mathbf{c} = A^\top \mathbf{t}$: a sum of independent normals, so $N\left(\mathbf{t}^\top \boldsymbol{\mu},\ |\mathbf{c}|^2\right)$ and $|\mathbf{c}|^2 = \mathbf{t}^\top AA^\top \mathbf{t} = \mathbf{t}^\top \Sigma \mathbf{t}$. Then $E[e^Y] = e^{m + v/2}$ for $Y \sim N(m, v)$ (§1.2). "Determines everything" is the MGF uniqueness theorem (quoted).

E1. Expectation as a Lebesgue Integral

E1.1 INTEGRABILITY

$\xi \in L^1 \iff E|\xi| = \int_\Omega |\xi|\, dP < \infty$

then $E(\xi) = \int_\Omega \xi\, dP$

> **Use when:** "is ξ integrable?" **Always test $E|\xi|$, never $E\xi$.** A valid distribution can still fail to be integrable.

> **Q:** St Petersburg, $P(\xi = 2^k) = 2^{-k}$: $E|\xi| = \sum 1 = \infty$. **Not integrable.**
> **Q:** $P(\xi = k) = c/k^2$, $c = 6/\pi^2$: $E|\xi| = c\sum 1/k = \infty$. **Not integrable**, though P is a perfectly good distribution.
> **Trick:** bounded ⇒ integrable, free.

E1.2 WHY THE ABSOLUTE VALUE

$\xi = \xi^+ - \xi^-$, $\xi^+ = \max(\xi, 0)$, $\xi^- = \max(-\xi, 0)$

$|\xi| = \xi^+ + \xi^-$, so $E|\xi| < \infty \iff$ both parts finite

> **Use when:** if both parts are ∞ then $E\xi$ is $\infty - \infty$, which is **undefined** — not 0, even when the distribution is symmetric.

> **Q:** $P(\xi = 2^k) = P(\xi = -2^k) = 2^{-k-1}$. Perfectly symmetric, yet $E\xi^+ = E\xi^- = \infty$ and $E\xi$ does not exist.

▼ **Proof**

> $|\xi| = \xi^+ + \xi^-$ with both parts ≥ 0, so $E|\xi| = E\xi^+ + E\xi^-$, finite iff both are finite.

E1.3 INDICATORS

$$E[\mathbb{1}_A] = P(A)$$

Use when: the hinge between probability and expectation. It turns any probability into an expectation and back, and it is step one of the construction ladder.

Trick: it needs no ω-by-ω sum, so it works identically on a die and on $[0,1]$. That generality is the point of the Lebesgue integral.

▼ **Proof**

$\mathbf{1}_A$ is the simple function with value 1 on A and 0 off it:
$E[\mathbf{1}_A] = 1 \cdot P(A) + 0 \cdot P(A^c) = P(A)$.

E1.4 SIMPLE FUNCTIONS

$$\eta = \sum_{i=1}^{n} \eta_i \, \mathbb{1}_{A_i}, \quad A_i \text{ pairwise disjoint}$$

$$E(\eta) = \sum_{i=1}^{n} \eta_i \, P(A_i)$$

Use when: disjointness is for the *representation* to be unambiguous; linearity of the integral holds regardless. The A_i need not cover $\Omega - \eta = 0$ off their union.

Q: die, $\eta = 10 \cdot \mathbb{1}_{\{1,2\}} + 5 \cdot \mathbb{1}_{\{3\}}$:
$E\eta = 10(\frac{2}{6}) + 5(\frac{1}{6}) = \frac{25}{6} \approx \mathbf{4.17}$ ✓

E1.5 THE CONSTRUCTION LADDER

indicators $\to$ simple $\to$ non-negative (MCT) $\to$ general ($\xi^+ - \xi^-$)

Use when: "construct / define the Lebesgue integral". Name all four rungs and say which theorem carries you up each one — MCT for rung three.

Q: approximate $U \sim \text{Unif}[0,1]$ by $\eta_n = \lfloor 2^n U \rfloor / 2^n$.
$E\eta_n = \frac{1}{2}(1 - 2^{-n}) \uparrow \frac{1}{2} = EU$ ✓

E1.6 MARKOV, CHEBYSHEV, JENSEN

$$P(|\xi| \geq a) \leq \frac{E|\xi|}{a}$$

$$P(|\xi - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

$$h \text{ convex} \Rightarrow E[h(\xi)] \geq h(E\xi)$$

Use when: a bound is asked instead of an exact value, or a comparison between $E[h(\xi)]$ and $h(E\xi)$.

Q: $U \sim \text{Unif}[0,1]$.
$E(U^2) = \frac{1}{3} > \frac{1}{4} = (EU)^2$ (convex ✓).
$E\sqrt{U} = \frac{2}{3} \approx 0.667 < \sqrt{0.5} \approx 0.707$ (concave, inequality flips).

▼ **Proof**

Markov: $a\,\mathbf{1}_{\{|\xi| \geq a\}} \leq |\xi|$ pointwise; take E.
Chebyshev: Markov on $(\xi - \mu)^2$ with level a^2.
Jensen: a convex h lies above its tangent line at $m = E\xi$: $h(x) \geq h(m) + c(x - m)$. Put $x = \xi$, take E: $E[h(\xi)] \geq h(m) + 0$.

E2. Change of Variables for Expectation

E2.1 THE THEOREM

$$E\big(h(\xi)\big) = \int_\Omega h(\xi(\omega))\,dP(\omega) = \int_{\mathbb{R}} h(x)$$

h Borel, $h(\xi)$ integrable.

> **Use when:** this is the theorem that lets you **stop working on Ω**. Everything downstream (densities, MGFs, the whole BM sheet) is this identity.

> **Q:** two tosses, ξ = #heads, $h(x) = x^2$.
> On Ω: $\frac{1}{4}(4+1+1+0) = 1.5$. On $\mathbb{R}$:
> $0(\frac{1}{4}) + 1(\frac{1}{2}) + 4(\frac{1}{4}) = 1.5$ ✓
> **Trick:** the $\mathbb{R}$ version merges HT and TH — fewer terms, same answer.

▼ **Proof**

> Climb the ladder (§E1.5). $h = \mathbf{1}_B$:
> $E[\mathbf{1}_B(\xi)] = P(\xi \in B) = P_\xi(B) = \int \mathbf{1}_B\,dP_\xi$.
> Linearity gives simple h; monotone limits (MCT) give $h \geq 0$; $h = h^+ - h^-$ gives general h.

E2.2 THE TWO WORKING FORMS

continuous: $E[h(\xi)] = \int_{-\infty}^{\infty} h(x) f_\xi(x)\,dx$

discrete: $E[h(\xi)] = \sum_i h(x_i)\,P(\xi = x_i)$

> **Use when:** every numerical expectation you will ever compute. The measure-theoretic statement collapses to one of these two lines.

> **Q:** $Z \sim N(0,1)$,
> $E[e^Z] = \int e^x \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx = e^{1/2} \approx \mathbf{1.649}$.
> **Trick:** complete the square — that single computation is §1.2 and powers all of GBM.

▼ **Proof**

> Continuous: $dP_\xi = f_\xi(x)\,dx$, so $\int h\,dP_\xi = \int h(x) f_\xi(x)\,dx$. Discrete: P_ξ puts mass $P(\xi = x_i)$ at each x_i, so the integral is $\sum_i h(x_i) P(\xi = x_i)$.

E2.3 $E[H(\Xi)] \neq H(E\Xi)$

apply h to each value **before** averaging, never after

> **Use when:** the single most common conceptual error in the subject. It is why GBM has an $e^{\sigma^2 t/2}$ and why the mean of a log-normal exceeds its median.

> **Q:** die,
> $E(1/\xi) = \frac{1}{6}(1 + \frac{1}{2} + \cdots + \frac{1}{6}) \approx \mathbf{0.408}$,
> whereas $1/E\xi = 1/3.5 \approx 0.286$. Not equal.

▼ **Proof**

> Counterexample: $Z \sim N(0,1)$, $h(x) = x^2$:
> $E[h(Z)] = 1$ but $h(EZ) = 0$. For convex h Jensen (§E1.6) gives the direction:
> $E[h] \geq h(E)$.

E2.4 EVALUATIONS WORTH MEMORIZING

$$E|Z| = \sqrt{2/\pi} \approx 0.798, \quad Z \sim N(0,1)$$

$\xi \geq 0$: $E\xi = \int_0^\infty P(\xi > t)\,dt$ (tail formula)

$\xi \sim \text{Poisson}(\lambda)$: $E[\xi(\xi-1)] = \lambda^2$

> **Use when:** the tail formula turns an awkward density integral into a survival-function integral — it is how $E[\tau] = \infty$ is shown in §12.5.

> **Q:** $\xi \sim \text{Exp}(1)$: $E\xi = \int_0^\infty e^{-t} dt = \mathbf{1}$ with no integration by parts.
> **Note:** $E|Z| = \sqrt{2/\pi}$ reappears as $E[M(t)] = \sqrt{2t/\pi}$ in §13.4, because $M(t) \overset{d}{=} |W(t)|$.

▼ **Proof**

> $E|Z| = 2\int_0^\infty x \frac{e^{-x^2/2}}{\sqrt{2\pi}} dx = \frac{2}{\sqrt{2\pi}}\big[-e^{-x^2/2}\big]_0^\infty =$.
> Tail: $\xi = \int_0^\infty \mathbf{1}_{\{\xi > t\}} dt$ for $\xi \geq 0$; take E and swap E and $\int$ (everything ≥ 0).
> Poisson:
> $\sum_k k(k-1) e^{-\lambda} \frac{\lambda^k}{k!} = \lambda^2 \sum_{k \geq 2} e^{-\lambda} \frac{\lambda^{k-2}}{(k-2)!} = \lambda^2$.

E3. **Conditional Probability & Independence**

E3.1 DEFINITION

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

> **Use when:** conditioning *rescales*: outcomes outside B get probability 0, and B becomes the new sample space of total mass 1.

> **Q:** die, told "even". $P(\{6\} \mid B) = \frac{1/6}{1/2} = \frac{1}{3}$, $P(\{\geq 4\} \mid B) = \frac{2/6}{1/2} = \frac{2}{3}$.

E3.2 SUBSET & DISJOINT SPECIAL CASES

$A \subseteq B$: $P(A \mid B) = \dfrac{P(A)}{P(B)}$ **(not 1)**, $P(B \mid A) = 1$

$A \cap B = \varnothing$: $P(A \mid B) = 0$

> **Use when:** the $A \subseteq B$ case is a standard trap — conditioning on the *larger* event makes A more likely, not certain.

> **Q:** $A = \{6\} \subseteq B = \{2, 4, 6\}$: $P(A \mid B) = \frac{1}{3}$ but $P(B \mid A) = 1$ (a six is always even).

▼ **Proof**

> $A \subseteq B$: $A \cap B = A$, so $P(A \mid B) = P(A)/P(B)$ and $P(B \mid A) = P(A)/P(A) = 1$. Disjoint: $P(A \cap B) = P(\varnothing) = 0$.

E3.3 TOTAL PROBABILITY & BAYES

$$P(A) = \sum_i P(A \mid B_i)\, P(B_i) \quad (\{B_i\} \text{ a partition})$$

$$P(B_j \mid A) = \frac{P(A \mid B_j)P(B_j)}{\sum_i P(A \mid B_i)P(B_i)}$$

> **Use when:** the problem describes a two-stage experiment ("pick a machine, then a part"). **Write the denominator as the full sum first** — that is the whole method.

> **Q:** prevalence 1%, sensitivity 99%, false-positive rate 5%.
> $P(+) = .99(.01) + .05(.99) = .0594$, so $P(\text{sick} \mid +) = .0099/.0594 \approx \mathbf{0.167}$.
> **Trick:** only ~17% — the base rate dominates, because healthy people vastly outnumber sick ones.

▼ **Proof**

> $A = \bigsqcup_i (A \cap B_i)$, so $P(A) = \sum_i P(A \cap B_i) = \sum_i P(A \mid B_i)P(B_i)$. Bayes: $P(B_j \mid A) = \frac{P(A \cap B_j)}{P(A)}$; write the top as $P(A \mid B_j)P(B_j)$ and the bottom by total probability.

E3.4 ☆ PAIRWISE ≠ MUTUAL INDEPENDENCE

two events: $P(A \cap B) = P(A)P(B)$

n events: the product rule for **every** sub-collection of size $2, \ldots, n$

that is $2^n - n - 1$ equations ($n{=}3 \to 4$, $n{=}4 \to 11$)

> **Use when:** the question says "mutually independent". Checking only the pairs is the intended mistake.

> **Q:** two dice. A = first even, B = second even, C = sum even. All three pairs give $\frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$ ✓, but $P(A \cap B \cap C) = \frac{1}{4} \neq \frac{1}{8}$ ×.
> **Trick:** any two of them determine the third — that is the giveaway.

▼ **Proof**

> Sub-collections of size ≥ 2: all 2^n subsets minus the empty one minus the n singletons, $2^n - n - 1$.
> Pairwise but not mutual: two fair coins, A = first is H, B = second is H, C = both the same. Each pair: $P = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$. But $P(A \cap B \cap C) = \frac{1}{4} \neq \frac{1}{8}$.

E3.5 DISJOINT IS THE OPPOSITE OF INDEPENDENT

$P(A), P(B) > 0$ and $A \cap B = \varnothing \Rightarrow$ **dependent**

$A \perp B \Rightarrow A \perp B^c, \;\; A^c \perp B, \;\; A^c \perp B^c$

> **Use when:** disjoint means knowing A *rules out* B — maximal dependence, not independence. Students swap these two words constantly.

> **Proof of the second line:**
> $P(A \cap B^c) = P(A) - P(A \cap B) = P(A)(1 -$
> ✓ One line.

▼ **Proof**

> Disjoint: $P(A \cap B) = 0 < P(A)P(B)$.
> Complement:
> $P(A \cap B^c) = P(A) - P(A \cap B) = P(A) - P$
> . Apply again to get the other two.

E4. Conditional Expectation & σ-Fields

E4.1 IT IS A RANDOM VARIABLE

$E(X \mid Y{=}y)$ — a **number**

$E(X \mid Y) = E(X \mid \sigma(Y))$ — a **random variable**

$E(X \mid \mathcal{G})$ — the general version, $\mathcal{G} \subseteq \mathcal{F}$

> **Use when:** know which one is being asked for. Martingale theory is entirely about the random-variable version.

> **Q:** two dice, X = sum, Y = first die. Number: $E(X \mid Y{=}4) = 4 + 3.5 = \mathbf{7.5}$. Random variable: $E(X \mid Y) = \mathbf{Y + 3.5}$, which equals 7.5 at $\omega = (4,1)$ and 5.5 at $\omega = (2,6)$.

E4.2 DEFINING PROPERTIES

(i) $E(X \mid \mathcal{G})$ is $\mathcal{G}$-measurable

(ii)

$$\int_G E(X \mid \mathcal{G})\, dP = \int_G X\, dP \quad \forall\, G \in \mathcal{G}$$

unique a.s.

> **Use when:** "define" or "prove" questions. Every rule below is derived from these two, and a proof is always: check (i), then check (ii).

> **Reading it:** (i) says it cannot see finer detail than $\mathcal{G}$ allows; (ii) says it must have the right average on every block. Together: **it is the block-average of X.**

E4.3 ⭐ PARTITION FORMULA

$\mathcal{G} = \sigma(A_1, \ldots, A_k)$ a partition, $P(A_i) > 0$:

$$E(X \mid \mathcal{G})(\omega) = \frac{E[X\, \mathbb{1}_{A_i}]}{P(A_i)} = \frac{\int_{A_i} X\, dP}{P(A_i)}$$

for $\omega \in A_i$

> **Use when: every finite or discrete problem.** No integration — it is the plain average of X over the atom containing ω.

> **Q:** die, $X = \omega$, $\mathcal{G} = \sigma(\{1,2,3\}, \{4,5,6\})$. Low: $\frac{1+2+3}{3} = \mathbf{2}$. High: $\frac{4+5+6}{3} = \mathbf{5}$.
> **Trick:** write the atoms, average X inside each, done. Derivation: put $Y = c_i$ on A_i into (ii) and divide by $P(A_i)$.

▼ **Proof**

> $Y = \sum_i c_i \mathbf{1}_{A_i}$ with $c_i = E[X\mathbf{1}_{A_i}]/P(A_i)$ is constant on atoms, so $\mathcal{G}$-measurable. On an atom: $E[Y\mathbf{1}_{A_j}] = c_j P(A_j) = E[X\mathbf{1}_{A_j}]$. Every $G \in \mathcal{G}$ is a union of atoms; add. So Y passes the defining test (§E4.2), and by uniqueness $Y = E(X \mid \mathcal{G})$.

E4.4 TOWER PROPERTY

$$E\big[E(X \mid \mathcal{G})\big] = E(X)$$

$$\mathcal{H} \subseteq \mathcal{G} \Rightarrow E\big[E(X \mid \mathcal{G}) \mid \mathcal{H}\big] = E(X \mid \mathcal{H})$$

> **Use when: the smaller σ-field always wins.** This one line proves the multi-step martingale property (M2.3) and constant mean (M2.4).

> **Q:** die low/high: $\frac{1}{2}(2) + \frac{1}{2}(5) = 3.5 = EX$ ✓
> **Q:** random coin, bias $p \sim \mathrm{Unif}[0,1]$, X = heads in 10 tosses. $E(X \mid p) = 10p$, so $E(X) = E[10p] = \mathbf{5}$.

▼ **Proof**

> Let $Y = E(X \mid \mathcal{G})$. Its defining property: $E[Y\mathbf{1}_A] = E[X\mathbf{1}_A]$ for every $A \in \mathcal{G}$.
> (1) Take $A = \Omega$: $E[Y] = E[X]$.
> (2) Let $Z = E(Y \mid \mathcal{H})$. For $A \in \mathcal{H} \subseteq \mathcal{G}$: $E[Z\mathbf{1}_A] = E[Y\mathbf{1}_A] = E[X\mathbf{1}_A]$. Z is $\mathcal{H}$-measurable and passes the defining test of $E(X \mid \mathcal{H})$, so by uniqueness $Z = E(X \mid \mathcal{H})$.

E4.5 ☆ THE THREE OPERATIONAL RULES

1. linearity:
$E(aX + bY \mid \mathcal{G}) = aE(X \mid \mathcal{G}) + bE(Y \mid$
2. taking out what is known: Z $\mathcal{G}$-measurable $\Rightarrow E(ZX \mid \mathcal{G}) = Z\,E(X \mid \mathcal{G})$
3. independence:
$X \perp \mathcal{G} \Rightarrow E(X \mid \mathcal{G}) = E(X)$

> **Use when: this trio solves every conditional-expectation computation on the paper**, including all the Brownian ones in §7.6. Name the rule you are using at each step.

> **Q:** two dice:
> $E(X_1X_2 \mid X_1) = X_1E(X_2) = \mathbf{3.5X_1}$ (rules 2 then 3).
> **Q:** $E\big[(X_1+X_2)^2 \mid X_1\big] = X_1^2 + 7X_1 + \frac{91}{6}$ (expand, then 1, 2, 3).

▼ **Proof**

Each time, check the candidate is $\mathcal{G}$-measurable and has the right integral over every $G \in \mathcal{G}$ (§E4.2).
1. $aE(X \mid \mathcal{G}) + bE(Y \mid \mathcal{G})$: integrals add linearly.
2. $Z = \mathbf{1}_C$, $C \in \mathcal{G}$:
$\int_G \mathbf{1}_C E(X \mid \mathcal{G}) = \int_{G\cap C} E(X \mid \mathcal{G}) = \int_{G\cap C} X =$. Extend to simple, then general Z.
3. The constant EX is $\mathcal{G}$-measurable and $\int_G X\,dP = E[X\mathbf{1}_G] = EX \cdot P(G) = \int_G EX\,d$ by independence.

E4.6 THE TWO EXTREMES

X is $\mathcal{G}$-measurable $\Rightarrow E(X \mid \mathcal{G}) = X$ (full information)
$\mathcal{G} = \{\varnothing, \Omega\} \Rightarrow E(X \mid \mathcal{G}) = E(X)$ (no information)

> **Use when:** every martingale proof is these two extremes plus E4.5. Split X_{t+1} into a known part (extreme 1) and an independent increment (extreme 2), and you are done.

> **Trick:** that split is exactly the increment trick of §6.3, seen from the σ-field side.

▼ **Proof**

Full information: X itself is $\mathcal{G}$-measurable and trivially $\int_G X = \int_G X$. No information: a $\{\varnothing, \Omega\}$-measurable variable is a constant c; $G = \Omega$ forces $c = EX$.

E4.7 CONDITIONAL JENSEN & VARIANCE DECOMPOSITION

h convex $\Rightarrow E[h(X) \mid \mathcal{G}] \geq h\big(E[X \mid \mathcal{G}]\big)$
$\mathrm{Var}(X) = E\big[\mathrm{Var}(X \mid \mathcal{G})\big] + \mathrm{Var}\left(E[X\right.$

> **Use when:** conditional Jensen is why $|X_t|$ and X_t^2 are **sub**-martingales whenever X_t is a martingale — a standard T/F.

> **Q:** X_t MG
> $\Rightarrow E[X_{t+1}^2 \mid \mathcal{F}_t] \geq (E[X_{t+1} \mid \mathcal{F}_t])^2 = X_t^2$.
> So X_t^2 is a sub-MG, and $X_t^2 - t$ is the correction that restores the martingale (§7.2).

▼ **Proof**

Jensen: a convex h is the max of lines below it, $h(x) = \sup(a + bx)$. For each line, $E[h(X) \mid \mathcal{G}] \geq a + bE[X \mid \mathcal{G}]$; take the sup.
Variance:
$\mathrm{Var}(X \mid \mathcal{G}) = E[X^2 \mid \mathcal{G}] - E[X \mid \mathcal{G}]^2$. So $E[\mathrm{Var}(X \mid \mathcal{G})] = EX^2 - E\big[E[X \mid \mathcal{G}]^2\big]$ and $\mathrm{Var}(E[X \mid \mathcal{G}]) = E\big[E[X \mid \mathcal{G}]^2\big] - (EX)^2$.
Add: $EX^2 - (EX)^2$.

M1. Filtrations & Adapted Processes

M1.1 FILTRATION

$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots \subseteq \mathcal{F}$

$\mathcal{F}_t$ = everything observable by time t

> **Use when:** information only accumulates — the inclusion never reverses. That nesting is what every tower-property argument leans on.

> **Q:** two tosses. $\mathcal{F}_0 = \{\varnothing, \Omega\}$ (2 events); $\mathcal{F}_1 = \{\varnothing, \{HH, HT\}, \{TH, TT\}, \Omega\}$ (4); $\mathcal{F}_2 = 2^\Omega$ (16).
> **Note:** $\{HT\} \notin \mathcal{F}_1$ — after one toss you cannot tell HT from HH.

M1.2 ADAPTED

(X_t) is adapted $\iff$ X_t is $\mathcal{F}_t$-measurable for every t

atom test: X_t is **constant on every atom of** $\mathcal{F}_t$

> **Use when:** "is X_t adapted?" This is F3.4 with $\mathcal{G} = \mathcal{F}_t$ — the same test, relabelled.

> **Q:** two tosses. X_1 = heads so far: $1, 1, 0, 0$ on HH, HT, TH, TT — constant on each atom ✓ **adapted**.
> Y_1 = heads in *both* tosses: $2, 1, 1, 0$ — differs across HH and HT, one atom ✗ **not adapted** (it needs toss 2).

M1.3 NATURAL FILTRATION

$\mathcal{F}_t = \sigma(X_0, X_1, \ldots, X_t)$

> **Use when:** no filtration is stated. It is the smallest one to which X is adapted, so "adapted" becomes automatic and the real content moves to the martingale equality.

> **Trick:** if the question says "let $\mathcal{F}_n = \sigma(\eta_1, \ldots, \eta_n)$", it has handed you the natural filtration and is telling you not to waste time on condition (i).

M1.4 FUNCTIONS PRESERVE ADAPTEDNESS

f Borel, (X_t) adapted $\Rightarrow$ $(f(X_t))$ adapted

> **Use when:** justifying that S_t^2, e^{S_t}, $|S_t|$ etc. are adapted. Same two-line pre-image proof as F4.3.

> **Trick:** f can *merge* atoms but can never *split* one, so information can only be lost. That is the whole content.

▼ **Proof**

> $\{f(X_t) \in B\} = \{X_t \in f^{-1}(B)\}$, and $f^{-1}(B)$ is Borel, so the event is in $\mathcal{F}_t$ (§F4.3).

M1.5 INCREMENTS ARE NOT $\mathcal{F}_T$-MEASURABLE

$Y_t := X_{t+1} - X_t \in \mathcal{F}_{t+1}$, **not** $\mathcal{F}_t$

> **Use when:** this is precisely why a betting stake must be *predictable* (M7.2) and why $E[X_{t+1} \mid \mathcal{F}_t]$ is a real computation rather than a substitution.

> **Q:** binomial $S_0{=}100$, $u{=}1.2$, $d{=}0.8$. On the atom $A_u = \{uu, ud\}$ of $\mathcal{F}_1$: $S_2 - S_1 = +24$ on uu but -24 on ud. Not constant on the atom ✗

▼ **Proof**

> $Y_t = X_{t+1} - X_t$ uses X_{t+1}, which is in $\mathcal{F}_{t+1}$. Counterexample for $\mathcal{F}_t$: coin-flip walk, Y_t = the $(t{+}1)$-th flip; $\{Y_t = 1\}$ has probability $\frac{1}{2}$ and is independent of $\mathcal{F}_t$, so it is not one of $\mathcal{F}_t$'s events (those are decided by the first t flips).

M2. Discrete-Time Martingales

M2.1 ⭐ DEFINITION — THREE CONDITIONS

(i) (X_t) adapted

(ii) $E|X_t| < \infty$ (integrable)

(iii) $E[X_{t+1} \mid \mathcal{F}_t] = X_t \quad P$-a.s.

> **Use when:** "Is X_n a martingale? (Yes/No)". State all three, but **(iii) is what carries the marks** — show the one-line conditional computation.

> **Paper:** CT-2'25 Q3 — $X_0{=}1$, $X_n = 2^n U_1 \cdots U_n$, $U_i \sim \text{Unif}(0,1)$. $E[X_{n+1} \mid \mathcal{F}_n] = 2X_n\, E[U_{n+1}] = 2X_n \cdot \frac{1}{2} = X$ ✓ **Yes**.
> **Trick:** take out X_n (E4.5 rule 2), then independence (rule 3). Two steps, every time.

M2.2 SUB- AND SUPER-MARTINGALES

sub-MG: $E[X_{t+1} \mid \mathcal{F}_t] \geq X_t$ (favourable to you)

super-MG: $E[X_{t+1} \mid \mathcal{F}_t] \leq X_t$ (unfavourable)

> **Use when:** the naming is counter-intuitive, so fix it once: **a SUPERmartingale goes DOWN.** Sub goes up.

> **Q:** roulette on red (18/37): $E\eta = -\frac{1}{37}$, a **super**-MG — the casino's game, and you lose about 2.7 paise per rupee staked.

M2.3 MULTI-STEP PROPERTY

$E[X_{t+s} \mid \mathcal{F}_t] = X_t$ for every $s \geq 1$

> **Use when:** conditioning across more than one step. Proof is the tower property applied repeatedly, peeling one step at a time.

> **Proof:**
> $E[X_{t+s} \mid \mathcal{F}_t] = E\big[E[X_{t+s} \mid \mathcal{F}_{t+s-1}] \mid \mathcal{F}_t\big] = E$ ✓

▼ **Proof**

> Tower:
> $E[X_{t+s} \mid \mathcal{F}_t] = E\big[E(X_{t+s} \mid \mathcal{F}_{t+s-1}) \mid \mathcal{F}_t\big] = E$. Repeat s times to reach X_t.

M2.4 ⭐ CONSTANT MEAN

$E[X_t] = E[X_0]$ for every t

> **Use when:** the recurring MCQ. Proof: take $E[\cdot]$ of M2.3 and use the tower property with $\mathcal{H} = \{\varnothing, \Omega\}$.

> **Paper:** CT-1'26 Q8, **all four sets** — "Which statement is TRUE?" The correct option is always worded "**the mean of a martingale process is constant with respect to time**"; only its letter moves: Set-1 **(c)**, Set-2 **(a)**, Set-3 **(d)**, Set-4 **(b)**.
> **Trick:** don't read the distractors — find that sentence.

▼ **Proof**

> $E[X_{t+1}] = E\big[E(X_{t+1} \mid \mathcal{F}_t)\big] = E[X_t]$ (tower, then the MG property). Repeat: $E[X_t] = E[X_{t-1}] = \cdots = E[X_0]$.

M2.5 THE STANDARD EXAMPLES

$S_n = \sum_{i \leq n} Y_i,\ E[Y] = 0$ → MG

$S_n - n\,E[Y]$ → MG for any i.i.d. Y (drift removed)

$S_n^2 - n\,\text{Var}(Y)$ → MG when $E[Y] = 0$

$\prod_{i \leq n} Y_i,\ E[Y] = 1$ → MG

> **Use when:** the first and last are §2.1 / §2.2; the middle two are the discrete twins of W_t and $W_t^2 - t$ (§7.1, §7.2).

> **Q:** biased walk $p = 0.6$: $E[Y] = 0.2$, so S_n is **not** a MG — but $S_n - 0.2n$ **is**.
> **Trick:** if it is not a martingale, subtract the drift and it becomes one. That is the whole idea behind the $-t$ in $W_t^2 - t$.

▼ **Proof**

> $\mathcal{F}_n = \sigma(Y_1 .. Y_n)$; Y_{n+1} is independent of $\mathcal{F}_n$, so $E[Y_{n+1} \mid \mathcal{F}_n] = EY$ (§E4.5).
> 1–2. $E[S_{n+1} \mid \mathcal{F}_n] = S_n + EY$; subtract $(n+1)EY$ and you get $S_n - nEY$.
> 3. $S_{n+1}^2 = S_n^2 + 2S_nY_{n+1} + Y_{n+1}^2$. Conditional mean: $S_n^2 + 2S_n \cdot 0 + \text{Var}\,Y$. So $S_{n+1}^2 - (n+1)\,\text{Var}\,Y \mapsto S_n^2 - n\,\text{Var}\,Y$.
> 4. $\Pi_{n+1} = \Pi_n Y_{n+1}$; take out Π_n: $\Pi_n\, EY = \Pi_n$.
> Integrability needs $E|Y| < \infty$ ($EY^2 < \infty$ for 3).

M3. The Binomial Market Model

M3.1 THE TREE

$S_t = v_t S_{t-1}, \quad v_t \in \{u, d\}, \quad 0 < d < u$
$\Omega = \{u, d\}^T$; after k up-moves:
$S_t = S_0\, u^k d^{\,t-k}$

> **Use when:** draw the tree with numbers before anything else. Almost every error in this topic is a mis-drawn tree, not a mis-applied formula.

> **Q:** S_0=100, u=1.2, d=0.8, T=2:
> $S_2 \in \{144, 96, 96, 64\}$ on uu, ud, du, dd.

M3.2 FILTRATION & RECOMBINATION

$\mathcal{F}_t = \sigma(S_0, \ldots, S_t)$; 2^t atoms at time t

> **Use when:** the tree *recombines* (ud and du give the same price) but the *filtration does not* — $\mathcal{F}_t$ records the whole path, not just the current price.

> **Q:** above, $S_2 = 96$ on both ud and du. $\sigma(S_2)$ alone cannot separate them;
> $\mathcal{F}_2 = \sigma(S_0, S_1, S_2)$ can, because S_1 differs (120 vs 80).
> **Trick:** that distinction is exactly why path-dependent options exist.

▼ **Proof**

> Each time-t history is a word of t letters from $\{u, d\}$; since $u \neq d$ the price path tells you the word, so all 2^t histories are distinguishable: 2^t atoms. (The *prices* recombine into $t+1$ values, but $\mathcal{F}_t$ remembers the path.)

M3.3 MARTINGALE CONDITION FOR THE STOCK ($R = 0$)

$E[S_{t+1} \mid \mathcal{F}_t] = S_t\big(pu + (1-p)d\big)$

MG

$$\iff pu + (1-p)d = 1 \iff p = \frac{1 -}{u -}$$

> **Use when:** "for which p is S_t a martingale?" Larger p → sub-MG; smaller → super-MG.

> **Q:** u=1.2, d=0.8: $p = \frac{0.2}{0.4} = \mathbf{1/2}$.
> u=2, d=0.5: $p = \frac{0.5}{1.5} = \mathbf{1/3}$.
> **Check:** $\frac{1}{3}(2) + \frac{2}{3}(0.5) = 1$ ✓

▼ **Proof**

> $S_{t+1} = v_{t+1} S_t$ with S_t known and v_{t+1} independent of $\mathcal{F}_t$:
> $E[S_{t+1} \mid \mathcal{F}_t] = S_t E[v_{t+1}] = S_t(pu + (1-p)d)$
> . Equals S_t iff $pu + (1-p)d = 1$ iff
> $p(u-d) = 1 - d$.

M3.4 ⭐ RISK-NEUTRAL PROBABILITY (WITH INTEREST R)

$$p^* = \frac{1 + r - d}{u - d}$$

$\dfrac{S_t}{(1+r)^t}$ is a martingale under p^*

> **Use when:** an interest rate is given. **p^* has nothing to do with the real probability** — the real p never enters a price. With $r = 0$ this reduces to M3.3.

> **Paper:** CT-2'25 Q1 — u=1.25, d=0.25, r=0.15: $p^* = \frac{1.15 - 0.25}{1.00} = \mathbf{0.9}$.

▼ **Proof**

> We need $E^*\left[\dfrac{S_{t+1}}{(1+r)^{t+1}} \,\middle|\, \mathcal{F}_t\right] = \dfrac{S_t}{(1+r)^t}$.
> Left side $= \dfrac{S_t\,(p\,u + (1-p)\,d)}{(1+r)^{t+1}}$, since S_{t+1} is uS_t or dS_t.
> Equal to the right side
> $\iff p\,u + (1-p)\,d = 1 + r \iff p = \frac{1 +}{u -}$
> . This is in $(0, 1)$ exactly when $d < 1 + r < u$.

M3.5 GROWTH OF THE MEAN

$$E[S_t] = S_0\big(pu + (1-p)d\big)^t$$

> **Use when:** "find $E[S_3]$". Compute $E[v] = pu + (1-p)d$ once, then raise it to the power t. Never expand the tree.

> **Q:** S_0=50, u=1.1, d=0.9. p=0.5: $E[v] = 1$, MG, $E[S_3] = \mathbf{50}$. p=0.7: $E[v] = 1.04$, $E[S_3] = 50(1.04)^3 = \mathbf{56.24}$.

▼ **Proof**

> $E[S_t] = E\big[E(S_t \mid \mathcal{F}_{t-1})\big] = (pu + (1-p)d)\,E$ by §M3.3; iterate t times.

M4. No-Arbitrage Pricing in the Binomial Model

M4.1 NO-ARBITRAGE CONDITION

$$d < 1 + r < u$$

> **Use when:** "show the market is arbitrage-free". If $1 + r \le d$ the stock beats the bond for sure; if $1 + r \ge u$ the bond beats the stock for sure. Equivalent to $0 < p^* < 1$.

> **Q:** Endsem'25 Q5(b)(i) — S_0=5, $S_1 \in \{6, 40/9\}$, r=1/9. Then $u = 6/5$, $d = 8/9$, $1 + r = 10/9$. Check $\frac{8}{9} < \frac{10}{9} < \frac{6}{5}$ ✓

▼ **Proof**

> If $1 + r \le d$: borrow S_0, buy the stock. At time 1 you hold $\ge dS_0 \ge (1+r)S_0$ = what you owe, with strict gain in the up state: arbitrage.
> If $1 + r \ge u$: short the stock, lend S_0. You receive $(1+r)S_0 \ge uS_0 \ge S_1$, strict in the down state: arbitrage.
> If $d < 1 + r < u$: $p^* \in (0,1)$ is an EMM, so no arbitrage (§D3.3).

M4.2 RISK-NEUTRAL PRICING FORMULA

$$V_0 = \frac{1}{(1+r)^T}\,E^*\big[\text{payoff}\big]$$

one period: $V_0 = \dfrac{p^*V_u + (1-p^*)V_d}{1+r}$

> **Use when:** pricing anything. Multi-period: work **backwards** node by node, discounting one step at a time.

> **Q:** CT-2'25, put with K=225, S_0=200, u=1.25, d=0.25, r=0.15.
> $V_u = (225 - 250)^+ = 0$,
> $V_d = (225 - 50)^+ = 175$, $p^* = 0.9$.
> $V_0 = \frac{0.9(0)+0.1(175)}{1.15} = \frac{17.5}{1.15} = \mathbf{15.22}$ ✓

▼ **Proof**

> One period: the replicating portfolio costs $\frac{p^*V_u+(1-p^*)V_d}{1+r}$ (§M4.4), and any other price is an arbitrage (§A1.5 proof). Many periods: repeat at every node, backwards (§D4.2); the tower property collapses the nested averages to $(1+r)^{-T}E^*[\text{payoff}]$.

M4.3 ⭐ REPLICATING PORTFOLIO

$$\Delta = \frac{V_u - V_d}{(u-d)\,S_0} \quad \text{(shares)}$$

$$B = \frac{u\,V_d - d\,V_u}{(1+r)(u-d)} \quad \text{(cash / bond)}$$

$$V_0 = \Delta S_0 + B$$

> **Use when:** "construct the replicating portfolio, of the form (Δ, B)". Δ is the hedge ratio; it is negative for a put.

> **Paper:** CT-2'25 Q1 — put, V_u=0, V_d=175, $u-d=1$, S_0=200, r=0.15.
> $\Delta = \frac{0-175}{1\cdot 200} = \mathbf{-0.875}$;
> $B = \frac{1.25(175)-0.25(0)}{1.15(1)} = \frac{218.75}{1.15} = \mathbf{190.22}$.
> Answer $(\Delta, B) = (-0.875,\ 190.22)$ ✓ and $V_0 = -0.875(200) + 190.22 = \mathbf{15.22}$ ✓

▼ **Proof**

> Match both states: $\Delta u S_0 + B(1+r) = V_u$ and $\Delta d S_0 + B(1+r) = V_d$. Subtract: $\Delta(u-d)S_0 = V_u - V_d$. Then $B(1+r) = V_u - u\frac{V_u - V_d}{u-d} = \frac{uV_d - dV_u}{u-d}$.

M4.4 TWO ROUTES, ONE ANSWER

$$\Delta S_0 + B \;=\; \frac{p^* V_u + (1-p^*)V_d}{1+r}$$

> **Use when:** they are algebraically the same. **Use replication when the question asks for (Δ, B); use risk-neutral when it asks only for the price** — and use the other one to check.

> **Trick:** in a two-part question ("price it", then "replicate it") do replication first — you get the price for free from $V_0 = \Delta S_0 + B$.

▼ **Proof**

> $\Delta S_0 + B = \frac{V_u - V_d}{u-d} + \frac{uV_d - dV_u}{(1+r)(u-d)} = \frac{(1+r-d)V_u + (u$
> , using $p^* = \frac{1+r-d}{u-d}$, $1-p^* = \frac{u-1-r}{u-d}$.

M4.5 ⭐ THE "NO SURE WIN" / ARBITRAGE-C QUESTION

S_0 today; at time 1 either S_u or S_d (already present values)

option to buy y shares at price Ky costs cy

1. $p = \dfrac{S_0 - S_d}{S_u - S_d}$
2. payoffs $(S_u - K)^+$ and $(S_d - K)^+$
3. $c^* = p\,(S_u - K)^+ + (1-p)(S_d - K)^+$

no sure win $\iff c = c^*$; arbitrage $\iff c \neq c^*$

> **Use when:** this exact question has appeared in **three** consecutive papers. Prices are given as present values, so **there is no discounting** — three lines, no r.

> **Paper — all three instances:**
> Midsem'25 Q1(c): $50 \to \{25, 100\}$, K=85.
> $p = \frac{25}{75} = \frac{1}{3}$, payoff 15 or 0, $c^* = \mathbf{5}$. Key: "$c \neq 5$" ✓
> CT-2'25 Q5: $40 \to \{25, 100\}$, K=75.
> $p = \frac{15}{75} = \frac{1}{5}$, payoff 25 or 0, $c = \mathbf{5}$ ✓
> Midsem'24 Q1(a): $75 \to \{50, 200\}$, K=120.
> $p = \frac{25}{150} = \frac{1}{6}$, payoff 80 or 0,
> $c = \frac{80}{6} = \mathbf{13.33}$

▼ **Proof**

> Step 1 makes the stock fair: $pS_u + (1-p)S_d = S_0$. Any position (x shares, y options) then has p-expected gain $x \cdot 0 + y(c^* - c)$.
> If $c = c^*$: every position has expected gain 0 with $p \in (0,1)$, so it cannot be positive in both states: no sure win.
> If $c \neq c^*$: choose x so the gain is the same in both states (§A1.2). A constant equals its own expectation $y(c^* - c)$; pick the sign of y to make it positive: sure win.

M4.6 PUT–CALL PARITY

$$C - P = S_0 - \frac{K}{(1+r)^T}$$

Same strike, same maturity, European.

Use when: the paper asks for both a call and a put on the same tree (Endsem'25 Q4 does). Price one properly, get the other in one line instead of rebuilding the tree.

Proof in one line: the portfolios "call + cash $K/(1+r)^T$" and "put + one share" have identical payoff $\max(S_T, K)$ at T, so they must cost the same today.

▼ **Proof**

Hold: 1 share + 1 put − 1 call. At T it is worth $S_T + (K - S_T)^+ - (S_T - K)^+$.
If $S_T \geq K$: $S_T + 0 - (S_T - K) = K$. If $S_T < K$: $S_T + (K - S_T) - 0 = K$. So it pays K for sure.
A sure K at T costs $K/(1+r)^T$ today (otherwise buy the cheaper of the two and sell the dearer: arbitrage). Hence $S_0 + P - C = K/(1+r)^T$.

M4.7 SELF-FINANCING

$$\Delta_t S_{t+1} + B_t(1+r) \;=\; \Delta_{t+1} S_{t+1} + B_{t+}$$

Use when: "verify the replicating strategy is self-financing". Meaning: at each rebalancing date the new portfolio costs exactly what the old one is now worth — no money in, none out.

Paper: Endsem'25 Q4(iii).
Trick: compute the old portfolio's value at the new prices, then the new portfolio's cost, and show they match at *every* node.

M5. **Stopping Times**

M5.1 DEFINITION

$\tau : \Omega \to \{0, 1, 2, \ldots\} \cup \{\infty\}$ with $\{\tau \le t\} \in \mathcal{F}_t$ for every t

> **Use when:** plain meaning: at time t you can already tell whether you have stopped, using only what you have seen. **No peeking at the future.** ∞ is allowed — the event may never happen.

> **Q:** three tosses, τ = first H.
> $\{\tau \le 1\} = H__ \in \mathcal{F}_1$ ✓,
> $\{\tau \le 2\} = H__ \cup TH_ \in \mathcal{F}_2$ ✓ **Yes**.
> **Q:** $\tau = 1$ if toss 2 is H else 2:
> $\{\tau \le 1\} = \{_H_\} \notin \mathcal{F}_1$ × **No**.

M5.2 EQUIVALENT CHARACTERISATION

$\{\tau \le t\} \in \mathcal{F}_t \; \forall t \iff \{\tau = t\} \in \mathcal{F}_t \; \forall t$

> **Use when:** $\{\tau = t\}$ is almost always the easier one to write down. Prove the equivalence with the two identities below — it is a standard 3-mark part.

> **(⇒)** $\{\tau = t\} = \{\tau \le t\} \cap \{\tau \le t-1\}^c$, both in $\mathcal{F}_t$ by nesting.
> **(⇐)** $\{\tau \le t\} = \bigcup_{s=0}^{t} \{\tau = s\}$, a finite union of $\mathcal{F}_t$-sets ✓

▼ **Proof**

> (⇒) $\{\tau = t\} = \{\tau \le t\} \setminus \{\tau \le t-1\}$, and $\{\tau \le t-1\} \in \mathcal{F}_{t-1} \subseteq \mathcal{F}_t$.
> (⇐) $\{\tau \le t\} = \bigcup_{k \le t} \{\tau = k\}$, each in $\mathcal{F}_k \subseteq \mathcal{F}_t$.

M5.3 FIRST ENTRY TIME IS A STOPPING TIME

$\tau = \min\{n : X_n \in B\}$ is a ST for adapted (X_n)

$\{\tau = n\} = \{X_1 \notin B\} \cap \cdots \cap \{X_{n-1} \notin B\} \cap \{X_n \in B\}$

> **Use when:** covers every τ on this paper — first hit of a level, first hit of a moving line, ruin time. Each piece of the intersection sits in $\mathcal{F}_n$ by adaptedness plus nesting.

> **Q:** walk with steps $+1, -1, +1, +1$, $B = \{2\}$:
> $X = 1, 0, 1, 2$ so $\tau = 4$. At time 4 you check only $X_1, \ldots, X_4$ ✓
> Steps $-1, -1, -1, \ldots$ never reach 2, so $\tau = \infty$ — which is why ∞ is in the codomain.

▼ **Proof**

> Each $\{X_k \notin B\} \in \mathcal{F}_k \subseteq \mathcal{F}_n$ (adapted) and $\{X_n \in B\} \in \mathcal{F}_n$, so the intersection $\{\tau = n\} \in \mathcal{F}_n$. Done by §M5.2.

M5.4 ⭐ LAST EXIT TIME IS NOT A STOPPING TIME

$\tau = \max\{n : X_n \in B\}$ is **not** a ST

> **Use when:** the standard counterexample question. Also not stopping times: "the time at which X_n attains its maximum over $n \le N$", "the day before the crash". All need the future.

> **Q:** three-step walk, τ = last $n \le 3$ with $X_n = 1$.
> Steps $(+1, -1, -1)$: $X = 1, 0, -1$, so $\tau = 1$.
> Steps $(+1, -1, +1)$: $X = 1, 0, 1$, so $\tau = 3$.
> Both begin $+1$, so they lie in the **same atom of $\mathcal{F}_1$**, yet τ differs $\Rightarrow \{\tau = 1\} \notin \mathcal{F}_1$ ×
> **Trick:** exhibit two paths agreeing up to time t with different τ. That is a complete proof.

▼ **Proof**

> Counterexample: two coin flips, X_n = number of heads after n flips, $B = \{0\}$, τ = last $n \le 2$ with $X_n = 0$. Then $\{\tau = 0\} = \{\text{first flip is H}\}$, which is not in $\mathcal{F}_0 = \{\varnothing, \Omega\}$: at time 0 you cannot know it.

M5.5 CLOSURE OPERATIONS

$\tau_1 \wedge \tau_2$ ✓ $\tau_1 \vee \tau_2$ ✓ $\tau + 1$ ✓ $\tau \wedge c$ ✓
$\tau - 1$ × (stopping one step *before* something you cannot yet see)

> **Use when:** $\tau \wedge c$ is the workhorse — it is how you manufacture the **bounded** stopping time that the Optional Stopping Theorem requires (M6.3).

> **Proofs, one line each:**
> $\{\tau_1 \wedge \tau_2 \le t\} = \{\tau_1 \le t\} \cup \{\tau_2 \le t\}$;
> $\{\tau_1 \vee \tau_2 \le t\} = \{\tau_1 \le t\} \cap \{\tau_2 \le t\}$;
> $\{\tau + 1 \le t\} = \{\tau \le t - 1\} \in \mathcal{F}_{t-1} \subseteq \mathcal{F}_t$.

▼ **Proof**

> $\{\tau_1 \wedge \tau_2 \le t\} = \{\tau_1 \le t\} \cup \{\tau_2 \le t\}$;
> $\{\tau_1 \vee \tau_2 \le t\} = \{\tau_1 \le t\} \cap \{\tau_2 \le t\}$;
> $\{\tau + 1 \le t\} = \{\tau \le t - 1\} \in \mathcal{F}_{t-1} \subseteq \mathcal{F}_t$;
> $\{\tau \wedge c \le t\}$ is Ω if $t \ge c$, else $\{\tau \le t\}$.
> $\tau - 1$ fails: $\{\tau - 1 \le t\} = \{\tau \le t + 1\}$ needs $\mathcal{F}_{t+1}$. E.g. τ = first head: $\{\tau - 1 = 0\} = \{$flip 1 is H$\} \notin \mathcal{F}_0$.

M6. Stopped Process & Optional Stopping

M6.1 THE STOPPED PROCESS

$$Z_t = X_{t\wedge\tau} = \begin{cases} X_t & \tau > t \text{ (running)} \\ X_\tau & \tau \le t \text{ (frozen)} \end{cases}$$

$$Z_{t+1} = \sum_{k=0}^{t} X_k \mathbb{1}_{\{\tau=k\}} + X_{t+1}\mathbb{1}_{\{\tau>t\}}$$

> **Use when:** the explicit formula is the one you differentiate the proof from — every term in it is $\mathcal{F}_t$-measurable except X_{t+1}.

> **Q:** steps $+1, +1, -1, +1$ so $X = 0, 1, 2, 1, 2$; τ = first hit of 2, so $\tau = 2$.
> $Z = 0, 1, 2, 2, 2$. After τ, Z ignores X: $X_3 = 1$ but $Z_3 = 2$.

▼ **Proof**

> If $\tau = k \le t$, the process froze at X_k: only the term $X_k \mathbf{1}_{\{\tau=k\}}$ survives. If $\tau > t$ it is still running and equals X_{t+1}. Exactly one indicator is 1.

M6.2 PRESERVATION THEOREM

(X_t) MG $\Longrightarrow$ $(X_{t\wedge\tau})$ MG
sub-MG $\to$ sub-MG; super-MG $\to$ super-MG (type is inherited)

> **Use when:** the key step is $\mathbb{1}_{\{\tau>t\}} \in \mathcal{F}_t$, which holds because $\{\tau \le t\} \in \mathcal{F}_t$ — **this is the only place the stopping-time property is used**. Say that sentence.

> **Proof:** pull the $\mathcal{F}_t$-measurable pieces out of M6.1, then $E[X_{t+1} \mid \mathcal{F}_t] = X_t$ gives $\sum_{k\le t} X_k \mathbb{1}_{\{\tau=k\}} + \mathbb{1}_{\{\tau>t\}} X_t = Z_t$ ✓

▼ **Proof**

> The stopped process moves only while running: $Z_{t+1} - Z_t = \mathbf{1}_{\{\tau>t\}}(X_{t+1} - X_t)$, and $\{\tau > t\} \in \mathcal{F}_t$. So $E[Z_{t+1} - Z_t \mid \mathcal{F}_t] = \mathbf{1}_{\{\tau>t\}} E[X_{t+1} - X_t \mid \mathcal{F}_t]$ $: = 0$ for a MG, ≥ 0 for a sub-MG, ≤ 0 for a super-MG (the indicator is ≥ 0). Integrable: $|Z_t| \le |X_0| + \cdots + |X_t|$.

M6.3 ⭐ OPTIONAL STOPPING THEOREM

(X_t) MG, τ a **bounded** ST ($\tau \le c$ a.s.) $\Longrightarrow E[X_\tau] = E[X_0]$

Use when: no stopping rule beats a fair game. Its power on this paper: it converts an unknown distribution of τ into one line of algebra.

Four-line proof: 1. $Z_t = X_{t\wedge\tau}$ is a MG (M6.2). 2. So $E[Z_t] = E[Z_0] = E[X_0]$. 3. Put $t = c$; since $\tau \le c$, $Z_c = X_\tau$. 4. Hence $E[X_\tau] = E[X_0]$ ∎

▼ **Proof**

Write the stopped value as a sum of steps: $X_\tau = X_0 + \sum_{k=1}^{c} \mathbf{1}_{\{\tau \ge k\}}(X_k - X_{k-1})$. $\{\tau \ge k\} = \{\tau \le k-1\}^c \in \mathcal{F}_{k-1}$, so it is known one step early: $E\big[\mathbf{1}_{\{\tau\ge k\}}(X_k - X_{k-1})\big] = E\big[\mathbf{1}_{\{\tau\ge k\}}\, E(X_k - X$

Sum the c terms: $E[X_\tau] = E[X_0]$.

M6.4 ⚠ WHEN OST FAILS

boundedness of τ is **essential**

Use when: if your algebra returns something impossible — a negative expected time, or $E[X_\tau] \ne E[X_0]$ — the hypothesis failed, not the arithmetic. Check before you write.

Counterexample: symmetric walk from 0, τ = first hit of $+1$. Then $X_\tau = 1$ always, so $E[X_\tau] = 1 \ne 0$. τ is finite a.s. but $E[\tau] = \infty$. **Q:** $\tau = \min\{t : W_t = 1 + 2t\}$ would give $E[\tau] = -\frac{1}{2} < 0$. Impossible — the line escapes upward and $P(\tau = \infty) > 0$.

▼ **Proof**

Counterexample: fair ± 1 walk from 0, τ = first time it hits 1. $\tau < \infty$ a.s., but $E[X_\tau] = 1 \ne 0 = E[X_0]$. τ is unbounded and $X_{t\wedge\tau}$ is not bounded below, so neither the bounded nor the dominated version (§M6.7) applies.

M6.5 OST RECIPE #1 — HITTING A MOVING LINE

$\tau = \min\{t : W_t = a + bt\}$

$E[W_\tau] = 0$ and

$W_\tau = a + b\tau \;\Rightarrow\; E[\tau] = -\dfrac{a}{b}$

Needs $ab < 0$.

Use when: three lines: apply OST, substitute the barrier equation, solve. See also §10.7.

Paper: Endsem'25 Q1(d) — $W_t = 2 - 3t$: $0 = 2 - 3E[T]$, so $E[T] = \mathbf{2/3}$. **Q:** $W_t = 3 - 6t$ gives $E[\tau] = \mathbf{1/2}$.

▼ **Proof**

OST on the MG W (justified as in §M6.7; when the line slopes back towards 0, $E\tau < \infty$): $E[W_\tau] = 0$. At τ the path is on the line: $W_\tau = a + b\tau$. So $0 = a + bE\tau$, $E\tau = -a/b$. If $ab > 0$ this would be negative, impossible: then the line may never be hit.

M6.6 ⭐ OST RECIPE #2 — GAMBLER'S RUIN WITH TWO MARTINGALES

walk from 0, absorbed at $+A$ or $-B$:

use X_n:

$$Ap - B(1-p) = 0 \Rightarrow p = \frac{B}{A+B}$$

use $X_n^2 - n$:

$$E[\tau] = E[X_\tau^2] = A^2p + B^2(1-p) = A$$

> **Use when:** both parts of the ruin question in one framework. $E[\tau]$ **always collapses to the product** AB — compute it either way as a check.

> **Paper:** CT-1'26 Q10 — stop at $+10$ or -5, so A=10, B=5.
> $p = \frac{5}{15} = \mathbf{1/3}$ ✓ |
> $E[\tau] = 100(\frac{1}{3}) + 25(\frac{2}{3}) = \mathbf{50}$ ✓ $= 10 \times 5$ ✓
> Set-2 $(20, 5)$: $\frac{1}{5}$, 100. Set-3 $(30, 10)$: $\frac{1}{4}$, 300. Set-4 $(6, 5)$: $\frac{5}{11}$, 30. All match §3.1–3.2.

▼ **Proof**

> X_n is a MG trapped in $[-B, A]$, so OST holds (§M6.7): $0 = E[X_\tau] = Ap - B(1-p)$, $p = \frac{B}{A+B}$.
> $X_n^2 - n$ is a MG (§M2.5): $E[X_{\tau\wedge n}^2] = E[\tau \wedge n]$. Let $n \to \infty$ (left side bounded, right side increasing):
> $E\tau = E[X_\tau^2] = A^2p + B^2(1-p) = \frac{A^2B+B^2A}{A+B}$:
> .

M6.7 JUSTIFYING AN UNBOUNDED T

τ unbounded but $X_{t\wedge\tau}$ **bounded** (trapped between barriers)

$\Rightarrow$ dominated convergence lets you pass $t \to \infty$ in $E[X_{t\wedge\tau}] = E[X_0]$

> **Use when:** the gambler's-ruin τ is *not* bounded, so M6.3 does not literally apply. Write this one sentence and the marks are safe.

> **Trick:** the path is confined to $[-B, A]$, which is exactly the domination you need. Contrast M6.4, where the walk can sink arbitrarily low — no bound, no theorem.

▼ **Proof**

> $\tau \wedge t$ is bounded, so $E[X_{\tau\wedge t}] = E[X_0]$ (§M6.3). As $t \to \infty$, $X_{\tau\wedge t} \to X_\tau$ (since $\tau < \infty$), and $|X_{\tau\wedge t}| \le C$. Dominated convergence:
> $E[X_\tau] = \lim_t E[X_{\tau\wedge t}] = E[X_0]$.

M7. **Gambling Strategies & Fairness**

M7.1 THE GAME PROCESS

η_n = winnings per unit stake in game n;
$\Xi_n = \sum_{k \le n} \eta_k$; $\mathcal{F}_n = \sigma(\eta_1, \dots, \eta_n)$
fair $\iff$ MG | favourable $\iff$ sub-MG | unfavourable $\iff$ super-MG

Use when: the dictionary between the gambling words and the martingale words. Translate first, then it is M2.

Q: roulette on red, 18 of 37 pockets: $E\eta = \frac{18}{37} - \frac{19}{37} = -\frac{1}{37} \approx -0.027$. **Super-MG** — you lose 2.7 paise per rupee staked, whatever you do.

M7.2 ⭐ PREDICTABLE STRATEGY

the stake α_n must be $\mathcal{F}_{n-1}$-measurable (**predictable**)

Use when: the whole subject rests on this. $\mathcal{F}_n$-measurable would mean placing the bet *after* seeing the outcome. Say why, not just what.

Allowed ✓: $\alpha_n = 2$ if you won game $n-1$ else 1; $\alpha_n = n$; doubling after each loss; $\alpha_n = |\Xi_{n-1}| + 1$.
Forbidden ✗: $\alpha_n = \mathbb{1}\{\eta_n = +1\}$ ("bet only on games you will win") — it gives $E[Z_1] = \frac{1}{2} > 0$ in a fair game. Predictability is exactly what rules it out.

M7.3 WEALTH PROCESS (MARTINGALE TRANSFORM)

$$Z_n = \sum_{k=1}^{n} \alpha_k \eta_k = \sum_{k=1}^{n} \alpha_k (\Xi_k - \Xi_{k-1})$$

recursive: $Z_n = Z_{n-1} + \alpha_n(\Xi_n - \Xi_{n-1})$

Use when: this is the discrete Itô integral — a predictable integrand against a martingale increment. §7.5 is the same statement in continuous time.

Q: doubling after each loss, results $+1, -1, -1, +1$: stakes $1, 1, 2, 4$, gains $+1, -1, -2, +4$, so $Z = 1, 0, -2, \mathbf{2}$.

▼ **Proof**

Telescoping:
$Z_n - Z_{n-1} = \alpha_n \eta_n = \alpha_n(\Xi_n - \Xi_{n-1})$, since $\eta_n = \Xi_n - \Xi_{n-1}$. Sum from 1 to n.

M7.4 PRESERVATION THEOREM

Ξ MG, α predictable and bounded $\Rightarrow$ Z MG

Ξ sub-MG, α predictable, bounded **and** $\alpha_n \ge 0 \Rightarrow$ Z sub-MG

Use when: "why is $\alpha_n \ge 0$ needed?" is a standard follow-up. Because a *negative* stake against a favourable game turns a positive expected gain into a loss.

Proof:
$E(Z_n \mid \mathcal{F}_{n-1}) = Z_{n-1} + \alpha_n \underbrace{E(\Xi_n - \Xi_{n-1} \mid \mathcal{F}_n}_{=0}$
✓
Both Z_{n-1} and α_n pull out because both are $\mathcal{F}_{n-1}$-measurable — that is where predictability is used.

▼ **Proof**

$E[Z_n - Z_{n-1} \mid \mathcal{F}_{n-1}] = \alpha_n E[\Xi_n - \Xi_{n-1} \mid \mathcal{F}_{n-}$
(take out what is known, α_n is predictable).
This is 0 for a MG, and ≥ 0 for a sub-MG when $\alpha_n \ge 0$. Integrable:
$|\alpha_k| \le C \Rightarrow |Z_n| \le C \sum_k |\eta_k|$.

M7.5 YOU CANNOT BEAT A FAIR GAME

$E[Z_n] = 0$ for every predictable bounded strategy

> **Use when:** a betting system changes the **shape** of the payoff distribution, never its **mean**. That sentence is the answer to most conceptual parts here.

> **Q:** doubling until the first win, capped at 3 games. Win $+1$ with probability $\frac{7}{8}$; lose $1+2+4 = 7$ with probability $\frac{1}{8}$.
> $E = \frac{7}{8} - \frac{7}{8} = \mathbf{0}$ ✓
> **Trick:** a frequent small win paid for by a rare huge loss. Uncapped it appears to beat the game — but then τ is unbounded and M6.3 does not apply.

▼ **Proof**

> $Z_n = \sum_{k=1}^{n} \alpha_k (X_k - X_{k-1})$ with α_k known at time $k-1$ and bounded.
> $E[\alpha_k (X_k - X_{k-1})] = E\big[\alpha_k \, E(X_k - X_{k-1} \mid \mathcal{F}_$
> . Sum over k: $E[Z_n] = 0$. (Same step as the OST proof, §M6.3.)

M7.6 AMERICAN VS EUROPEAN (SHORT-NOTE)

European: exercise only at T. American: exercise at any ST $\tau \le T$.

American $\ge$ European; **equal** for a call on a non-dividend stock

> **Use when:** the short-note question. Say why they are equal for a call: the discounted price is a MG, early exercise throws away time value, and OST says no stopping rule improves a fair game. For a **put** early exercise can be optimal.

> **Paper:** Midsem'25 Q2(a) — "short note on American and European call options" [2 marks].
> **Trick:** definition, the inequality, the call-equality with its one-line reason, and the put caveat. Four sentences.

▼ **Proof**

> American $\ge$ European: $\tau = T$ is one of the allowed stopping times.
> Equal for a call ($r \ge 0$, no dividends): $Y_t = (1+r)^{-t}(S_t - K)^+$ is a sub-MG under P^*. By conditional Jensen,
> $E^*[Y_{t+1} \mid \mathcal{F}_t] \ge (1+r)^{-(t+1)}\big((1+r)S_t - K\big)$
> .
> For a sub-MG and $\tau \le T$, $E^*[Y_\tau] \le E^*[Y_T]$ (OST, sub-MG version). So the best stopping time is T.

Part II — Random Walks & Brownian Motion

Sections 1–14. Jump here from the routing table above.

1. Normal & MVN Toolkit — the plug-in layer

1.1 STANDARD NORMAL CDF

$$\Phi(-x) = 1 - \Phi(x), \qquad \Phi(0) = 0.5$$
$$P(a < Z \le b) = \Phi(b) - \Phi(a)$$

Use when: always. Standardize first: $X \sim N(\mu, \sigma^2) \Rightarrow P(X \le x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$.

Table values worth memorizing: $\Phi(1)$=.8413, $\Phi(1.5)$=.9332, $\Phi(2)$=.9772, $\Phi(2.5)$=.9938, $\Phi(3)$=.9987.

▼ **Proof**

The density is symmetric: $\Phi(-x) = P(Z \le -x) = P(Z \ge x) = 1 - \Phi(x$. At $x = 0$: $\Phi(0) = 1 - \Phi(0)$, so $\frac{1}{2}$. Interval: §F5.2.

1.2 EXPONENTIAL MOMENT (LOG-NORMAL MEAN)

$$X \sim N(\mu, \sigma^2) \implies E[e^X] = e^{\mu + \sigma^2/2}$$
$$E[e^{aZ}] = e^{a^2/2}, \qquad Z \sim N(0,1)$$

Use when: an exponential of a normal appears — GBM means, $\mathrm{Cov}(e^{W_s}, e^{W_t})$, exponential martingale, Black–Scholes.

Trick: the whole GBM section is this one identity. $E[e^{W_t}] = e^{t/2}$ because $W_t \sim N(0, t)$.

▼ **Proof**

$E[e^{aZ}] = \int e^{az} \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz$. Complete the square: $az - \frac{z^2}{2} = \frac{a^2}{2} - \frac{(z-a)^2}{2}$. So it is $e^{a^2/2} \int \frac{e^{-(z-a)^2/2}}{\sqrt{2\pi}} dz = e^{a^2/2}$ (a density integrates to 1).
$X = \mu + \sigma Z$: $E[e^X] = e^{\mu} E[e^{\sigma Z}] = e^{\mu + \sigma^2/2}$.

1.3 LOG-NORMAL VARIANCE

$$\mathrm{Var}(e^X) = e^{2\mu + \sigma^2}\left(e^{\sigma^2} - 1\right)$$

$X \sim N(\mu, \sigma^2)$.

Use when: asked for Var or Cov of a GBM / of e^{W_t}.

Q: $Y = e^{W_2}$, find $\mathrm{Var}(Y)$. Here μ=0, σ^2=2: $e^2(e^2 - 1) = \mathbf{47.21}$.
Trick: $\mathrm{Var} = (\text{mean})^2(e^{\sigma^2} - 1)$ — compute the mean first, square it.

▼ **Proof**

$\mathrm{Var}(e^X) = E[e^{2X}] - (E[e^X])^2$.
$2X \sim N(2\mu, 4\sigma^2)$, so $E[e^{2X}] = e^{2\mu + 2\sigma^2}$ (§1.2). Result: $e^{2\mu+2\sigma^2} - e^{2\mu+\sigma^2} = e^{2\mu+\sigma^2}(e^{\sigma^2} - 1)$.

1.4 EVEN MOMENTS OF A NORMAL

$$E[Z^{2n}] = (2n-1)!! = 1 \cdot 3 \cdot 5 \cdots (2n -$$
$$E[Z]=0,\ E[Z^2]=1,\ E[Z^3]=0,\ E[Z^4]=3$$
$$E[W_t^{2n}] = (2n-1)!!\, t^{\,n}, \qquad E[W_t^4] = :$$

Use when: any odd power → 0 by symmetry. Any 4th moment → $3t^2$.

Paper: CT-1 prints "$E(Z^3) = 0,\ E(Z^4) = 3$" at the top of the sheet — they expect you to need it in Q2.

▼ **Proof**

Use $z\varphi(z) = -\varphi'(z)$ and integrate by parts: $E[Z^{2n}] = \int z^{2n-1}\, z\varphi(z)\, dz = (2n-1) \int z^{2n-2}$. Iterate down to $E[Z^0] = 1$. Odd moments are 0 by symmetry. $W_t = \sqrt{t}\, Z$ gives the factor t^n.

1.5 MVN — DEFINITION BY LINEAR COMBINATIONS

$\mathbf{X}$ is MVN $\iff$ $\mathbf{a}^\top \mathbf{X} = \sum_i a_i X_i$ is univariate normal for *every* $\mathbf{a}$

> **Use when:** proving something is a Gaussian process. This *is* the definition — do not try to write a joint density.

> **Q:** CT-1'26 Q3 — is $e^{-t}W(e^{2t})$ a Gaussian process?
> **Trick:** it is a fixed linear combination of BM values, so every $\sum a_i X(t_i)$ is normal. **Yes**, with zero computation.

1.6 VARIANCE OF A LINEAR COMBINATION

$$\mathrm{Var}\Big(\sum_i a_i X_i\Big) = \sum_i \sum_j a_i a_j \,\mathrm{Cov}(X_i,$$

> **Use when:** the master identity for every "$aW_s + bW_t \sim$?" question. Plug $\mathrm{Cov}(W_s, W_t) = \min(s,t)$.

> **Q:** $\mathrm{Var}(W_1 - W_2 + 3W_4)$.
> **Trick:** write the $\min$ matrix $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 4 \end{pmatrix}$, sandwich with $\mathbf{a} = (1, -1, 3)$ → **31**.

▼ **Proof**

> $\mathrm{Var}\left(\sum a_i X_i\right) = E\Big[\big(\sum_i a_i(X_i - \mu_i)\big)^2\Big] = \sum$. Expand the square, then use linearity.

1.7 MVN LINEAR TRANSFORMATION

$$\mathbf{X} \sim N_n(\boldsymbol{\mu}, \Sigma) \Rightarrow A\mathbf{X} + \mathbf{b} \sim N_m(A\boldsymbol{\mu} +$$

> **Use when:** a process is built as a fixed matrix times BM values — e.g. the Brownian bridge $B(s) = W(s) - \frac{s}{t}W(t)$.

> **Q:** show $B(s) = W(s) - \frac{s}{t}W(t)$ is Gaussian with $\mathrm{Cov} = \min(u,v) - \frac{uv}{t}$.
> **Trick:** B is a *fixed* matrix times $(W(s), W(t))$, so MVN is automatic — then just compute $A\Sigma A^\top$.

▼ **Proof**

> Any combination $\mathbf{c}^\top(A\mathbf{X} + \mathbf{b}) = (A^\top \mathbf{c})^\top \mathbf{X} + \mathbf{c}^\top \mathbf{b}$ is a combination of $\mathbf{X}$ plus a constant, so normal: MVN (§1.5). Mean $A\boldsymbol{\mu} + \mathbf{b}$; covariance $E[A(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top A^\top] = A\Sigma A^\top$.

1.8 MVN JOINT MGF

$$M_{\mathbf{X}}(\mathbf{t}) = \exp\Big(\mathbf{t}^\top \boldsymbol{\mu} + \tfrac{1}{2}\mathbf{t}^\top \Sigma\, \mathbf{t}\Big)$$

> **Use when:** you need $E[e^{W_s + W_t}]$ — set $\mathbf{t} = (1,1)$. Fastest route to $\mathrm{Cov}(e^{W_s}, e^{W_t})$.

> **Q:** $E[e^{W_1 + W_2}]$.
> $W_1 + W_2 \sim N(0,\ 1 + 2 + 2(1)) = N(0,5)$ → $\mathbf{e^{5/2}}$.
> **Trick:** set $\mathbf{t} = (1,1)$; the MGF hands you $\exp(\frac{1}{2}\sum_i \sum_j \Sigma_{ij})$ in one step.

▼ **Proof**

> $\mathbf{t}^\top \mathbf{X} \sim N(\mathbf{t}^\top \boldsymbol{\mu},\ \mathbf{t}^\top \Sigma \mathbf{t})$ (§1.5, §1.6), and $E[e^Y] = e^{m + v/2}$ (§1.2).

1.9 MVN CONDITIONAL DISTRIBUTION

$$\mathbf{X} \mid \mathbf{Y}=\mathbf{y} \sim N\big(\boldsymbol{\mu}_X + \Sigma_{XY}\Sigma_{YY}^{-1}(\mathbf{y} - \boldsymbol{\mu}_Y)$$

Use when: any "given $W(t) = b$, find the law of $W(s)$". Conditional variance *never* depends on the observed value.

Trick: for BM this collapses to §9.4 — don't invert matrices, quote the bridge result.

▼ **Proof**

Let $C = \Sigma_{XY}\Sigma_{YY}^{-1}$ and $\mathbf{R} = \mathbf{X} - \boldsymbol{\mu}_X - C(\mathbf{Y} - \boldsymbol{\mu}_Y)$. Then $\operatorname{Cov}(\mathbf{R}, \mathbf{Y}) = \Sigma_{XY} - C\Sigma_{YY} = 0$; jointly normal and uncorrelated means independent. So given $\mathbf{Y} = \mathbf{y}$: $\mathbf{X} = \boldsymbol{\mu}_X + C(\mathbf{y} - \boldsymbol{\mu}_Y) + \mathbf{R}$, with $\mathbf{R} \sim N(0, \operatorname{Cov}\mathbf{R})$ and $\operatorname{Cov}\mathbf{R} = \Sigma_{XX} - C\Sigma_{YX} - \Sigma_{XY}C^\top + C\Sigma_{YY}C^\top$.

2. Random Walks — Additive & Geometric

2.1 ADDITIVE RANDOM WALK

$$X_0 = 0, \quad X_n = Y_1 + \cdots + Y_n, \quad Y_i \text{ i.i.d}$$

$\{X_n\}$ is a martingale $\iff E[Y_1] = 0$

Needs $E|Y_1| < \infty$.

Use when: the walk is built by *adding* i.i.d. steps. Sub-MG if $E[Y] > 0$, super-MG if $E[Y] < 0$.

Paper: Midsem'24 Q1(b) — "$\{X_n\}$ is a martingale if $E(Y_1) = _$" → 0.

▼ **Proof**

$E[X_{n+1} \mid \mathcal{F}_n] = X_n + E[Y_{n+1}] = X_n + EY_1$ (Y_{n+1} independent of the past). Equal to X_n iff $EY_1 = 0$.

2.2 GEOMETRIC RANDOM WALK

$$X_n = X_0 \textstyle\prod_{i=1}^n Y_i, \quad Y_i > 0 \text{ i.i.d.}$$

$\{X_n\}$ is a martingale $\iff E[Y_1] = 1$

Use when: the walk *multiplies*. Proof is one line: $E[X_{n+1} \mid \mathcal{F}_n] = X_n E[Y_{n+1}]$ by independence + taking out what is known.

Paper: CT-2'25 Q3 — $X_n = 2^n U_1 \cdots U_n$, $U_i \sim \text{Unif}(0,1)$. Step is $Y_i = 2U_i$, $E[2U] = 2 \cdot \frac{1}{2} = 1 \Rightarrow$ **Yes, MG.**

▼ **Proof**

$E[X_{n+1} \mid \mathcal{F}_n] = X_n E[Y_{n+1}] = X_n EY_1$ (take out X_n). Since $X_n > 0$, this equals X_n iff $EY_1 = 1$.

2.3 LOG-NORMAL STEPS — DRIFT CORRECTION

$$Y_i = e^{Z_i},\ Z_i \sim N(\mu, \sigma^2) \Rightarrow E[Y_i] = e^{\mu+}$$

martingale $\iff \mu = -\dfrac{\sigma^2}{2}$

> **Use when:** geometric walk with normal log-steps. The $-\sigma^2/2$ is the *same* correction that appears in GBM (§11.4) and Black–Scholes.

> **Q:** log-price steps $Z_i \sim N(\mu, 0.04)$; for what μ is the price a martingale? → $\mu = \mathbf{-0.02}$.
> **Trick:** this $-\sigma^2/2$ reappears in GBM (§11.4) and as d_2 in Black–Scholes. Same correction, three places.

▼ **Proof**

> $E[e^Z] = e^{\mu+\sigma^2/2}$ (§1.2). By §2.2 MG $\iff e^{\mu+\sigma^2/2} = 1 \iff \mu = -\sigma^2/2$.

2.4 MOMENTS OF A SYMMETRIC ± 1 WALK

$$E[X_n] = 0, \qquad \mathrm{Var}(X_n) = n$$
$$\mathrm{Cov}(X_m, X_n) = \min(m, n)$$

> **Use when:** discrete analogue of $\mathrm{Cov}(W_s, W_t) = \min(s, t)$. Same proof: overlap of independent increments.

> **Q:** $\mathrm{Cov}(X_3, X_7)$ for a ± 1 walk $= \mathbf{3}$.
> **Trick:** only the shared first $\min(m, n)$ steps contribute; everything after is independent and mean zero.

▼ **Proof**

> Each step has mean 0 and variance 1, independent: $EX_n = 0$, $\mathrm{Var}\, X_n = n$. For $m \le n$:
> $\mathrm{Cov}(X_m, X_n) = \mathrm{Cov}\left(X_m, X_m + (X_n - X_m)\right.$
> .

3. Gambler's Ruin — probability *and* duration

3.0 SETTING UP THE WORD PROBLEM

"stops when up W or down L"
$\Longrightarrow\ i = L, \quad N = L + W$

Use when: the exam always phrases it as "won W rupees or lost L rupees", never as "start at i, barriers 0 and N". **Do this translation first** — it is where marks are lost.

Paper: CT-1'26 Set-1 — "won 10 or lost 5" ⇒ $i = 5$, $N = 15$. Set-3 — "won 30 or lost 10" ⇒ $i = 10$, $N = 40$.

3.1 RUIN PROBABILITY — FAIR GAME

$$P_i = P(\text{hit } N \text{ before } 0 \mid X_0 = i) = \frac{i}{N}$$

$p = \frac{1}{2}$.

Use when: "equal chances", "fair coin", "wins or loses one rupee with equal probability".

Paper: CT-1'26 all four sets. Set-1: $5/15 = \mathbf{1/3}$. Set-2 (won 20, lost 5): $5/25 = \mathbf{1/5}$. Set-3: $10/40 = \mathbf{1/4}$. Set-4 (won 6, lost 5): $\mathbf{5/11}$.

▼ **Proof**

First step: from i you go to $i \pm 1$ with probability $\frac{1}{2}$ each, so $P_i = \frac{1}{2}P_{i+1} + \frac{1}{2}P_{i-1}$, with $P_0 = 0$, $P_N = 1$.
Rearranged: $P_{i+1} - P_i = P_i - P_{i-1}$, so the gaps are equal and P_i is a straight line through $(0, 0)$ and $(N, 1)$: $P_i = i/N$.
(OST route: X_n is a MG, so $i = E[X_\tau] = N \cdot P_i + 0$.)

3.2 EXPECTED DURATION — FAIR GAME

$$D_i = E[\text{number of rounds}] = i\,(N - i)$$

$p = \frac{1}{2}$. *Equivalently $D = L \cdot W$ in the word-problem variables.*

Use when: part (b) of the ruin question — "expected number of rounds before stopping". It is *always* asked alongside 3.1.

Paper: CT-1'26 Set-1: $5 \times 10 = \mathbf{50}$. Set-2: $5 \times 20 = \mathbf{100}$. Set-3: $10 \times 30 = \mathbf{300}$. Set-4: $5 \times 6 = \mathbf{30}$. **Trick:** it is literally $L \times W$ — no formula needed.

▼ **Proof**

$X_n^2 - n$ is a martingale for the fair ± 1 walk (§M6.6). OST at the exit time τ:
$E[X_\tau^2] - E[\tau] = i^2$. Now $X_\tau = N$ with probability i/N and 0 otherwise, so
$E[X_\tau^2] = N^2 \cdot \frac{i}{N} = iN$.
Hence $D_i = E[\tau] = iN - i^2 = i(N - i)$.

3.3 RUIN PROBABILITY — UNFAIR GAME

$$P_i = \frac{1 - (q/p)^i}{1 - (q/p)^N}, \qquad q = 1 - p,\ p \neq \frac{1}{2}$$

Use when: win probability is given as anything other than $1/2$ (e.g. $2/3$ and $1/3$). Let $r = q/p$ and evaluate numerically — do not expand.

Q: $p = 2/3$, stop at $+3$ or -2. Translate: i=2, N=5, $r = q/p = \frac{1}{2}$.
$$P = \frac{1 - (1/2)^2}{1 - (1/2)^5} = \frac{0.75}{0.96875} = \mathbf{0.774}.$$
Trick: compute r once, then it is two powers and a division.

▼ **Proof**

First step (§3.6): $P_i = pP_{i+1} + qP_{i-1}$, i.e. $p(P_{i+1} - P_i) = q(P_i - P_{i-1})$. With $\rho = q/p$ the gaps grow geometrically:
$P_{i+1} - P_i = \rho^i P_1$. Add up:
$P_i = P_1(1 + \rho + \cdots + \rho^{i-1}) = P_1 \frac{1-\rho^i}{1-\rho}$.
$P_N = 1$ fixes P_1, giving $P_i = \frac{1-\rho^i}{1-\rho^N}$.

3.4 EXPECTED DURATION — UNFAIR GAME

$$D_i = \frac{i}{q-p} - \frac{N}{q-p}\cdot\frac{1-(q/p)^i}{1-(q/p)^N}$$
$$= \frac{1}{q-p}\left(i - NP_i\right)$$

> **Use when:** duration with $p \neq \frac{1}{2}$. **Trick:** compute P_i first (3.3), then this is one subtraction.

> **Q:** same numbers. $q-p = -\frac{1}{3}$, so $D = \dfrac{2-5(0.774)}{-1/3} = \mathbf{5.61}$ rounds.
> **Trick:** always get P_i from 3.3 first — never expand the full expression.

▼ **Proof**

> $X_n - n(p-q)$ is a MG (§M2.5, drift removed). OST at the exit time: $E[X_\tau] - (p-q)E\tau = i$. Since $E[X_\tau] = N\,P_i + 0$: $D_i = E\tau = \frac{i-NP_i}{q-p}$.

3.5 INFINITE BANKROLL LIMIT

$$\lim_{N\to\infty} P_i = \begin{cases} 1-(q/p)^i & p > \frac{1}{2} \\ 0 & p \leq \frac{1}{2} \end{cases}$$

> **Use when:** "plays forever against an infinitely rich opponent" / "$N \to \infty$". Ruin is a.s. certain unless there is a genuine edge.

> **Q:** "plays forever against the house with $p = 0.49$, starting with 10". $p < \frac{1}{2}$ → ruin probability $\mathbf{1}$.
> **Trick:** any fair-or-worse game against an infinite bankroll ends in ruin almost surely. No arithmetic needed.

▼ **Proof**

> $\rho = q/p$. If $p > \frac{1}{2}$: $\rho < 1$, $\rho^N \to 0$, limit $1-\rho^i$. If $p < \frac{1}{2}$: $\rho > 1$, $\frac{\rho^i - 1}{\rho^N - 1} \to 0$. If $p = \frac{1}{2}$: $\frac{i}{N} \to 0$.

3.6 DERIVATION HOOK (FIRST-STEP ANALYSIS)

$$P_i = p\,P_{i+1} + q\,P_{i-1}, \quad P_0 = 0,\ P_N = 1$$
$$D_i = 1 + p\,D_{i+1} + q\,D_{i-1}, \quad D_0 = D_N$$

> **Use when:** the question says "derive" or "using first-step analysis" (Midsem'25 Q2(c) asks for a short note on gambler's ruin). The $+1$ in the duration recursion is the round just played.

> **Paper:** Midsem'25 Q2(c) — "write a short note on gambler's ruin" [2 marks].
> **Trick:** state the recursion, state the boundary conditions, then quote 3.1/3.3 as its solution. That is the whole answer.

▼ **Proof**

> Condition on the first round: with probability p you move to $i+1$, with q to $i-1$, and the game starts afresh from there. For D, that first round itself costs 1. Boundaries: at 0 or N the game is over.

4. Random Walk → Brownian Motion (scaling limit)

4.1 DIFFUSIVE SCALING — DRIFTLESS

steps $\pm\Delta x$ every Δt, $\Delta x = \sigma\sqrt{\Delta t}$

$$X_{\lfloor t/\Delta t \rfloor} \xrightarrow[\Delta t \to 0]{d} \sigma W(t)$$

> **Use when:** a limit of a symmetric walk is taken. The **only** scaling that survives is space $\sim \sqrt{\text{time}}$ — that is what forces $\mathrm{Var}(W_t) = t$.

> **Q:** "why is $\mathrm{Var}(W_t) = t$ and not t^2?"
> **Trick:** $n = t/\Delta t$ steps of size Δx give variance $t(\Delta x)^2/\Delta t$. That is finite *and* non-zero only if $\Delta x \propto \sqrt{\Delta t}$. Everything else degenerates.

▼ **Proof**

> After $n = t/\Delta t$ steps: a sum of n i.i.d. $\pm\Delta x$, mean 0, variance $n\Delta x^2 = \frac{t}{\Delta t}\sigma^2\Delta t = \sigma^2 t$. CLT: $\to N(0, \sigma^2 t)$. Disjoint time intervals use disjoint steps, so increments are independent: the limit is σW.

4.2 SCALING WITH DRIFT

$$p = \tfrac{1}{2}\left(1 + \frac{\mu}{\sigma}\sqrt{\Delta t}\right), \quad \text{steps } \pm\sigma\sqrt{\Delta t}$$

$$\implies X(t) \to \mu t + \sigma W(t)$$

> **Use when:** the question hands you a walk with p depending on $\sqrt{\Delta t}$ and says "argue the limit is BM with drift μ".

> **Paper:** Midsem'24 Q3(b) — steps $\pm\sqrt{\Delta t}$, $p = \frac{1}{2}(1 + \mu\sqrt{\Delta t})$ (so $\sigma = 1$).
> **Two-line proof:** per-step mean $= \sqrt{\Delta t}(2p-1) = \mu\,\Delta t$; per-step variance $= \Delta t\big(1 - (2p-1)^2\big) \to \Delta t$. Over $n = t/\Delta t$ steps: mean $\to \mu t$, variance $\to t$. CLT finishes it.

▼ **Proof**

> One step has mean $\sigma\sqrt{\Delta t}(2p-1) = \sigma\sqrt{\Delta t}\cdot\frac{\mu}{\sigma}\sqrt{\Delta t} = \mu\Delta t$ and variance $\sigma^2\Delta t - \mu^2\Delta t^2 \approx \sigma^2\Delta t$. Over $t/\Delta t$ steps: mean μt, variance $\to \sigma^2 t$. CLT as in §4.1: $\mu t + \sigma W(t)$.

5. Wiener Process — Axioms & Identification Tests

5.1 WIENER PROCESS AXIOMS

(i) $W(0) = 0$

(ii) independent increments

(iii) $W(t) - W(s) \sim N(0,\, t-s)$ for $s < t$ (stationary)

(iv) $t \mapsto W(t)$ continuous a.s.

> **Use when:** the four-item checklist for "short note" questions and for verifying a constructed process.

> **Paper:** Midsem'25 Q2 / Endsem — "short note on the Wiener process".
> **Trick:** write the four axioms, then add $\mathrm{Cov} = \min(s,t)$ and "continuous but nowhere differentiable" for the extra mark.

5.2 IS $X(T)$ A WIENER PROCESS? — 4-POINT TEST

1. $X(0) = 0$?
2. X Gaussian (all f.d.d. normal) ?
3. $E[X(t)] = 0$?
4. $\mathrm{Cov}(X(s), X(t)) = \min(s,t)$?

> **Use when:** T/F questions of the form "$V(t) = \ldots$ is also a Wiener process". Checking $\min(s,t)$ is usually the deciding step; $X(0) = 0$ is the usual trap.

> **Paper:** Midsem'25 Q1(e) — $V(t) = \frac{1}{c}W(c^2 t)$, $c > 0$. $V(0) = 0$ ✓;
> $\mathrm{Cov} = \frac{1}{c^2}\min(c^2 s, c^2 t) = \min(s,t)$ ✓ ⇒ **True**.

▼ **Proof**

> A Gaussian process is fixed by its mean and covariance. If X is Gaussian with $X(0) = 0$, mean 0, covariance $\min(s,t)$, it has the same finite-dimensional laws as BM. Independent increments follow: for $s \le t$, $\mathrm{Cov}(X_t - X_s, X_s) = s - s = 0$, and uncorrelated jointly normal variables are independent. Add continuity and it is BM.

5.3 IS $X(T)$ A GAUSSIAN PROCESS? — WEAKER TEST

Yes $\iff$ every finite $\sum a_i X(t_i)$ is normal

$f(t)\,W(g(t))$ with f, g deterministic ⇒ **always** Gaussian

> **Use when:** the question asks Gaussian process, *not* Wiener. A deterministic scaling and a deterministic time change can never destroy jointly-normal structure — the answer is Yes without computation.

> **Paper:** CT-1'26 Q3, all four sets — $e^{-t}W(e^{2t})$, $e^{-2t}W(e^{2t})$, $e^{-t}W(e^{t})$, $e^{-3t}W(e^{2t})$. Every answer is **Yes**. Not Wiener though: $X(0) = W(1) \neq 0$.

▼ **Proof**

> Gaussian process = every finite combination is normal (definition). For $f(t)W(g(t))$: $\sum a_i f(t_i) W(g(t_i))$ is a fixed linear combination of BM values, which is normal (§6.2).

5.4 THE FOUR INVARIANCES OF BM

Scaling: $\frac{1}{c}W(c^2 t)$

Symmetry: $-W(t)$

Time inversion: $t\,W(1/t)$ for $t > 0$, 0 at $t = 0$

Time reversal on $[0,T]$: $W(T) - W(T-t)$

> **Use when:** all four *are* standard BM. Anything else in a T/F question is probably not — run §5.2.

> **Q:** "Is $-W(t)$ a Wiener process? Is $tW(1/t)$?" — both **Yes**.
> **Trick:** memorize these four outright. A T/F on anything *else* means you must run the §5.2 checklist.

▼ **Proof**

> Each is a linear function of W, so Gaussian with mean 0 and value 0 at $t = 0$. Check covariance $= s$ for $s \le t$ (§5.2):
> Scaling: $\frac{1}{c^2}\min(c^2 s, c^2 t) = s$. Symmetry: unchanged. Inversion: $st\min(\frac{1}{s}, \frac{1}{t}) = st \cdot \frac{1}{t} = s$. Reversal: $\mathrm{Cov} = T - \min(T, T-t) - \min(T-s, T) +$.
> (Inversion is continuous at 0 because $W(u)/u \to 0$, law of large numbers.)

5.5 ORNSTEIN–UHLENBECK FROM BM

$X(t) = e^{-\lambda t} W\big(e^{2\lambda t}\big)$

$\mathrm{Cov}(X(s), X(t)) = e^{-\lambda|t-s|}$

> **Use when:** covariance depends on $|t-s|$ only ⇒ **stationary** Gaussian process (OU), *not* Wiener. Wiener needs $\min(s,t)$.

> **Trick:**
> $\mathrm{Cov} = e^{-\lambda(s+t)}\min(e^{2\lambda s}, e^{2\lambda t}) = e^{-\lambda(s+t)}e^{2\lambda \mathrm{mi}}$.

▼ **Proof**

> For $s \le t$:
> $\mathrm{Cov} = e^{-\lambda(s+t)}\min(e^{2\lambda s}, e^{2\lambda t}) = e^{-\lambda(s+t)}e^{2\lambda s}$.

5.6 LÉVY'S MARTINGALE CHARACTERIZATION

X continuous, $X(0) = 0$, X_t and $X_t^2 - t$ both martingales $\Longrightarrow$ X is standard BM

Use when: you must prove BM-ness without touching distributions. Two martingale checks replace the whole axiom list.

Q: "prove the scaled random-walk limit is Brownian motion".
Trick: Lévy lets you skip distributions entirely — show continuity, $X_0 = 0$, and that X_t and $X_t^2 - t$ are both martingales.

▼ **Proof**

Sketch. $X^2 - t$ a MG gives $(dX)^2 = dt$, so Itô applies as for BM: $e^{uX_t - u^2 t/2}$ has zero drift and is a MG. Hence $E\left[e^{u(X_t - X_s)} \mid \mathcal{F}_s\right] = e^{u^2(t-s)/2}$, the MGF of $N(0, t-s)$, and it does not depend on the past. So increments are $N(0, t-s)$ and independent of $\mathcal{F}_s$: X is BM.

5.7 PATH PROPERTIES (SHORT-NOTE AMMO)

Continuous everywhere, differentiable nowhere (a.s.)

Quadratic variation on $[0, t] = t$, i.e. $(dW)^2 = dt$

Total variation $= \infty$; Markov; strong Markov

Use when: "write a short note". $(dW)^2 = dt$ is the single fact that powers all of Itô's formula.

Q: Endsem-style — "explain quadratic variation".
Trick: say the sentence "$(dW)^2 = dt$ is why Itô's formula carries the $\frac{1}{2}f''$ term at all". That one link is what they are marking.

▼ **Proof**

Quadratic variation: $E\sum(\Delta W)^2 = \sum \Delta t = t$ and
$\operatorname{Var}\sum(\Delta W)^2 = \sum 2\Delta t^2 \le 2t \cdot \text{mesh} \to 0$.
Total variation:
$\sum(\Delta W)^2 \le \max|\Delta W| \cdot \sum|\Delta W|$. Left $\to t > 0$, $\max|\Delta W| \to 0$ (continuity), so $\sum|\Delta W| \to \infty$.
Not differentiable (idea): $\frac{W_{t+h} - W_t}{h} \sim N(0, \frac{1}{h})$, which blows up as $h \to 0$. Markov: future increments are independent of the past.

6. Covariance & Moments — the $\min(s,t)$ machine

6.1 COVARIANCE OF BM

$$\mathrm{Cov}(W_s, W_t) = E[W_s W_t] = \min(s,t)$$

Use when: the single most-used identity in the paper. Derivation: for $s < t$, $E[W_s W_t] = E[W_s(W_s + (W_t - W_s))] = s +$ (
.

Paper: Midsem'24 Q1(d) — $\mathrm{Var}(W_4) \times \mathrm{Cov}(W_1, W_4) = 4 \times 1 = \mathbf{4}$.

▼ **Proof**

$s \le t$:
$E[W_s W_t] = E[W_s(W_s + (W_t - W_s))] = E[W$
.

6.2 LINEAR COMBINATION OF BM VALUES

$\sum_i a_i W_{t_i} \sim N\Big(0,\ \sum_i \sum_j a_i a_j \min(t_i, t_j$

two terms:

$$\mathrm{Var}(aW_s + bW_t) = a^2 s + b^2 t + 2ab\,\mathrm{min}$$

Use when: "$4W_1 + 6W_3 \sim$?". **Trick:** build the $\min$ matrix, then it is one dot product. Mean is always 0.

Paper: CT-1'26 Q4 — $4W_1 + 6W_3$: $16(1) + 36(3) + 2(24)(1) = 16 + 108 + 48 =$. Set-2 $3W_1{+}6W_3$: $\mathbf{153}$; Set-3 $4W_1{+}5W_3$: $\mathbf{131}$; Set-4 $4W_1{+}7W_3$: $\mathbf{219}$. Midsem'25 Q1(d) — $W_1 - W_2 + 3W_4$: $1 + 2 + 36 - 2(1) + 6(1) - 6(2) = \mathbf{N(0, 31)}$.

▼ **Proof**

BM values are jointly normal, so a linear combination is normal with mean 0. Variance by §1.6 with $\mathrm{Cov}(W_{t_i}, W_{t_j}) = \min(t_i, t_j)$ (§6.1).

6.3 INCREMENT SPLITTING — THE UNIVERSAL TRICK

$$s < t: \quad W_t = W_s + \underbrace{(W_t - W_s)}_{\Delta \sim N(0,\, t-s),\ \perp \mathcal{F}_s}$$

Use when: *any* mixed moment $E[g(W_s)h(W_t)]$. Substitute, expand, kill odd powers of Δ, use independence. This replaces every 2-D integral.

Q: $E[W_2 W_3^3]$. Put $W_3 = W_2 + \Delta$, expand: $E[W_2^4] + 3E[W_2^2]E[\Delta^2] = 3(4) + 3(2)(1) = \mathbf{1}$.
General: $E[W_s W_t^3] = 3st$ for $s \le t$. **Trick:** every odd power of Δ dies.

6.4 MIXED FOURTH MOMENT

$$E[W_s^2 W_t^2] = 2s^2 + st = s(2s + t),$$

Use when: exactly this question. **Order matters** — s is the *smaller* index. Derivation: $E[W_s^2(W_s + \Delta)^2] = E[W_s^4] + E[W_s^2]E[\Delta^2] =$.

Paper: CT-1'26 Q2 — $E(W_2^2 W_3^2) = 2(4) + 6 = \mathbf{14}$; $E(W_1^2 W_3^2) = 2 + 3 = \mathbf{5}$; $E(W_2^2 W_4^2) = 8 + 8 = \mathbf{16}$; $E(W_1^2 W_4^2) = 2 + 4 = \mathbf{6}$.

▼ **Proof**

$W_t = W_s + \Delta$, $\Delta \sim N(0, t-s)$ independent of W_s.
$E[W_s^2(W_s + \Delta)^2] = E[W_s^4] + 2E[W_s^3]E[\Delta] +$.

6.5 COVARIANCE OF EXPONENTIALS

$$\text{Cov}\left(e^{W_s}, e^{W_t}\right) = e^{(s+t)/2}\left(e^{\min(s,t)} - 1\right)$$

Use when: $Y_t := e^{W_t}$, asked for $\text{Cov}(Y_s, Y_t)$.
Trick: $W_s + W_t \sim N(0,\ s + t + 2\min(s,t))$, then $E[e^{W_s+W_t}] = e^{(s+t+2\min)/2}$, minus $e^{s/2}e^{t/2}$.

Paper: CT-1′26 Q9 —
$\text{Cov}(Y_1, Y_2) = e^{3/2}(e-1)$;
$\text{Cov}(Y_2, Y_3) = e^{5/2}(e^2-1)$;
$\text{Cov}(Y_2, Y_4) = e^3(e^2-1) \approx \mathbf{948.43}$;
$\text{Cov}(Y_1, Y_4) = e^{5/2}(e-1) \approx \mathbf{20.93}$.

▼ **Proof**

$s \le t$:
$W_s + W_t = 2W_s + \Delta \sim N(0, 4s + t - s) = N$
, so $E[e^{W_s+W_t}] = e^{(3s+t)/2}$. And
$E[e^{W_s}]E[e^{W_t}] = e^{(s+t)/2}$. Subtract:
$e^{(s+t)/2}(e^s - 1)$.

6.6 CORRELATION OF BM

$$\text{Corr}(W_s, W_t) = \frac{\min(s,t)}{\sqrt{st}} = \sqrt{\frac{s}{t}}, \quad s$$

Use when: a correlation (not covariance) is requested, or when setting up a bivariate normal for conditioning.

Q: $\text{Corr}(W_1, W_4) = \sqrt{1/4} = \mathbf{0.5}$.
Trick: correlation is $\sqrt{s/t}$ — it decays like a square root, so distant times are far less correlated than the covariance $\min(s,t)$ suggests.

▼ **Proof**

$\frac{\text{Cov}}{\sqrt{\text{Var}\,\text{Var}}} = \frac{s}{\sqrt{st}} = \sqrt{\frac{s}{t}}$ for $s \le t$.

7. Martingales of Brownian Motion & the Drift Test

7.1 CANONICAL MARTINGALE 1 — LINEAR

$W(t)$ is a martingale; $E[W_t \mid \mathcal{F}_s] = W_s$

> **Use when:** the base case, and the engine behind every Optional Stopping argument (§10.7).

> **Paper:** Endsem'25 Q1(d) needs exactly one fact: $E[W_T] = 0$ at a stopping time.
> **Trick:** W_t being a martingale is the *only* input to every OST question in §10.7. Nothing else is required.

▼ **Proof**

> $E[W_t \mid \mathcal{F}_s] = W_s + E[W_t - W_s] = W_s$ (increment independent of $\mathcal{F}_s$, mean 0). Integrable: $E|W_t| < \infty$.

7.2 CANONICAL MARTINGALE 2 — QUADRATIC

$W(t)^2 - t$ is a martingale;
$E[W_t^2 \mid \mathcal{F}_s] = W_s^2 + (t - s)$

> **Use when:** you need $E[\tau]$ for an exit problem, or a T/F on $|W_t|^2 - t$.

> **Paper:** Midsem'25 Q1(f) — "$|W_t|^2 - t$ is **NOT** a martingale (T/F)". It *is* one, so the statement is **False**. Watch the negation.

▼ **Proof**

> $E[(W_s + \Delta)^2 \mid \mathcal{F}_s] = W_s^2 + 2W_s E\Delta + E\Delta^2$ = . Subtract t from both sides: $W_s^2 - s$.

7.3 CANONICAL MARTINGALE 3 — EXPONENTIAL

$M_\lambda(t) = \exp\left(\lambda W_t - \frac{\lambda^2}{2} t\right)$ is a martingale, every $\lambda \in \mathbb{R}$

> **Use when:** hitting-time Laplace transforms, Girsanov, and checking GBM martingality. $E[M_\lambda(t)] = 1$ for all t.

> **Q:** find $E[e^{-\lambda \tau_a}]$ (§12.7).
> **Trick:** choose the λ in the martingale that makes the exponent match the barrier, then apply OST. $E[M_\lambda(t)] = 1$ finishes it.

▼ **Proof**

> $E[e^{\lambda W_t} \mid \mathcal{F}_s] = e^{\lambda W_s} E[e^{\lambda \Delta}] = e^{\lambda W_s + \lambda^2 (t-s)/2}$ (§1.2). Multiply by $e^{-\lambda^2 t/2}$: $M_\lambda(s)$.

7.4 MARTINGALE TEST FOR $F(T, W_T)$ — ITÔ DRIFT $= 0$

$$df = \left(\partial_t f + \tfrac{1}{2}\partial_{ww} f\right) dt + \partial_w f \, dW$$

martingale $\iff \partial_t f + \frac{1}{2}\partial_{ww} f = 0$

special case $f = e^{\lambda t} g(w)$: MG $\iff \lambda g + \frac{1}{2} g'' = 0$

> **Use when:** every "Is $Y_t := \ldots$ a martingale? (Yes/No)". For $g = \sin$ or $\cos$, $g'' = -g$, so the **only** working exponent is $\lambda = \frac{1}{2}$.

> **Paper:** CT-1'26 Q5 — $e^{t/2} \sin W_t$ **Yes**; $e^t \cos W_t$ **No**; $e^t \sin W_t$ **No**; $e^{t/2} \cos W_t$ **Yes**.
> **One-second rule:** exponent $\frac{1}{2}$ → Yes, anything else → No.

▼ **Proof**

> Itô Form 2 (§I3.3) gives df; a process is a MG iff its dt coefficient is 0 (§I5.1). Special case $f = e^{\lambda t} g(w)$: $\partial_t f = \lambda e^{\lambda t} g$, $\partial_{ww} f = e^{\lambda t} g''$, so the drift is $e^{\lambda t}(\lambda g + \frac{1}{2} g'')$.

7.5 ITÔ INTEGRAL — ALWAYS A MARTINGALE, ALWAYS NORMAL

$$\int_0^T f(t)\, dW_t \sim N\left(0, \ \int_0^T f(t)^2 \, dt\right)$$

Itô isometry:

$$E\left[\left(\int_0^T H_t dW_t\right)^2\right] = E\int_0^T H_t^2 \, dt$$

Normality needs f deterministic; the martingale/isometry part holds for adapted H.

> **Use when:** "compute the distribution of $\int_0^1 t^k dW_t$". Mean is 0 free; variance is a one-line calculus integral.

> **Paper:** CT-1'26 Q1 — $\int_0^1 t^k dW_t \sim N\left(0, \frac{1}{2k+1}\right)$: $k{=}1 \to \mathbf{N(0, 1/3)}$, $k{=}2 \to \mathbf{1/5}$, $k{=}3 \to \mathbf{1/7}$, $k{=}4 \to \mathbf{1/9}$. CT-2'25 Q7: $\int_0^1 t^2 dW_t \sim N(0, 1/5)$.

▼ **Proof**

> Normal: §I1.5. Isometry: §I1.4. Martingale: §I1.3 (open their proofs).

7.6 CONDITIONAL EXPECTATIONS YOU CAN QUOTE

$$E[W_t \mid \mathcal{F}_s] = W_s$$
$$E[W_t^2 \mid \mathcal{F}_s] = W_s^2 + (t-s)$$
$$E[e^{W_t} \mid \mathcal{F}_s] = e^{W_s + (t-s)/2}$$
$$E[e^{\lambda W_t} \mid \mathcal{F}_s] = e^{\lambda W_s + \lambda^2 (t-s)/2}$$

Use when: "filtration expectation" questions. All four come from §6.3 + §1.2 in one step.

Q: find $E[W_3^2 \mid \mathcal{F}_1]$ → $\mathbf{W_1^2 + 2}$.
Trick: all four lines are §6.3 (split the increment) followed by §1.2 (exponential moment). Derive them, don't memorize them.

▼ **Proof**

Split $W_t = W_s + \Delta$ with $\Delta \sim N(0, t-s)$ independent of $\mathcal{F}_s$: lines 1–2 are §7.1–7.2.
Lines 3–4:
$E[e^{\lambda W_t} \mid \mathcal{F}_s] = e^{\lambda W_s} E[e^{\lambda \Delta}] = e^{\lambda W_s + \lambda^2 (t-s)/2}$,
with $\lambda = 1$ for line 3.

8. Density, Transition Kernel & the Diffusion Equation

8.1 DENSITY OF $W(T)$

$$f_{W(t)}(x) = \frac{1}{\sqrt{2\pi t}}\, e^{-x^2/(2t)}$$
$$P(W(t) \le x) = \Phi\left(\frac{x}{\sqrt{t}}\right)$$

Use when: W alone, no maximum, no barrier. **Always divide by $\sqrt{t}$, never t.**

Q: $P(W(4) > 3) = 1 - \Phi(3/2) = \mathbf{0.0668}$.
Trick: $\sqrt{t} = 2$, **not** 4. Dividing by t instead of $\sqrt{t}$ is the single most common lost mark on this paper.

▼ **Proof**

$W(t) \sim N(0, t)$: that is the normal density with variance t.
$P(W(t) \le x) = P(\sqrt{t} Z \le x) = \Phi(x/\sqrt{t})$.

8.2 TRANSITION DENSITY

$$p(t; x, y) = \frac{1}{\sqrt{2\pi t}} \exp\left(- \frac{(y-x)^2}{2t} \right)$$

Use when: BM started at $x \neq 0$. Everything else is unchanged — recentre and reuse every formula.

Q: BM started at 5;
$P(W(1) < 7) = \Phi(2) = \mathbf{0.9772}$.
Trick: subtract the starting point first — then every other formula on this sheet applies unchanged.

▼ **Proof**

Given $W(s) = x$:
$W(s+t) = x + (W(s+t) - W(s))$, and the increment is $N(0, t)$, independent of the past. So $W(s+t) \sim N(x, t)$.

8.3 DIFFUSION (HEAT) EQUATION

$$\frac{\partial p}{\partial t} = \frac{1}{2}\frac{\partial^2 p}{\partial y^2}$$

Use when: "show the density satisfies the heat equation". Differentiate 8.1 twice; the $\frac{1}{2}$ is the same $\frac{1}{2}$ as in Itô's formula.

Q: "show the transition density satisfies the heat equation".
Trick: ∂_{yy} brings down $\left(\frac{(y-x)^2}{t^2} - \frac{1}{t}\right)$; ∂_t brings down exactly half of it. One line each, then compare.

▼ **Proof**

Put $u = y - x$, $p = (2\pi t)^{-1/2} e^{-u^2/(2t)}$. Then $\partial_t p = p\left(-\frac{1}{2t} + \frac{u^2}{2t^2}\right)$ and $\partial_y p = -\frac{u}{t}p$, $\partial_{yy} p = p\left(\frac{u^2}{t^2} - \frac{1}{t}\right)$. Compare: $\partial_t p = \frac{1}{2}\partial_{yy} p$.

8.4 CHAPMAN–KOLMOGOROV

$$p(t+s; x, z) = \int_{\mathbb{R}} p(t; x, y)\, p(s; y, z)\, dy$$

Use when: justifying the Markov property, or short-note questions on why independent increments force Gaussian transitions.

Q: "why *must* the increments be Gaussian?"
Trick: Chapman–Kolmogorov plus stationary independent increments plus continuity forces the normal family. That sentence is the short-note answer.

▼ **Proof**

Go from x to z in time $t+s$ via the position y at time t: condition on $W(t) = y$, then the next stretch is an independent increment. Total probability over y gives the integral. (Both sides are the $N(x, t+s)$ density at z.)

9. Conditioning & the Brownian Bridge

9.1 WHICH DIRECTION ARE YOU CONDITIONING?

Forward (know past, want future) → 9.2

Backward (know endpoint, want interior) → 9.4

> **Use when: decide this first.** Forward is trivial (martingale, variance grows). Backward is the bridge (variance shrinks to 0 at both ends).

> **Q:** "Given $W(1) = 2$, find $E[W(3)]$" → $\mathbf{2}$ (martingale). "Given $W(3) = 2$, find $E[W(1)]$" → $\frac{1}{3}(2) = \mathbf{2/3}$ (bridge).
> **Trick:** identify which time is *known* before you write a single symbol.

9.2 FORWARD CONDITIONAL

$$W(t) \mid \mathcal{F}_s \sim N\big(W(s),\, t-s\big), \qquad t > s$$

> **Use when:** future given present. Independent increments, nothing more.

> **Q:** $W(5) \mid W(2) = 3 \sim \mathbf{N(3,\, 3)}$.
> **Trick:** the mean stays exactly where you left it; the variance is the elapsed time and nothing else.

▼ **Proof**

> $W(t) = W(s) + (W(t) - W(s))$: $W(s)$ is known, the increment is $N(0, t-s)$ and independent of $\mathcal{F}_s$.

9.3 STANDARD BROWNIAN BRIDGE

$$B(s) = W(s) - \frac{s}{t}W(t), \quad 0 \le s \le t$$

$$E[B(s)] = 0, \qquad \mathrm{Cov}(B(u), B(v)) = \mathrm{m}$$

$$B \perp W(t)$$

> **Use when:** the process is pinned at both ends. The $-uv/t$ term is exactly the information removed by knowing the endpoint.

> **Q:** $\mathrm{Var}(B(t/2))$ on $[0,t] = \frac{t}{2} - \frac{(t/2)^2}{t} = \mathbf{t/4}$.
> **Trick:** bridge variance is zero at both ends and peaks at the midpoint — a 2-second sanity check on any bridge answer.

▼ **Proof**

> Mean: linear in W. For $u, v \le t$:
> $\mathrm{Cov}(B_u, B_v) = \min(u,v) - \frac{v}{t}\min(u,t) - \frac{u}{t}$ ı .
> $\mathrm{Cov}(B(s), W(t)) = s - \frac{s}{t}\,t = 0$; jointly normal, so independent.

9.4 BACKWARD INTERPOLATION — THE WORKHORSE

$$W(s) \mid W(t){=}b \;\sim\; N\!\left(\frac{s}{t}\,b,\; \frac{s(t-s)}{t}\right)$$

> **Use when:** "given $W(t) = b$, find $P(W(s) < x)$ / $E[W(s)]$ / Var". Mean is the straight line from $(0,0)$ to (t,b); variance vanishes at both ends and peaks at the midpoint.

> **Trick:** derive it in 10 seconds from §1.9 with $\Sigma = \begin{pmatrix} s & s \\ s & t \end{pmatrix}$: mean $= \frac{s}{t}b$, var $= s - \frac{s^2}{t}$.

▼ **Proof**

> $W(s) = B(s) + \frac{s}{t}W(t)$ with $B(s) \perp W(t)$ (§9.3). Given $W(t) = b$: mean $\frac{s}{t}b$, variance $\mathrm{Var}\, B(s) = s - \frac{s^2}{t} = \frac{s(t-s)}{t}$.

9.5 BRIDGE PINNED $A \to B$

$$X(s) \sim N\left(a + \frac{s}{t}(b-a),\ \frac{s(t-s)}{t}\right)$$

$X(0) = a,\ X(t) = b.$

> **Use when:** the process does not start at 0. Only the mean line shifts; the variance is untouched.

> **Q:** $W(0) = 2,\ W(10) = 8$; find the law of $W(4)$. Mean $= 2 + \frac{4}{10}(6) = \mathbf{4.4}$, Var $= \frac{4(6)}{10} = \mathbf{2.4}$.
> **Trick:** the mean is plain linear interpolation between the two pins.

▼ **Proof**

> Write $X(s) = a + W(s)$ conditioned on $W(t) = b - a$. By §9.4: mean $a + \frac{s}{t}(b-a)$, variance $\frac{s(t-s)}{t}$.

9.6 SCALED BM ($Y = \Sigma W$)

$$Y(s) \mid Y(t){=}b \sim N\left(\frac{s}{t}b,\ \sigma^2\frac{s(t-s)}{t}\right)$$

> **Use when:** volatility $\sigma \neq 1$. **Mean is unchanged** (ratios cancel); only variance picks up σ^2.

> **Q:** $Y = 3W$ with $Y(10) = 12$; find the law of $Y(4)$. Mean $= \frac{4}{10}(12) = \mathbf{4.8}$, Var $= 9 \cdot \frac{4(6)}{10} = \mathbf{21.6}$.
> **Trick:** σ^2 scales the variance and leaves the mean untouched.

▼ **Proof**

> $Y(t) = b \iff W(t) = b/\sigma$. Then $Y(s) = \sigma W(s)$ has mean $\sigma \cdot \frac{s}{t} \cdot \frac{b}{\sigma} = \frac{s}{t}b$ and variance $\sigma^2 \frac{s(t-s)}{t}$ (§9.4).

10. BM with Drift — Exit Probabilities & Optimal Stopping

10.1 DEFINITION

$X(t) = \mu t + \sigma W(t) \sim N(\mu t,\ \sigma^2 t)$

$X(t) - X(s) \sim N\big(\mu(t-s),\ \sigma^2(t-s)\big)$, independent increments

> **Use when:** "drift coefficient / drift parameter μ and variance parameter σ^2". Careful: some questions give σ^2, some give σ.

> **Paper:** Midsem'25 Q1(a) says "drift coefficient -1" → $\mu{=}-1$, $\sigma{=}1$. CT-1'26 Q6 says "volatility parameter $\sigma = 0.3$" → that is σ, not σ^2.
> **Trick:** check whether they handed you σ or σ^2 — the exit formulas all want σ^2.

▼ **Proof**

> $\sigma W(t) \sim N(0, \sigma^2 t)$, shift by μt. Increments: $X(t) - X(s) = \mu(t-s) + \sigma(W_t - W_s)$, inheriting independence from W.

10.2 WHICH EXIT FORMULA? — DECISION

$\mu = 0$, two barriers → 10.4 (distance ratio)

$\mu \neq 0$, two barriers → 10.3 (exponentials)

$\mu < 0$, *one* upper barrier, no lower → 10.5

$\mu \geq 0$, one upper barrier → probability $= 1$

> **Use when:** run this fork before writing anything. Most lost marks are from using 10.3 when $\mu = 0$ (it is $0/0$).

> **Paper:** Midsem'24 Q1(c) gives $\mu = 0$, so 10.4 applies and 10.3 does not.
> **Trick:** at $\mu = 0$ the exponential formula is literally $0/0$. Check the drift *before* choosing a formula, every time.

10.3 TWO-BARRIER EXIT — DRIFT $\neq 0$

$P(\text{hit } A \text{ before } -B \mid X(0){=}0) = \dfrac{1 -}{1 - e}$

from $X(0) = x$, barriers A (up) and $-B$ (down): replace $B \to x + B$

$A, B > 0$, $\mu \neq 0$.

> **Use when:** continuous-time gambler's ruin. It is §3.3 with $(q/p)^i \to e^{-2\mu x/\sigma^2}$.

> **Q:** $\mu = 0.1$, $\sigma = 1$, barriers $+2$ and -3 from 0.
> $P = \dfrac{1 - e^{-0.6}}{1 - e^{-1}} = \dfrac{0.4512}{0.6321} = \mathbf{0.714}$.
> **Trick:** set $r = 2\mu/\sigma^2$ once, then it reads $\frac{1-e^{-rB}}{1-e^{-r(A+B)}}$.

▼ **Proof**

> With $\theta = -2\mu/\sigma^2$, $e^{\theta X_t}$ has Itô drift $e^{\theta X}(\theta\mu + \frac{1}{2}\theta^2\sigma^2) = 0$: a MG, bounded between the barriers. OST (§M6.7) with $p = P(\text{hit } A \text{ first})$:
> $1 = p\,e^{\theta A} + (1-p)e^{-\theta B} \Rightarrow p = \frac{1-e^{-\theta B}}{e^{\theta A} - e^{-\theta B}}$.
> Multiply top and bottom by $e^{\theta B}$:
> $\frac{e^{\theta B}-1}{e^{\theta(A+B)}-1} = \frac{1-e^{-2\mu B/\sigma^2}}{1-e^{-2\mu(A+B)/\sigma^2}}$. Starting at x: shift everything by x.

10.4 TWO-BARRIER EXIT — ZERO DRIFT

$P(\text{hit } A \text{ before } -B \mid X(0){=}0) = \dfrac{B}{A +}$

$P(\text{hit } -B \text{ first}) = \dfrac{A}{A+B}$

> **Use when:** $\mu = 0$. **Trick:** the probability of hitting a barrier is proportional to the distance to the *opposite* barrier. σ cancels and never appears.

> **Paper:** Midsem'24 Q1(c) — $B(t)$, $\mu{=}0$, $\sigma^2{=}1$, "reaches -2 before 1". Up-barrier $A{=}1$, down-barrier $B{=}2$.
> $P(\text{down first}) = \frac{A}{A+B} = \frac{1}{3} \Rightarrow \mathbf{1/3}$.

▼ **Proof**

> $X = \sigma W$ is a MG trapped between the barriers. OST: $0 = pA - (1-p)B$, so $p = \frac{B}{A+B}$.

10.5 NEGATIVE DRIFT — MAXIMUM EVER REACHED

$$P\Big(\max_{t\geq 0} X(t) \geq a\Big) = e^{2\mu a/\sigma^2} = e^{-2|\mu|}$$

$$\max_{t\geq 0} X(t) \sim \text{Exp}\left(\frac{2|\mu|}{\sigma^2}\right)$$

Use when: only an upper target, drift pulls down, infinite horizon. If $\mu \geq 0$ the probability is 1 (the process reaches every level a.s.).

Trick: memoryless ⇒ the all-time max is Exponential. That is what makes the optimization in 10.6 a one-line calculus problem.

▼ **Proof**

Take §10.3 with upper barrier a and lower barrier $-B$, then $B \to \infty$. With $c = 2|\mu|/\sigma^2 > 0$ ($\mu < 0$): $p = \frac{e^{cB}-1}{e^{c(a+B)}-1} \to e^{-ca}$. So $P(\max \geq a) = e^{-ca}$, which is the tail of $\text{Exp}(c)$.

10.6 OPTIMAL EXERCISE THRESHOLD (DRIFT $-\mu$)

maximize $g(x) = (x-K)\,e^{-2\mu x/\sigma^2}$ over $x \geq K$

$$x^* = K + \frac{\sigma^2}{2\mu}, \qquad g(x^*) = \frac{\sigma^2}{2\mu}\,e^{-2\mu K/c}$$

$\sigma = 1$ case: $x^* = K + \frac{1}{2\mu}$.

Use when: "option to buy at fixed price K; price follows BM with drift coefficient $-\mu$; when, if ever, should we exercise?"
Set-up: payoff $x - K$, probability of ever reaching x is $e^{-2\mu x/\sigma^2}$ (10.5). Differentiate, done.

Paper: Midsem'25 Q1(a) — $K = 100$, drift -1 (so μ=1, σ=1), start 0. $x^* = 100 + \frac{1}{2} = \mathbf{100.5}$. Exercise the first time the price hits 100.5.

▼ **Proof**

Waiting until X first reaches x pays $x - K$ and happens with probability e^{-cx}, $c = 2\mu/\sigma^2$ (§10.5). So maximise $g(x) = (x-K)e^{-cx}$: $g'(x) = e^{-cx}\big(1 - c(x-K)\big) = 0 \Rightarrow x^* = K +$ (g' changes from $+$ to $-$: a max). $g(x^*) = \frac{1}{c}e^{-cK-1}$.

10.7 HITTING A SLOPING LINE — VIA OST

$$T = \min\{t : W_t = a + bt\}$$

$$E[W_T] = 0 \Rightarrow a + b\,E[T] = 0 \Rightarrow E[T] =$$

Needs $ab < 0$ (line slopes back toward 0), else $E[T] = \infty$.

Use when: the barrier *moves*. Apply OST to the martingale W_t at the stopping time, substitute the barrier equation, solve for $E[T]$.

Paper: Endsem'25 Q1(d) — $T = \min\{t : W_t = 2 - 3t\}$. $0 = 2 - 3E[T] \Rightarrow E[T] = \mathbf{2/3}$.

▼ **Proof**

Same as §M6.5: $E[W_T] = 0$ and $W_T = a + bT$ on the line, so $0 = a + bE[T]$.

10.8 EXPECTED EXIT TIME (TWO BARRIERS)

$$\mu = 0: \quad E[T] = \frac{A\,B}{\sigma^2} \quad \text{(from } X(0) = 0\text{)}$$

$$\mu \neq 0: \quad E[T] = \frac{A\,p - B(1-p)}{\mu}, \quad p \text{ from 10.3}$$

Use when: duration asked as well as probability. μ=0 case uses the martingale $X_t^2 - \sigma^2 t$; $\mu \neq 0$ uses $X_t - \mu t$ (Wald).

Check: $\mu = 0$, $\sigma = 1$, $A = B = n$ gives $E[T] = n^2$ — the discrete $i(N-i)$ of §3.2 with $i = n$, $N = 2n$.

▼ **Proof**

$\mu = 0$: $X^2 - \sigma^2 t$ is a MG, so $\sigma^2 E[T] = E[X_T^2] = A^2 p + B^2(1-p) = AB$ with $p = \frac{B}{A+B}$.
$\mu \neq 0$: $X_t - \mu t$ is a MG, so $E[X_T] = \mu E[T]$, and $E[X_T] = Ap - B(1-p)$.

11. Geometric Brownian Motion

11.1 ⚠ TWO CONVENTIONS FOR "DRIFT PARAMETER μ"

(R) Ross / log-drift:

$$Y(t) = Y(0)\, e^{\mu t + \sigma W_t}$$

(S) SDE:

$$dY = \mu Y\, dt + \sigma Y\, dW \;\Rightarrow\; Y(t) = Y(0$$

The paper accepts BOTH. CT-1'26 Q6 official answer key literally reads "$5e^{0.55}$ **or** $5e^{0.1}$". **Write both, label them** — that is the full-marks answer.

Tell them apart: "logarithm of the price is BM with drift μ" ⇒ (R). "$dS = \mu S\, dt + \ldots$" or "expected return μ" ⇒ (S).

Paper: CT-1'26 Q6, all four sets, and Midsem'25 Q1(b).
Trick: write both, labelled: "log-drift convention: . . . ; SDE convention: . . .". Two boxed expressions. The key accepts either, and labelling proves you know why there are two.

▼ **Proof**

(S): apply Itô to the candidate $Y_0 e^{(\mu-\sigma^2/2)t+\sigma W_t}$: drift $(\mu - \frac{\sigma^2}{2}) + \frac{\sigma^2}{2} = \mu$, diffusion σ. It solves $dY = \mu Y\, dt + \sigma Y\, dW$ (§I4.6).

11.2 MEAN OF GBM

(R) $E[Y(t)] = Y(0)\, e^{(\mu+\sigma^2/2)\, t}$

(S) $E[Y(t)] = Y(0)\, e^{\mu t}$

Use when: the single most repeated GBM question. Both are §1.2 applied to $\mu t + \sigma W_t \sim N(\mu t, \sigma^2 t)$.

Paper: CT-1'26 Q6 — μ=.01, σ=.3, Y_0=5: $5e^{(.01+.045)10} = \mathbf{5e^{0.55}}$ or $\mathbf{5e^{0.1}}$. Set-2 μ=.02, σ=.4, Y_0=10: $\mathbf{10e^{1}}$ or $\mathbf{10e^{0.2}}$. Set-3: $\mathbf{5e^{0.5}}$ / $5e^{0.3}$. Set-4: $\mathbf{5e^{0.3}}$ / $5e^{0.1}$. Midsem'25 Q1(b) — μ=.01, σ=.2, Y_0=10, t=100: $\mathbf{10e^{3}}$ or $10e^{1}$.

▼ **Proof**

(R): $E[e^{\mu t+\sigma W_t}] = e^{\mu t+\sigma^2 t/2}$ (§1.2). (S): the exponent is $(\mu - \sigma^2/2)t + \sigma W_t$, so the $\sigma^2/2$ cancels: $e^{\mu t}$.

11.3 FILTRATION EXPECTATION OF GBM

(R) $E[Y(t) \mid \mathcal{F}_s] = Y(s)\, e^{(\mu+\sigma^2/2)(t-s)}$

(S) $E[Y(t) \mid \mathcal{F}_s] = Y(s)\, e^{\mu(t-s)}$

Use when: conditional version. **Trick:** same formula with $Y(0) \to Y(s)$ and $t \to t - s$ — GBM is multiplicatively memoryless.

Q: $E[Y(10) \mid \mathcal{F}_4]$ with (R), μ=0.01, σ=0.2 → $Y(4)\, e^{(0.01+0.02)(6)} = \mathbf{Y(4)e^{0.18}}$.
Trick: same formula as 11.2 with $Y(0) \to Y(s)$ and $t \to t - s$.

▼ **Proof**

$Y(t) = Y(s)\, e^{\mu(t-s)+\sigma(W_t-W_s)}$. Take out $Y(s)$; the increment is independent: $E[e^{\sigma\Delta}] = e^{\sigma^2(t-s)/2}$. (S) is the same with $\mu - \sigma^2/2$ in place of μ.

11.4 GBM MARTINGALE CONDITION

(R) MG $\iff \mu = -\dfrac{\sigma^2}{2}$ (S) MG $\iff \mu = 0$

Use when: "is the discounted/undiscounted price a martingale?". Same $-\sigma^2/2$ as the geometric random walk (§2.3).

Q: "Is $e^{W_t - t/2}$ a martingale?" → **Yes**: it is (R) with $\mu = -\frac{1}{2} = -\sigma^2/2$, $\sigma = 1$.
Trick: it is precisely the exponential martingale of §7.3 at $\lambda = 1$. Same object, two names.

▼ **Proof**

From §11.3: $E[Y(t) \mid \mathcal{F}_s] = Y(s)$ iff the exponent is 0: $\mu + \sigma^2/2 = 0$ (R), $\mu = 0$ (S).

11.5 HIGHER MOMENTS & VARIANCE

(R) $E[Y(t)^n] = Y(0)^n\, e^{(n\mu+n^2\sigma^2/2)t}$

$\mathrm{Var}(Y(t)) = E[Y(t)]^2\left(e^{\sigma^2 t}-1\right)$

Use when: variance or second moment requested. Note Var is convention-free once written in terms of $E[Y(t)]$.

Q: (R) with μ=0, σ=1, Y_0=1, t=1: $E[Y] = e^{1/2}$, so $\mathrm{Var} = e(e-1) = \mathbf{4.67}$.
Trick: get $E[Y(t)]$ first, square it, multiply by $(e^{\sigma^2 t}-1)$. Convention-free once written that way.

▼ **Proof**

$Y^n = Y_0^n e^{n\mu t + n\sigma W_t}$, and $E[e^{n\sigma W_t}] = e^{n^2\sigma^2 t/2}$.
Variance:
$E[Y^2] - (EY)^2 = Y_0^2 e^{2\mu t + 2\sigma^2 t} - Y_0^2 e^{2\mu t+\sigma^2 t} =$
.

11.6 MEDIAN VS MEAN

(R) median $= Y(0)e^{\mu t}$, mean $= Y(0)e^{(\mu+\sigma^2/2)t}$

Use when: conceptual questions on why a GBM can have growing mean but drift to 0 almost surely: for $\mu < 0 < \mu + \sigma^2/2$, median $\to 0$ while mean $\to \infty$.

Q: (R) with $\mu = -0.1$, $\sigma = 0.6$: does the stock grow? The mean grows ($\mu + \sigma^2/2 = 0.08 > 0$) while the median $\to 0$.
Trick: that gap *is* the log-normal. Expect a conceptual mark on it.

▼ **Proof**

$\ln Y(t) \sim N(\ln Y_0 + \mu t, \sigma^2 t)$ is symmetric, so its median is its mean. e^x is increasing, so the median of Y is $e^{\text{median of } \ln Y} = Y_0 e^{\mu t}$. The mean has the extra $e^{\sigma^2 t/2}$ (§11.2).

11.7 DISTRIBUTION & PROBABILITIES

$\ln \dfrac{Y(t)}{Y(0)} \sim N\left(\mu t,\ \sigma^2 t\right)$ (R)

$P(Y(t) > K) = 1 - \Phi\left(\dfrac{\ln(K/Y_0) - \mu t}{\sigma\sqrt{t}}\right.$

Use when: "probability the price exceeds K". Take logs first, *then* standardize. This is the d_2 of Black–Scholes.

Q: $S_0 = 100$, (R) $\mu = 0.05$, $\sigma = 0.2$, $t = 1$. $P(S_1 > 110) = 1 - \Phi\left(\frac{\ln 1.1 - 0.05}{0.2}\right) = 1 - \Phi(0.2$
.
Trick: take logs *first*, then standardize. That expression is Black–Scholes' d_2.

▼ **Proof**

$\ln \frac{Y(t)}{Y_0} = \mu t + \sigma W_t \sim N(\mu t, \sigma^2 t)$.
$P(Y > K) = P\left(\mu t + \sigma\sqrt{t}Z > \ln \frac{K}{Y_0}\right) = P\left(Z\right.$
.

12. Hitting Times & the Reflection Principle

12.1 HITTING TIME & THE KEY EQUIVALENCE

$$\tau_a = \inf\{t \geq 0 : W(t) = a\}$$

$$\{\tau_a \leq t\} = \{M(t) \geq a\} \quad \text{for } a > 0$$

Use when: this equivalence is the bridge between §12 and §13 — every maximum question is a hitting-time question and vice versa. For $a < 0$ use $\tau_a \overset{d}{=} \tau_{|a|}$ by symmetry.

Q: $P(M(4) \geq 3)$ and $P(\tau_3 \leq 4)$ are the same number.
Trick: whenever you see one, silently rewrite it as the other and use whichever formula you actually remember. They are interchangeable by definition.

▼ **Proof**

If $M(t) \geq a > 0 = W(0)$, the continuous path climbs from 0 to $\geq a$ by time t, so by the intermediate value theorem it equals a at some time $\leq t$: $\tau_a \leq t$. Conversely $W(\tau_a) = a$ gives $M(t) \geq a$.

12.2 REFLECTION PRINCIPLE

$$P\big(W(t) \leq a - y,\ M(t) > a\big) = P\big(W(t)$$

Requires $a > 0$ ***and*** $y \geq 0$.

Use when: the engine for all of §13–14. Reflect the path after τ_a across the line $W = a$; the strong Markov property makes the reflected path an equally likely BM.
Why $a > 0$**:** it forces $\tau_a \leq t$ automatically, so that condition can be dropped.

Q: $P(W(4) \leq 1,\ M(4) > 2)$: read off $a = 2$, $y = 2 - 1 = 1 \to 1 - \Phi(3/2) = \mathbf{0.0668}$.
Trick: a is the barrier, y is barrier minus the W-bound. If $y < 0$ you are in the third branch of §14.1 instead.

▼ **Proof**

Let τ_a be the first time W hits a. On $\{M(t) > a\}$ we have $\tau_a < t$.
Reflect the path after τ_a: $\widetilde{W}(s) = W(s)$ for $s \leq \tau_a$, $\widetilde{W}(s) = 2a - W(s)$ after. By symmetry and the fresh start at τ_a, $\widetilde{W}$ is again a BM.
Reflection swaps $\{W(t) \leq a - y\}$ with $\{\widetilde{W}(t) \geq a + y\}$, and every path ending at $\geq a + y$ has already passed a. So $P(W(t) \leq a - y,\ M(t) > a) = P(W(t) \geq a +$.

12.3 CDF OF T_A

$$P(\tau_a \le t) = 2\Big(1 - \Phi\Big(\frac{a}{\sqrt{t}}\Big)\Big) = 2\,\Phi\Big(-$$

$a > 0$; for general a replace a by $|a|$.

Use when: "find the cdf of τ_a". **Trick:** set $y = 0$ in 12.2 and double for the symmetric half $\{W(t) > a\}$.

Paper: Midsem'25 Q5(a) — "find the cdf and pdf of τ_a". Quote 12.3 then differentiate for 12.4.

▼ **Proof**

$\{\tau_a \le t\} = \{M(t) \ge a\}$. Split on where $W(t)$ ends:
$P(M(t) \ge a) = P(W(t) \ge a) + P\big(W(t) < a,$; the first term needs no M (ending above a means you crossed it).
Reflection (§12.2 with $y = 0$): the second term $= P(W(t) > a)$. Total
$= 2P(W(t) \ge a) = 2\big(1 - \Phi(a/\sqrt{t})\big)$.

12.4 PDF OF T_A (LÉVY DISTRIBUTION)

$$f_{\tau_a}(t) = \frac{|a|}{\sqrt{2\pi}\,t^{3/2}}\,e^{-a^2/(2t)}, \qquad t > 0$$

Use when: follows from 12.3 by $\frac{d}{dt}$. **Trick:** differentiate $2\Phi(-at^{-1/2})$ using the chain rule; the $t^{-3/2}$ comes from $\frac{d}{dt}t^{-1/2}$.

Paper: Midsem'25 Q5(a), second half.
$f_{\tau_2}(4) = \frac{2}{\sqrt{2\pi}\,8}e^{-1/2} = \mathbf{0.0605}$.
Trick: differentiate 12.3 — never try to integrate the joint density for this.

▼ **Proof**

Differentiate the cdf $2\Phi(-a\,t^{-1/2})$:
$\frac{d}{dt} = 2\varphi(a/\sqrt{t}) \cdot \frac{a}{2}t^{-3/2} = \frac{a}{\sqrt{2\pi}\,t^{3/2}}e^{-a^2/(2t)}$.

12.5 CERTAIN HITTING, INFINITE MEAN

$$P(\tau_a < \infty) = \lim_{t\to\infty} 2\Phi\Big(-\frac{a}{\sqrt{t}}\Big) = 2\mathfrak{d}$$

$E[\tau_a] = \infty$ (the $t^{-3/2}$ tail is not integrable)

Use when: "$P(T_a < \infty) = ?$" → **1**. BM hits every level with certainty, but takes infinitely long on average. Both facts are asked.

Paper: Midsem'24 Q1(f) — "T_a = time to hit a; find $P(T_a < \infty)$" ⇒ **1**.

▼ **Proof**

$P(\tau_a < \infty) = \lim_t P(\tau_a \le t) = 2\Phi(0) = 1$.
Mean: $E\tau_a = \int_0^\infty P(\tau_a > t)\,dt$, and $P(\tau_a > t) = 2\Phi(a/\sqrt{t}) - 1 \approx \frac{2a}{\sqrt{2\pi t}}$ for large t. $\int^\infty t^{-1/2}dt = \infty$.

12.6 SCALING OF T_A

$$\tau_a \overset{d}{=} a^2\,\tau_1$$

Use when: comparing hitting times of different levels, or sanity-checking: doubling the barrier quadruples the typical time.

Q: "$E[\tau_1] = \infty$ — what about τ_{10}?" Also ∞, and typically $100\times$ longer.
Trick: $\tau_a \overset{d}{=} a^2\tau_1$ is the same $\sqrt{t}$ scaling as §4.1, seen from the other side.

▼ **Proof**

$\widetilde{W}(u) = \frac{1}{a}W(a^2u)$ is a BM (scaling, §5.4). W hits a at time a^2u exactly when $\widetilde{W}$ hits 1 at time u. So $\tau_a = a^2\widetilde{\tau}_1$, and $\widetilde{\tau}_1 \overset{d}{=} \tau_1$.

12.7 LAPLACE TRANSFORM OF T_A

$$E\left[e^{-\lambda\tau_a}\right] = e^{-a\sqrt{2\lambda}}, \qquad \lambda > 0,\ a > 0$$

Use when: a discounted payoff at a hitting time. Derived by OST on the exponential martingale (§7.3) with $\lambda_{\text{mg}} = \sqrt{2\lambda}$.

Q: value of Rs 1 paid the instant the price first hits a, discounted at rate λ → $\mathbf{e^{-a\sqrt{2\lambda}}}$.
Trick: apply OST to $\exp(\theta W_t - \frac{\theta^2}{2}t)$ with $\theta = \sqrt{2\lambda}$, so the t-term becomes exactly the discount factor.

▼ **Proof**

$M_t = e^{\theta W_t - \theta^2 t/2}$ ($\theta > 0$) is a MG (§7.3) and stays $\le e^{\theta a}$ up to τ_a. OST: $1 = E\left[e^{\theta a - \theta^2\tau_a/2}\right]$, so $E[e^{-(\theta^2/2)\tau_a}] = e^{-\theta a}$. Put $\lambda = \theta^2/2$, $\theta = \sqrt{2\lambda}$.

13. The Running Maximum $M(t)$ — alone

13.1 DEFINITION & HARD CONSTRAINTS

$$M(t) = \max_{0\le u\le t} W(u)$$
$$M(t) \ge W(t) \text{ and } M(t) \ge 0 \textbf{ always}$$

Use when: check these two first. Any event forcing $W(t) > M(t)$, or $M(t) < 0$, has probability **zero** — write "0" and move on.

Q: $P(2 < W(4) < 3,\ 0 < M(4) < 1) = \mathbf{0}$.
Trick: it demands $W > M$, which is impossible. Spot it, write 0, move on — this is a deliberate trap and free marks.

▼ **Proof**

The max over $[0,t]$ is at least the value at $u = t$, which is $W(t)$, and at $u = 0$, which is 0 .

13.2 DISTRIBUTION OF $M(T)$ — HALF-NORMAL

$$M(t) \overset{d}{=} |W(t)|$$
$$P(M(t) \ge a) = 2\left(1 - \Phi\left(\frac{a}{\sqrt{t}}\right)\right), \qquad a$$
$$P(M(t) \le a) = 2\Phi\left(\frac{a}{\sqrt{t}}\right) - 1$$

Use when: the question involves M but **not** W. One line, no reflection needed.

Interval:
$P(c < M(t) \le d) = 2\left[\Phi(d/\sqrt{t}) - \Phi(c/\sqrt{t})\right]$. For t=4, $(c,d) = (2,3)$:
$2[\Phi(1.5) - \Phi(1)] = 2[.9332 - .8413] = \mathbf{0.183}$.

▼ **Proof**

$P(M \ge a) = P(W \ge a) + P(W < a, M \ge a)$ (reflection, see §12.3 proof) $= P(|W(t)| \ge a)$. The second line is $1-$ the first.

13.3 DENSITY OF $M(T)$

$$f_{M(t)}(a) = \frac{2}{\sqrt{2\pi t}}\, e^{-a^2/(2t)}, \qquad a \geq 0$$

Use when: an expectation over M is needed. It is twice the $N(0,t)$ density on the positive half-line.

Q: "compute $E[M(4)]$ by integrating the density".
Trick: don't — quote §13.4. Reach for the density only when the functional is non-standard (e.g. $E[e^{M(t)}]$).

▼ **Proof**

Differentiate $P(M \leq a) = 2\Phi(a/\sqrt{t}) - 1$: $\frac{2}{\sqrt{t}}\varphi(a/\sqrt{t}) = \frac{2}{\sqrt{2\pi t}} e^{-a^2/(2t)}$.

13.4 MOMENTS OF $M(T)$

$$E[M(t)] = \sqrt{\frac{2t}{\pi}}, \qquad E[M(t)^2] = t$$

$$\mathrm{Var}(M(t)) = t\left(1 - \frac{2}{\pi}\right)$$

Use when: "expected maximum". Note $E[M(t)^2] = E[W(t)^2] = t$ because $M \overset{d}{=} |W|$.

Q: "expected highest level reached in 4 units of time" $= \sqrt{8/\pi} = \mathbf{1.596}$.
Trick: $E[M(t)] = \sqrt{2t/\pi} \approx 0.8\sqrt{t}$ — useful for checking an answer's order of magnitude.

▼ **Proof**

$M(t) \overset{d}{=} |W(t)| = \sqrt{t}|Z|$. $E|Z| = \sqrt{2/\pi}$ (§E2.4), so $EM = \sqrt{2t/\pi}$;
$EM^2 = EW^2 = t$; $\mathrm{Var} = t - \frac{2t}{\pi}$.

13.5 RUNNING MINIMUM

$$m(t) = \min_{0\leq u\leq t} W(u) \overset{d}{=} -M(t)$$

$$P(m(t) \leq -a) = 2\left(1 - \Phi\left(\frac{a}{\sqrt{t}}\right)\right), \quad a$$

Use when: a minimum or a lower barrier appears. Flip signs everywhere ($W \to -W$) and reuse §13–14 verbatim.

Q: $P\left(\min_{u\leq 4} W(u) \leq -3\right) = 2(1 - \Phi(1.5)) = \mathbf{0}$.
Trick: reflect the entire problem ($W \to -W$) and reuse §13.2 with $+3$. Never derive a separate minimum formula.

▼ **Proof**

$\min W = -\max(-W)$ and $-W$ is a BM (symmetry). So $m(t) \overset{d}{=} -M(t)$ and $P(m \leq -a) = P(M \geq a)$.

14. Joint Law of $\big(W(t), M(t)\big)$ — the master recipe

14.1 ⚠ THE GUARD EVERYONE GETS WRONG

$$G(b,c) := P\big(W(t) \le b,\ M(t) \ge c\big)$$

$$G(b,c) = \begin{cases} \Phi\left(\frac{b}{\sqrt{t}}\right) & c \le \\ 1 - \Phi\left(\frac{2c-b}{\sqrt{t}}\right) & 0 < \\ 1 + \Phi\left(\frac{b}{\sqrt{t}}\right) - 2\Phi\left(\frac{c}{\sqrt{t}}\right) & 0 < \end{cases}$$

The $b > c$ case is NOT zero. The reflection formula $1 - \Phi(\frac{2c-b}{\sqrt{t}})$ is only valid while $b \le c$. If $b > c$, split at c: once $W(t) > c$ the constraint $M(t) \ge c$ is automatic, so $G = \underbrace{[1 - \Phi(\frac{c}{\sqrt{t}})]}_{W \le c} + \underbrace{[\Phi(\frac{b}{\sqrt{t}}) - \Phi(\frac{c}{\sqrt{t}})]}_{c < W \le b}$.

Sanity checks: at $b = c$ both branches give $1 - \Phi(c/\sqrt{t})$ ✓. As $b \to \infty$: $G \to 2(1 - \Phi(c/\sqrt{t})) = P(M \ge c)$ ✓ (§13.2).

Q: this guard alone decides Midsem'25 Q5(b) — see 14.4.
Trick: before evaluating any G, **write down the comparison of b with c** on your script. $b \le c$ → reflection formula; $b > c$ → the three-term form.

▼ **Proof**

$c \le 0$: $M \ge 0 \ge c$ always, so $G = P(W \le b)$.
$0 < c$, $b \le c$: reflection at level c (§12.2, $y = c - b$):
$P(W \le b, M \ge c) = P(W \ge 2c - b)$.
$0 < c < b$: split $\{W \le c\}$ and $\{c < W \le b\}$. The first is the previous case with $b = c$: $1 - \Phi(\frac{c}{\sqrt{t}})$. On the second $W > c$ already forces $M \ge c$: $\Phi(\frac{b}{\sqrt{t}}) - \Phi(\frac{c}{\sqrt{t}})$. Add.

14.2 JOINT DENSITY

$$f_{W,M}(w,m) = \frac{2(2m-w)}{t\sqrt{2\pi t}} \exp\left(-\frac{(2m\right.$$

Support: $m \ge \max(0, w)$. Zero elsewhere.

Use when: the region is not a rectangle, or an expectation like $E[M(t)W(t)]$ is asked. For rectangles use 14.3 — it is far faster.

Q: $E[M(4)W(4)]$, or any non-rectangular region.
Trick: the substitution $u = 2m - w$ turns every integral of this density into a standard normal one — that substitution is exactly how §14.1 is derived.

▼ **Proof**

For $m > 0$, $w \le m$:
$H(w,m) := P(W \le w, M \ge m) = 1 - \Phi(u)$, $u = \frac{2m-w}{\sqrt{t}}$. Density $= -\partial_m \partial_w H$.
$\partial_w H = \varphi(u)/\sqrt{t}$; then ∂_m gives $\varphi'(u)\frac{2}{t} = -u\varphi(u)\frac{2}{t}$. So $f = \frac{2u}{t}\varphi(u) = \frac{2(2m-w)}{t\sqrt{2\pi t}} e^{-(2m-w)^2/(2t)}$.

14.3 ⭐ MASTER RECIPE — ANY RECTANGLE, ANY POSITION

$$P\big(a < W(t) \le b,\ c < M(t) \le d\big)$$
$$= \big[G(b,c) - G(a,c)\big] - \big[G(b,d) - G(a,$$

G from 14.1. Strict vs non-strict inequalities are irrelevant (continuous laws).

Use when: always. This one line replaces the whole "is I above / below / overlapping J" case tree — the branching lives inside G, so you never classify anything.
Four evaluations of G, one subtraction. That is the entire problem.

Trick: tabulate $G(b,c), G(a,c), G(b,d), G(a,d)$ in a 2×2 box, then (top-left − bottom-left) − (top-right − bottom-right).

▼ **Proof**

$P(a < W \le b,\ M \ge c) = G(b,c) - G(a,c)$.
Then $\{c < M \le d\} = \{M > c\} \setminus \{M > d\}$, so subtract the same thing at level d. (M is continuous, so $\ge$ vs $>$ makes no difference.)

14.4 ⭐ WORKED: MIDSEM'25 Q5(B)

$$P\big(1 < W(4) < 3,\ 2 < M(4) < 3\big)$$

$$t = 4,\ \sqrt{t} = 2; \quad a{=}1,\ b{=}3,\ c{=}2,\ d{=}3$$

$G(3,2)$: $0 < c{=}2 < b{=}3$ → branch 3:
$1 + \Phi(1.5) - 2\Phi(1) = 1 + .9332 - 1.6827 = .$

$G(1,2)$: $b{=}1 \le c{=}2$ → branch 2:
$1 - \Phi(\frac{4-1}{2}) = 1 - \Phi(1.5) = .0668$
$G(3,3)$: $b{=}3 \le c{=}3$ → branch 2:
$1 - \Phi(\frac{6-3}{2}) = 1 - \Phi(1.5) = .0668$
$G(1,3)$: $b{=}1 \le c{=}3$ → branch 2:
$1 - \Phi(\frac{6-1}{2}) = 1 - \Phi(2.5) = .0062$

$$\big[.2505 - .0668\big] - \big[.0668 - .0062\big] = .1837 -$$

Note: $G(3,2)$ hits the $b > c$ branch. Using the naive reflection formula there gives $1 - \Phi(\frac{4-3}{2}) = 1 - \Phi(.5) = .3085$ and a wrong final answer of 0.1811. **This single guard is the whole question.**

14.5 SHORTCUT — I ENTIRELY BELOW J

$$b \le c,\ c > 0:$$

$$P = \Phi\Big(\tfrac{2d-b}{\sqrt{t}}\Big) - \Phi\Big(\tfrac{2c-b}{\sqrt{t}}\Big) + \Phi\Big(\tfrac{2c-a}{\sqrt{t}}\Big) -$$

Use when: $I = (a, b]$ lies entirely below $J = (c, d]$. This is 14.3 with all four G's on branch 2 — write it directly if you spot $b \le c$.

Q: $P(1 < W(4) \le 2,\ 2 < M(4) \le 3)$: here $b{=}2 \le c{=}2$ ✓, so $\Phi(2) - \Phi(1) + \Phi(1.5) - \Phi(2.5) = \mathbf{0.0753}$.
Trick: this is exactly the lower half of Midsem'25 Q5(b); the upper half $(2, 3)$ needs the $b > c$ branch.

▼ **Proof**

$b \le c < d$, so all four G terms in §14.3 are in case 2 of §14.1. Substitute $G(x, y) = 1 - \Phi(\frac{2y-x}{\sqrt{t}})$; the 1's cancel and the four Φ terms remain.

14.6 SHORTCUT — I ENTIRELY ABOVE J

$$a \ge d \implies P\big(W(t) \in I,\ M(t) \in J\big) =$$

Use when: would need $W(t) > M(t)$ — impossible (§13.1). Also $d \le 0 \Rightarrow 0$, since $M \ge 0$. Spot these before computing anything.

Q: $P(3 < W(4) < 4,\ 1 < M(4) < 2) = \mathbf{0}$, since $a{=}3 \ge d{=}2$.
Trick: test $a \ge d$ and $d \le 0$ first — two comparisons that can retire the whole question instantly.

▼ **Proof**

$W \in I$ means $W > a \ge d$; but $M \ge W > d$, so $M \in J = (c, d]$ is impossible. Empty event.

14.7 USEFUL SPECIAL CASES OF G

$P(W(t) \le b,\ M(t) \ge c)$ with $b = c$:
$1 - \Phi\big(\frac{c}{\sqrt{t}}\big)$

$P(M(t) \ge c) = G(\infty, c) = 2\big(1 - \Phi(\frac{c}{\sqrt{t}}$

$P(W(t) \le b,\ M(t) < c) = \Phi\big(\frac{b}{\sqrt{t}}\big) - G(i$

Use when: the question gives a half-line instead of an interval — set the missing endpoint to $\pm\infty$ and use $\Phi(\pm\infty) = 1$ or 0.

Q:
$P(W(4) \le 1,\ M(4) < 2) = \Phi(0.5) - G(1, 2)$
.
Trick: "the max stayed below c" is the marginal of W minus G. Half-lines are just $\pm\infty$ endpoints.

▼ **Proof**

$b = c$ in case 2: $1 - \Phi(\frac{2c-c}{\sqrt{t}})$. $G(\infty, c)$: case 3 with $b \to \infty$, $1 + 1 - 2\Phi$. Last:
$P(W \le b) = P(W \le b, M \ge c) + P(W \le b, i$
.

14.8 BARRIER-OPTION TRANSLATION

knock-out / knock-in payoffs = rectangles in (W, M)

$P(\text{never hit } c,\ W(t) \in I) = P(W(t) \in i$ terms

> **Use when:** a question is dressed as a barrier option. Translate "price never exceeded c" into $\{M(t) < c\}$ and run 14.3.

> **Q:** up-and-out call, barrier 3, $S = e^W$ → $P\big(M(4) < \ln 3,\ W(4) \in I\big)$.
> **Trick:** "the price never touched the barrier" $\iff \{M(t) < c\}$. Translate, then run §14.3 verbatim.

Part III — Stochastic Calculus

Itô integral → Itô's formula → stochastic differential equations. Sections I1–I8.

I1. The Itô Integral

I1.1 DEFINITION — LEFT ENDPOINT

$$\int_a^b f(t)\,dW_t = \lim_{\text{mesh}\to 0} \sum_{i=1}^{n} f(t_{i-1})\big(W(t_i)$$

> **Use when:** the **left** endpoint t_{i-1} is the whole definition. It makes the integrand *predictable* — you commit the stake before seeing the increment (§M7.2). A midpoint rule would give a different answer (Stratonovich).

> **Trick:** for ordinary Riemann integrals the evaluation point does not matter in the limit. Here it does, because $\Delta W \sim \sqrt{\Delta t}$ is too big for the difference to vanish. Say that sentence in any "define the Itô integral" question.

I1.2 PATHWISE FORMULA (DETERMINISTIC F)

$$\int_a^b f(t)\,dW_t = f(b)W(b) - f(a)W(a) -$$

f non-random with continuous derivative.

> **Use when:** integration by parts, and note **no dW remains on the right** — it is ordinary calculus path by path. Only works because f is deterministic.

> **Q:** $\displaystyle\int_0^1 t\,dW_t = W(1) - \int_0^1 W(t)\,dt.$
> **Trick:** $f \equiv 1$ gives $\int_a^b dW = W(b) - W(a)$ — the sum telescopes.

▼ **Proof**

> Product rule with f deterministic (no correction, $df\,dW = 0$):
> $d(fW) = f'(t)W_t\,dt + f(t)\,dW_t$. Integrate from a to b and rearrange.

I1.3 MEAN ZERO & MARTINGALE

$$E\Big[\int_0^t H_s\,dW_s\Big] = 0$$

$\Big(\int_0^t H_s\,dW_s\Big)_{t\geq 0}$ is a martingale (H adapted, square-integrable)

> **Use when: every** expectation computation in Part III starts by killing the dW term. Each summand has $E[f(t_{i-1})\Delta W_i] = f(t_{i-1})\cdot 0 = 0$ by independence.

> **Trick:** "an Itô integral is a martingale" is the continuous-time version of "you cannot beat a fair game" (§M7.4). Predictable integrand against martingale increments — identical statement.

▼ **Proof**

> Simple H: $E[h_k\Delta W_k \mid \mathcal{F}_{t_k}] = h_k E[\Delta W_k] = 0$, so each piece has conditional mean 0; summing gives mean 0. For $s<t$, the pieces after s have zero conditional mean given $\mathcal{F}_s$: MG. General H: limits of simple ones.

I1.4 ⭐ ITÔ ISOMETRY

$$E\Big[\Big(\int_0^T H_s\,dW_s\Big)^2\Big] = \int_0^T E\big[H_s^2\big]\,ds$$

> **Use when:** any variance of a stochastic integral. It turns a *stochastic* second moment into an **ordinary** integral. Why it works: increments over disjoint intervals are independent, so all cross terms vanish and variances just add.

> **Q:** Endsem'25 Q1(a) — $\mathrm{Var}\left(\int_0^1 t^4 dW_t\right) = \int_0^1 t^8 dt = \mathbf{1/9}$.
> **Trick:** square the integrand, integrate dt, done. No stochastic work at all.

▼ **Proof**

> Simple integrand: $H = h_k$ on $(t_k, t_{k+1}]$, h_k known at t_k. Then $I = \sum_k h_k\,\Delta W_k$.
> $E[I^2] = \sum_k E[h_k^2\Delta W_k^2] + 2\sum_{j<k} E[h_j h_k \Delta W$
> .
> Cross terms: condition on $\mathcal{F}_{t_k}$; ΔW_k is independent of it with mean 0, so each is 0.
> Diagonal: $E[h_k^2]\,E[\Delta W_k^2] = E[h_k^2](t_{k+1}-t_k)$. The sum is $\int_0^T E[H_s^2]\,ds$. General H: limit of simple ones.

I1.5 ⭐ DISTRIBUTION (DETERMINISTIC INTEGRAND)

$$\int_0^T f(t)\,dW_t \;\sim\; N\Big(0,\ \int_0^T f(t)^2\,dt\Big)$$

*f **must be non-random.** Sums of independent normals are normal, and L^2 limits of normals are normal.*

> **Use when:** "compute the distribution of $\int_0^1 t^k dW_t$". Mean 0 is free; variance is a one-line calculus integral. Same card as §7.5.

> **Paper:** $\int_0^1 t^k dW_t \sim N\big(0, \frac{1}{2k+1}\big)$.
> CT-1'26: $k{=}1 \to \mathbf{N(0,\frac{1}{3})}$, $k{=}2 \to \frac{1}{5}$, $k{=}3 \to \frac{1}{7}$, $k{=}4 \to \frac{1}{9}$. CT-2'25 Q7: $\mathbf{N(0,\frac{1}{5})}$. Endsem'25 Q1(a): $\frac{1}{9}$.

▼ **Proof**

> Riemann sums $\sum f(t_i)\Delta W_i$ are sums of independent $N(0, f(t_i)^2\Delta t)$, so $N\big(0, \sum f^2\Delta t\big)$. As the mesh $\to 0$ the variance $\to \int_0^T f^2$, and an L^2 limit of normals is normal.

I1.6 ⚠ RANDOM INTEGRAND ⇒ NOT NORMAL

$$\int_0^t W_s\,dW_s \text{ is } \textbf{not } N\Big(0, \textstyle\int_0^t W_s^2\,ds\Big)$$

That "variance" would itself be a random variable. Normality in I1.5 needs a **deterministic** integrand. The isometry (I1.4) survives, normality does not.

Use when: check before writing "$\sim N$": **does the integrand contain W or X?** If yes, you may still take mean and variance, but not the distribution.

Q: what *is* it? By I1.7, $\int_0^t W\,dW = \frac{1}{2}W_t^2 - \frac{t}{2}$, so it is a shifted and scaled χ_1^2 — visibly skewed, not normal.

▼ **Proof**

$\int_0^t W\,dW = \frac{1}{2}(W_t^2 - t) \geq -\frac{t}{2}$ always. A normal variable takes arbitrarily negative values with positive probability. So it is not normal.

I1.7 ⭐ THE CORRECTION TERM

$$\int_0^t W_s\,dW_s = \tfrac{1}{2}W_t^2 - \tfrac{1}{2}t$$

Use when: the signature result of the subject. Ordinary calculus would give $\frac{1}{2}W_t^2$; the extra $-\frac{t}{2}$ is there because squared increments do **not** vanish.

Derivation:
$a(b-a) = \frac{1}{2}(b^2-a^2) - \frac{1}{2}(b-a)^2$; sum, telescope, and use $\sum(\Delta W)^2 \to t$.
Check: $E[\frac{1}{2}W_t^2 - \frac{t}{2}] = \frac{t}{2} - \frac{t}{2} = 0$ ✓ — as an Itô integral must have. Without the $-\frac{t}{2}$ the mean would be $\frac{t}{2} \neq 0$.

▼ **Proof**

Itô on $f(w) = w^2$:
$d(W_t^2) = 2W_t\,dW_t + \frac{1}{2}\cdot 2\,(dW_t)^2 = 2W_t\,dW_t$
.
Integrate from 0 to t: $W_t^2 = 2\int_0^t W_s\,dW_s + t$,
so $\int_0^t W_s\,dW_s = \frac{1}{2}W_t^2 - \frac{1}{2}t$.
(From sums:
$W_k(W_{k+1} - W_k) = \frac{1}{2}(W_{k+1}^2 - W_k^2) - \frac{1}{2}(\Delta W$
, and $\sum(\Delta W_k)^2 \to t$, §I1.8.)

I1.8 QUADRATIC VARIATION

$$\sum_j (\Delta W_j)^2 \longrightarrow t \quad \text{(in } L^2\text{, by the LLN)}$$

$$[W]_t = t, \quad \text{i.e. } (dW)^2 = dt$$

Use when: the **one fact** that generates all of Itô calculus. Proof: $\Delta W_j \overset{d}{=} \sqrt{t/n}\,Z_j$, so the sum is $t \cdot \frac{1}{n}\sum Z_j^2 \to t\,E[Z^2] = t$.

Numeric feel: $t{=}1$, $n{=}4$,
$Z = 1.2, -0.5, 0.8, -1.1$:
$\frac{1.44+0.25+0.64+1.21}{4} = 0.885 \approx 1$ ✓
Trick: BM has infinite total variation but finite quadratic variation. That is exactly why the second-order Taylor term survives.

▼ **Proof**

$E\sum(\Delta W_j)^2 = \sum \Delta t_j = t$ and
$\operatorname{Var}\sum(\Delta W_j)^2 = \sum 2\Delta t_j^2 \leq 2t \cdot \text{mesh} \to 0$.
A variable with mean t and variance $\to 0$ converges to t.

I2. Itô Processes & the Multiplication Table

I2.1 ITÔ PROCESS

$$X_t = X_0 + \int_0^t a_s\,ds + \int_0^t b_s\,dW_s$$

shorthand: $dX_t = a_t\,dt + b_t\,dW_t$

a (drift) and b (diffusion) adapted; the differential is notation for the integral equation, nothing more.

Use when: identify a and b first, always. Every subsequent formula is stated in terms of them.

Q: $X = W$: $a{=}0$, $b{=}1$. $X_t = \int_0^t s\,dW_s$: $a{=}0$, $b_s{=}s$.
From I1.7: $W_t^2 = \int_0^t 1\,ds + \int_0^t 2W_s\,dW_s$, so $d(W_t^2) = dt + 2W_t\,dW_t$ — $a{=}1$, $b{=}2W_t$.

I2.2 ⭐ MULTIPLICATION TABLE

$$dt \cdot dt = 0$$
$$dt \cdot dW = 0$$
$$dW \cdot dW = dt$$

Use when: memorize these three lines and Itô's formula is just Taylor. Reason: ΔW has size $\sqrt{\Delta t}$, so $(\Delta W)^2 \sim \Delta t$ survives while $(\Delta t)^2$ and $\Delta t^{3/2}$ die.

Orders: $\sum(\Delta t)^2 = t^2/n \to 0$;
$\sum \Delta t\,\Delta W \approx \frac{t}{n} W_t \to 0$; $\sum(\Delta W)^2 \to t$ ✓
Trick: third order and beyond all vanish, which is why the Taylor expansion stops at two terms.

▼ **Proof**

Sum over a partition of $[0,t]$. $\sum(\Delta W)^2 \to t$ (§I1.8), so $dW\,dW = dt$.
$E\big(\sum \Delta t\,\Delta W\big)^2 = \sum \Delta t^3 \le \text{mesh}^2 t \to 0$, so $dt\,dW = 0$. $\sum \Delta t^2 \le \text{mesh} \cdot t \to 0$, so $dt\,dt = 0$.

I2.3 SQUARE OF AN ITÔ PROCESS

$$(dX_t)^2 = (a\,dt + b\,dW)^2 = b_t^2\,dt$$

Use when: the plug-in for the correction term. **The drift never enters** $(dX)^2$ — only the diffusion coefficient does, and it gets squared.

Q: $dX = dt + 2\sqrt{X}\,dW \Rightarrow (dX)^2 = 4X\,dt$.
$dN = rN\,dt + \alpha N\,dW \Rightarrow (dN)^2 = \alpha^2 N^2\,dt$.

▼ **Proof**

$(a\,dt + b\,dW)^2 = a^2 dt^2 + 2ab\,dt\,dW + b^2 dW^2$ by the table.

I3. Itô's Formula — the three forms

I3.1 ⭐ THE RECIPE

1. Taylor-expand to **second** order
2. apply $dt\,dt = 0$, $dt\,dW = 0$, $dW\,dW = dt$
3. collect dt terms (drift) and dW terms (diffusion)

> **Use when: every** Itô question. You do not need to memorize the three forms below — they are all this recipe. Memorize the recipe and the table.

> **Paper:** Midsem'25 Q2(b) asks you to *derive* Itô's formula [2 marks].
> **Trick:** state the Taylor expansion, state the table with the $\sqrt{\Delta t}$ reason, note that third order vanishes. That is the derivation.

▼ **Proof**

> Taylor:
> $f(t+dt, X+dX) - f(t,X) = \partial_t f\,dt + \partial_x f\,d$
>
> Sizes: $dW \approx \sqrt{dt}$, so $(dW)^2$ has mean dt and variance $2dt^2$: it behaves like dt (not random to first order). $dt\,dW \approx dt^{3/2}$ and dt^2 are smaller than dt and vanish.
> Keep only dt and dW terms:
> $df = \partial_t f\,dt + \partial_x f\,dX + \frac{1}{2}\partial_{xx} f\,(dX)^2$. The $\frac{1}{2}f''$ term is the correction ordinary calculus lacks.

I3.2 FORM 1 — $F(X_T)$

$$df(X_t) = f'(X_t)\,dX_t + \tfrac{1}{2}f''(X_t)\,b_t^2\,dt$$

for $X = W$:

$$df(W_t) = f'(W_t)\,dW_t + \tfrac{1}{2}f''(W_t)\,dt$$

> **Use when:** no explicit t in f. The second line is the one you use most — the whole dictionary in §I4 comes from it.

> **Q:** $f(x) = x^3$, $X = W$:
> $d(W_t^3) = 3W_t^2\,dW_t + 3W_t\,dt$.
> **Trick:** compare with ordinary calculus, which would give only $3W^2 dW$. The extra $\frac{1}{2}f''$ term *is* Itô calculus.

▼ **Proof**

> Taylor in x only (§I3.1):
> $df = f'\,dX + \frac{1}{2}f''(dX)^2$, and $(dX)^2 = b^2 dt$ (§I2.3). For $X = W$: $b = 1$.

I3.3 ⭐ FORM 2 — $F(T, X_T)$

$$df = \partial_t f\,dt + \partial_x f\,dX_t + \tfrac{1}{2}\,\partial_{xx} f\,b_t^2\,dt$$

for $X = W$:

$$df = \left(\partial_t f + \tfrac{1}{2}\partial_{xx} f\right)dt + \partial_x f\,dW_t$$

> **Use when: the exam's standard form** — almost every question has an $e^{\lambda t}$ out front. Note there is **no** ∂_{tt} term: only x carries the dW.

> **Q:** $f = x^2 - t$, $X = W$:
> $dY = -dt + 2W\,dW + dt = 2W_t\,dW_t$. Zero drift ✓ consistent with $W_t^2 - t$ being a martingale (§7.2).

▼ **Proof**

> §I3.1 with $(dX)^2 = b^2 dt$; the cross terms $dt\,dX$ vanish. For $X = W$: $b = 1$, $dX = dW$.

I3.4 FORM 3 — PRODUCT RULE

$$d(X_t Y_t) = X_t\,dY_t + Y_t\,dX_t + dX_t\,dY_t$$

$$dX\,dY = b_1 b_2\,dt \quad \text{(the Itô correction)}$$

> **Use when:** the product of two processes. It is the ordinary product rule **plus one cross term**. If either factor is deterministic, $b_1 b_2 = 0$ and the correction disappears.

> **Q:** $d(tW_t) = W_t\,dt + t\,dW_t$ — no correction, since t is deterministic.
> **Q:** $X{=}Y{=}W$: $d(W^2) = 2W\,dW + dt$ ✓ recovers I2.1.

▼ **Proof**

> Two-variable form (§I3.6) with $f = xy$: $f_x = y$, $f_y = x$, $f_{xy} = 1$, $f_{xx} = f_{yy} = 0$. So $d(XY) = Y\,dX + X\,dY + dX\,dY$.

I3.5 FORM 4 — QUOTIENT RULE

$$d\left(\frac{X_t}{Y_t}\right) = \frac{dX_t}{Y_t} - \frac{X_t}{Y_t^2}\,dY_t + \frac{X_t}{Y_t^3}(dY_t)^2$$

Use when: rarely asked directly. Usually faster to write $X/Y = X \cdot Y^{-1}$ and use I3.2 on Y^{-1} then the product rule.

Q: $X{=}1$, $Y = e^{W_t}$:
$d(e^{-W_t}) = \frac{1}{2}e^{-W_t}\,dt - e^{-W_t}\,dW_t$.
Check: that is GBM (§I4) with $\mu{=}0$, $\sigma{=}-1$ ✓

▼ **Proof**

§I3.6 with $f = x/y$: $f_x = \frac{1}{y}$, $f_y = -\frac{x}{y^2}$, $f_{xx} = 0$, $f_{xy} = -\frac{1}{y^2}$, $f_{yy} = \frac{2x}{y^3}$. The second-order part $\frac{1}{2}[2f_{xy}dXdY + f_{yy}(dY)^2] = -\frac{dXdY}{y^2} + \frac{x(dY)^2}{y^3}$.

I3.6 GENERAL 2-VARIABLE FORM

$$df(X,Y) = \partial_x f\,dX + \partial_y f\,dY + \frac{1}{2}\left[\partial_{xx} f\right.$$

Use when: I3.4 and I3.5 are this with $f = xy$ and $f = x/y$. Derive them rather than memorizing, if you prefer — it is two derivatives each.

Check: $f = xy$ gives $\partial_{xy} f = 1$, $\partial_{xx} f = \partial_{yy} f = 0$, so the bracket is $2\,dX\,dY$, halved to $dX\,dY$ ✓ exactly I3.4.

▼ **Proof**

Second-order Taylor in two variables, then drop every term smaller than dt using the table (§I2.2), exactly as in §I3.1.

I4. ⭐ The Differential Dictionary — memorize these

I4.1 POWERS OF W

$$d(W_t^2) = 2W_t\,dW_t + dt$$
$$d(W_t^3) = 3W_t^2\,dW_t + 3W_t\,dt$$
$$d(W_t^n) = nW_t^{n-1}dW_t + \tfrac{1}{2}n(n-1)W_t^{n-}$$

> **Use when:** $f = x^n$ in I3.2. The dt coefficient is $\frac{1}{2}f''$, nothing more.

> **Bonus — moment recursion:** taking E kills the dW term (I1.3), leaving $E[W_t^n] = \frac{1}{2}n(n-1)\int_0^t E[W_s^{n-2}]\,ds$. $n{=}2 \to t$; $n{=}4 \to 6\int_0^t s\,ds = \mathbf{3t^2}$; $n{=}6 \to 15\int_0^t 3s^2 ds = \mathbf{15t^3}$ ✓ (§1.4)

▼ **Proof**

> Form 1 with $f = w^n$: $f' = nw^{n-1}$, $f'' = n(n-1)w^{n-2}$, so the drift is $\frac{1}{2}n(n-1)W^{n-2}$.

I4.2 EXPONENTIAL OF W

$$d(e^{W_t}) = e^{W_t}\,dW_t + \tfrac{1}{2}e^{W_t}\,dt$$
$$d\big(e^{\sigma W_t}\big) = \sigma e^{\sigma W_t}\,dW_t + \tfrac{\sigma^2}{2}e^{\sigma W_t}\,dt$$
$$d\big(e^{\sigma W_t - \sigma^2 t/2}\big) = \sigma\, e^{\sigma W_t - \sigma^2 t/2}\,dW_t \quad \textbf{(zero drift)}$$

> **Use when:** the third line is the exponential martingale of §7.3 — the $-\sigma^2 t/2$ exists precisely to cancel the Itô correction.

> **Paper:** Midsem'24 Q4(b) — $X_t = e^{W_t - t/2}$. $\partial_t f = -\frac{1}{2}f$, $\partial_x f = \partial_{xx} f = f$, so $dX = -\frac{1}{2}X\,dt + X\,dW + \frac{1}{2}X\,dt = \mathbf{X_t\,dW_t}$. Zero drift ⇒ **Yes, a martingale**.

▼ **Proof**

> $f = e^{\sigma w}$: $f' = \sigma f$, $f'' = \sigma^2 f$. Form 1: $df = \sigma f\,dW + \frac{\sigma^2}{2}f\,dt$.

I4.3 ⭐ SINE AND COSINE

$$d\big(\sin W_t\big) = \cos W_t\,dW_t - \tfrac{1}{2}\sin W_t\,dt$$
$$d\big(\cos W_t\big) = -\sin W_t\,dW_t - \tfrac{1}{2}\cos W_t\,d$$

> **Use when:** both have $f'' = -f$, so the drift is always $-\frac{1}{2}f$. **That single fact answers every trig question on this paper.**

> **Paper:** CT-2'25 Q8 — "compute $d\sin(W_t)$" → $\cos\mathbf{W_t}\,\mathbf{dW_t} - \frac{1}{2}\sin\mathbf{W_t}\,\mathbf{dt}$ ✓
> Midsem'24 Q1(e) — "compute $d\cos(W_t)$" → $-\sin\mathbf{W_t}\,\mathbf{dW_t} - \frac{1}{2}\cos\mathbf{W_t}\,\mathbf{dt}$ ✓

▼ **Proof**

> sin: $f' = \cos$, $f'' = -\sin$. cos: $f' = -\sin$, $f'' = -\cos$. Plug into $df = f'\,dW + \frac{1}{2}f''dt$.

I4.4 ⭐ THE E^{MT} FAMILY

$$d\big(e^{\lambda t}g(W_t)\big) = e^{\lambda t}\Big(\lambda g + \tfrac{1}{2}g''\Big)dt \;+\; e^{\lambda t}$$
$$\text{martingale} \iff \lambda g + \tfrac{1}{2}g'' = 0$$

> **Use when:** the exam's favourite shape. For $g = \sin$ or $\cos$ ($g'' = -g$) the condition is $\lambda = \frac{1}{2}$ — **exponent $\frac{1}{2}$ ⇒ martingale, anything else ⇒ not.**

> **Paper:** CT-1'26 Q5 — $e^{t/2}\sin W_t$ **Yes**; $e^t\cos W_t$ **No**; $e^t\sin W_t$ **No**; $e^{t/2}\cos W_t$ **Yes**.
> Midsem'25 Q4(b)(i) — $X_t = e^{t/2}\cos W_t$: $dX_t = \mathbf{-e^{t/2}\sin W_t\,dW_t}$, drift zero.

▼ **Proof**

> Form 2 with $f = e^{\lambda t}g(w)$: $\partial_t f = \lambda e^{\lambda t}g$, $\partial_w f = e^{\lambda t}g'$, $\partial_{ww} f = e^{\lambda t}g''$. MG iff the dt coefficient is 0 (§I5.1).

I4.5 DETERMINISTIC FACTORS (NO CORRECTION)

$$d\big(g(t)X_t\big) = g'(t)X_t\,dt + g(t)\,dX_t$$
$$d(tW_t) = W_t\,dt + t\,dW_t$$
$$d(e^tW_t) = e^tW_t\,dt + e^t\,dW_t$$

Use when: $\partial_{xx}f = 0$, so **no Itô correction** — it is the ordinary product rule. This is also the integrating factor of §I7.1.

Paper: Endsem'25 Q1(b) — "compute $d(e^tW_t)$" → $\mathbf{e^tW_t\,dt + e^t\,dW_t}$ ✓
Endsem'25 Q1(c) — "is $Y_t = e^tW_t$ a martingale?" Drift $= e^tW_t \neq 0 \Rightarrow$ **No**.

▼ **Proof**

Product rule with g deterministic: the correction $dg\,dX = g'(t)\,dt \cdot dX = 0$.

I4.6 GEOMETRIC BROWNIAN MOTION (BOTH DIRECTIONS)

$$Z_t = e^{\mu t+\sigma W_t} \;\Rightarrow\; dZ_t = \left(\mu + \tfrac{\sigma^2}{2}\right)Z_t\,dt$$
$$dN_t = rN_t\,dt + \alpha N_t\,dW_t \;\Rightarrow\; N_t = N_0\,e$$

Use when: this $\frac{\sigma^2}{2}$ shuttling back and forth **is** the two-convention problem of §11.1. Read the direction the question gives you before writing anything.

Derivation of line 2: Itô with $f = \log x$: $d(\log N) = \frac{dN}{N} - \frac{(dN)^2}{2N^2} = \left(r - \frac{\alpha^2}{2}\right)dt + \alpha\,dW$, then exponentiate.
Check: $E[N_t] = N_0e^{(r-\alpha^2/2)t}\,e^{\alpha^2t/2} = \mathbf{N_0e^{rt}}$ ✓ and median $= N_0e^{(r-\alpha^2/2)t}$.

▼ **Proof**

$f = e^{\mu t+\sigma w}$: $\partial_t = \mu f$, $\partial_w = \sigma f$, $\partial_{ww} = \sigma^2 f$, so $dZ = (\mu + \frac{\sigma^2}{2})Z\,dt + \sigma Z\,dW$. Reverse: match $\sigma = \alpha$ and $\mu + \frac{\alpha^2}{2} = r$, i.e. $\mu = r - \frac{\alpha^2}{2}$; multiply by N_0.

I4.7 MIXED PRODUCTS

$$d\big(W_te^{W_t}\big) = \big(1 + \tfrac{1}{2}W_t\big)e^{W_t}\,dt + (1 + W$$
$$d\big(e^{-W_t}\big) = \tfrac{1}{2}e^{-W_t}\,dt - e^{-W_t}\,dW_t$$

Use when: two routes agree — the product rule I3.4, or I3.2 with $f(x) = xe^x$ ($f' = (1+x)e^x$, $f'' = (2+x)e^x$). **Use one and check with the other.**

Trick: for $f(x) = xe^x$ the drift is $\frac{1}{2}f'' = \frac{1}{2}(2+W)e^W = (1+\frac{1}{2}W)e^W$ ✓ Same answer, fewer steps.

▼ **Proof**

$f = we^w$: $f' = (1+w)e^w$, $f'' = (2+w)e^w$, drift $\frac{1}{2}(2+w)e^w = (1+\frac{w}{2})e^w$. $f = e^{-w}$: $f' = -e^{-w}$, $f'' = e^{-w}$.

I5. Martingale Test & Computing Expectations

I5.1 ⭐ ZERO DRIFT = MARTINGALE

$dX_t = a_t\,dt + b_t\,dW_t$ is a martingale $\iff a_t \equiv 0$

Strictly, zero drift gives a local martingale; a true martingale also needs integrability, which holds in every exam case.

Use when: every "Is Y_t a martingale? (Yes/No)". **Apply Itô, read the dt coefficient, done.** You never need the definition.

Paper: Endsem'25 Q1(c) — $Y = e^t W_t$, drift $e^t W_t \neq 0 \Rightarrow$ **No**.
Midsem'24 Q4(b) — $X = e^{W_t - t/2}$, drift $0 \Rightarrow$ **Yes**.
Midsem'25 Q1(f) — $|W_t|^2 - t$ has drift 0, so "is NOT a MG" is **False**.

▼ **Proof**

Integral form:
$X_t = X_0 + \int_0^t a_s\,ds + \int_0^t b_s\,dW_s$.
(⇐) If $a \equiv 0$: $X_t - X_s = \int_s^t b\,dW$, and an Itô integral has conditional mean 0 (§I1.3). So $E(X_t \mid \mathcal{F}_s) = X_s$.
(⇒) If X is a MG then
$\int_0^t a_s\,ds = X_t - X_0 - \int_0^t b\,dW$ is a MG with smooth (finite-variation) paths. A continuous MG with such paths is constant, so
$\int_0^t a_s\,ds = 0$ for all t, i.e. $a \equiv 0$.

I5.2 ⭐ MEAN OF A SOLUTION — THE DRIFT ODE

take E of the integral form; the dW term dies (I1.3):

$$m(t) := E[X_t] = x_0 + \int_0^t E\big[a(s, X_s)\big]\,ds$$

a linear in $x \Rightarrow$ $m'(t) = a\big(t, m(t)\big)$, $m(0) = x_0$

Use when: "find $E(X_t)$" for **any** SDE with linear drift — **you never need the solution**. Differentiate the integral form and solve an ordinary ODE.

Q: $a = 1 \to m' = 1,\ m = \mathbf{t}$. $a = -x \to m' = -m,\ m = \mathbf{x_0 e^{-t}}$. $a = \kappa(b - x) \to m = \mathbf{b + e^{-\kappa t}(x_0 - b)}$.
Trick: the diffusion coefficient b never appears in the mean.

▼ **Proof**

Integral form $X_t = x_0 + \int_0^t a\,ds + \int_0^t b\,dW$. Take E: the Itô integral has mean 0 (§I1.3); swap E and $\int ds$. If $a(t,x) = \alpha(t) + \beta(t)x$, then $E[a(s, X_s)] = a(s, m(s))$; differentiate in t.

I5.3 ⭐ VARIANCE OF A SOLUTION — ITÔ ON X^2

$d(X_t^2) = 2X_t\,dX_t + (dX_t)^2 = \big(2X_t a_t +$

$\frac{d}{dt}E[X_t^2] = E\big[2X_t a_t + b_t^2\big]$, then $\mathrm{Var} = E[X_t^2] - m(t)^2$

> **Use when:** "also find $\mathrm{Var}(X_t)$" — the part they circle. Same method as I5.2, applied to X^2 instead of X.

> **Paper:** Midsem'25 Q4(a) — $a{=}1$, $b{=}2\sqrt{X}$, $x_0{=}0$.
> $\frac{d}{dt}E[X^2] = E[2X + 4X] = 6E[X] = 6t$, so $E[X_t^2] = 3t^2$ and $\mathrm{Var} = 3t^2 - t^2 = \mathbf{2t^2}$ ✓
> **Cross-check:** $X_t = W_t^2$, so $\mathrm{Var} = E[W^4] - t^2 = 3t^2 - t^2 = 2t^2$ ✓

▼ **Proof**

> Itô Form 1 on $f(x) = x^2$ ($f' = 2x$, $f'' = 2$): $d(X^2) = 2X\,dX + (dX)^2 = (2Xa + b^2)dt + 2$. Take E; the dW part has mean 0.

I5.4 ⭐ $E[F(W_T)]$ VIA A MARTINGALE

find λ with $\lambda f + \frac{1}{2}f'' = 0$; then $e^{\lambda t}f(W_t)$ is a MG (I4.4)

$\Rightarrow e^{\lambda t}E[f(W_t)] = f(0) \Rightarrow E[f(W_t)] = f$

> **Use when:** "using part (i) or otherwise, evaluate $E[f(W_t)]$". **Constant mean of a martingale turns a hard expectation into one division.**

> **Paper:** Midsem'25 Q4(b)(ii) — $E[\cos W_t]$. Part (i) gave $e^{t/2}\cos W_t$ a MG, so $e^{t/2}E[\cos W_t] = \cos 0 = 1 \Rightarrow E[\cos W_t] = \mathbf{e^{-t}}$ ✓
> **Check:** $E[e^{iW_t}] = e^{-t/2}$, and taking real parts gives the same ✓ Also $E[\sin W_t] = 0$ by symmetry.

▼ **Proof**

> $e^{\lambda t}f(W_t)$ is a MG, so its mean is constant: $E[e^{\lambda t}f(W_t)] = e^0 f(W_0) = f(0)$.

I5.5 READING THE DISTRIBUTION OFF AN SDE

b **deterministic** (no X in the noise) $\Rightarrow X_t$ is **Gaussian**

b contains $X \Rightarrow$ generally **not** Gaussian

> **Use when:** decide this before claiming a distribution. Gaussian: BM with drift, OU, Vasicek. Not Gaussian: GBM (log-normal), CIR, $dX = dt + 2\sqrt{X}\,dW$.

> **Trick:** it is I1.5 restated — only a deterministic integrand keeps the Itô integral normal. Look at the dW coefficient and nothing else.

▼ **Proof**

> Linear drift and deterministic b: the integrating factor (§I7.1) gives $X_t =$ (deterministic) $+ \int_0^t (\text{deterministic})\,dW$, which is normal (§I1.5). With X in the noise this fails: GBM, $dN = rN\,dt + \alpha N\,dW$, is log-normal and always > 0.

I6. Stochastic Differential Equations — setup & verification

I6.1 ITÔ PROCESS VS SDE

Itô process: $dX_t = a_t\,dt + b_t\,dW_t$ — a_t, b_t **given**, nothing to solve

SDE: $dX_t = a(t, X_t)\,dt + b(t, X_t)\,dW_t$, $X_0 = x_0$ — coefficients built from the **unknown**

> **Use when:** tell them apart before starting. $dX = W_t\,dt + dW_t$ is an Itô process (just integrate); $dX = X_t^2\,dt + dW_t$ is a genuine equation.

> **Trick:** most SDEs have **no closed form**. The exam only asks the handful in §I7 — or hands you the answer and asks you to verify it (I6.3).

I6.2 WHAT "SOLVES" MEANS

$$X_t = x_0 + \int_0^t a(s, X_s)\,ds + \int_0^t b(s, X_s)\,dW_s$$

> **Use when:** a solution is an *Itô process satisfying the integral identity*. The differential form is shorthand and can never be manipulated as a fraction.

> **Q:** $dX = dt + 2\,dW$, $x_0 = 3$: RHS $= 3 + t + 2W_t$, so $X_t = 3 + t + 2W_t$ ✓
> **Q:** $dX = X\,dt$, $x_0 = 1$: RHS $= 1 + \int_0^t X_s\,ds$; candidate e^t gives $1 + (e^t - 1) = e^t$ ✓

I6.3 ⭐ VERIFICATION RECIPE — "SHOW $X_T = \ldots$ IS A SOLUTION"

1. apply Itô to the proposed X_t
2. rewrite the result in terms of X_t itself
3. compare with the given SDE, and check $X_0 = x_0$

> **Use when:** the exam asks this far more often than it asks you to *derive* a solution. **It is a one-way check — never try to solve.**

> **Paper:** Midsem'25 Q4(a) — show $X_t = W_t^2$ solves $dX_t = dt + 2\sqrt{X_t}\,dW_t$, $X_0 = 0$.
> Itô: $d(W_t^2) = dt + 2W_t\,dW_t$. Since $\sqrt{X_t} = |W_t|$, this matches ✓ and $X_0 = W_0^2 = 0$ ✓
> Then $E[X_t] = E[W_t^2] = \mathbf{t}$ and $\mathrm{Var}(X_t) = 3t^2 - t^2 = \mathbf{2t^2}$ (I5.3).

▼ **Proof**

> If Itô turns the candidate into $dX_t = a(t, X_t)\,dt + b(t, X_t)\,dW_t$ and $X_0 = x_0$, integrating gives exactly the integral equation of §I6.2. That is the definition of a solution.

I6.4 EXPRESS THE SOLUTION AS A STOCHASTIC INTEGRAL

$$dY_t = h_t\,dt + k_t\,dW_t \iff Y_t = Y_0 + \int$$

> **Use when:** the question says "express X_t as a stochastic integral with respect to W_t". Translate differentials into integrals — that *is* the answer, no evaluation needed.

> **Paper:** Midsem'24 Q4(a) — $dX_t = -X_t\,dt + dW_t$, $X_0 = x_0$.
> Answer: $X_t = \mathbf{x_0 e^{-t}} + \int_0^t \mathbf{e^{-(t-s)}}\,\mathbf{dW_s}$ (derived in I7.3).

I7. The Solvable SDEs

I7.1 ☆ INTEGRATING FACTOR (LINEAR, ADDITIVE NOISE)

$$dX_t = \big(\alpha(t) - rX_t\big)dt + \sigma(t)\,dW_t$$

multiply by e^{rt}:

$d\big(e^{rt}X_t\big) = re^{rt}X_t\,dt + e^{rt}dX_t$ (no correction, I4.5)

$\Rightarrow d\big(e^{rt}X_t\big) = e^{rt}\alpha(t)\,dt + e^{rt}\sigma(t)\,dW_t$

the X cancels

> **Use when:** the **only** general solving technique on this syllabus. Integrate, then divide by e^{rt}. Works exactly when the noise does *not* contain X.

> **Trick:** the whole method is "choose f so that Itô's formula deletes the unknown". If the noise contains X (as in CIR) nothing cancels — see I7.5.

I7.2 BROWNIAN MOTION WITH DRIFT

$$dX_t = \mu\,dt + \sigma\,dW_t, \quad X_0 = x_0$$

$$X_t = x_0 + \mu t + \sigma W_t \sim N\big(x_0 + \mu t,\ \sigma^2 t\big)$$

> **Use when:** the trivial case — both coefficients constant, so you just integrate. All of §10 is built on it.

> **Trick:** constant coefficients $\Rightarrow$ no Itô correction anywhere, because there is no f to differentiate twice.

▼ **Proof**

> Integrate both sides: $X_t = x_0 + \mu t + \sigma W_t$. $\sigma W_t \sim N(0, \sigma^2 t)$.

I7.3 ☆ ORNSTEIN–UHLENBECK

$$dX_t = -rX_t\,dt + \sigma\,dW_t, \quad X_0 = x_0, \quad r, \sigma > 0$$

$$X_t = x_0 e^{-rt} + \sigma \int_0^t e^{-r(t-s)}\,dW_s$$

$$X_t \sim N\Big(x_0 e^{-rt},\ \frac{\sigma^2}{2r}\big(1 - e^{-2rt}\big)\Big)$$

> **Use when:** drift proportional to $-X$ = **mean reversion**: the pull toward 0 strengthens with distance. Gaussian, because the noise is deterministic (I5.5).

> **Paper:** Midsem'24 Q4(a) — $dX_t = -X_t\,dt + dW_t$ ($r{=}\sigma{=}1$). "Whether it is an OU process? Explain." **Yes.** $E[X_t] = \mathbf{x_0 e^{-t}}$, $\mathrm{Var}(X_t) = \mathbf{\frac{1}{2}(1 - e^{-2t})}$. **Variance derivation:** isometry gives $\int_0^t e^{-2r(t-s)}ds = \frac{1-e^{-2rt}}{2r}$ ✓

▼ **Proof**

> §I7.1 with $\alpha = 0$: $d(e^{rt}X_t) = \sigma e^{rt}dW_t$, so $e^{rt}X_t = x_0 + \sigma\int_0^t e^{rs}dW_s$; divide by e^{rt}. Normal: deterministic integrand (§I1.5). Mean x_0e^{-rt} (§I1.3). Variance by isometry: $\sigma^2\int_0^t e^{-2r(t-s)}ds = \frac{\sigma^2}{2r}(1 - e^{-2rt})$.

I7.4 OU: STATIONARY LIMIT

$$X_t \xrightarrow{d} N\Big(0,\ \frac{\sigma^2}{2r}\Big) \text{ as } t \to \infty$$

> **Use when:** the contrast with BM is the marked point — **OU variance stays bounded; BM variance grows like t.** The starting value is forgotten exponentially fast.

> **Cross-ref:** §5.5 — $e^{-\lambda t}W(e^{2\lambda t})$ is the *same* process, built by time-changing BM. Its covariance $e^{-\lambda|t-s|}$ depends only on $|t-s|$: stationary, and therefore **not** a Wiener process.

▼ **Proof**

> Let $t \to \infty$ in §I7.3: mean $x_0e^{-rt} \to 0$, variance $\to \frac{\sigma^2}{2r}$. Normal laws with converging mean and variance converge to the limiting normal.

I7.5 VASICEK & CIR (INTEREST RATES)

Vasicek: $dr_t = \kappa(b - r_t)dt + \sigma\, dW_t$

$r_t \sim N\Big(b + e^{-\kappa t}(r_0 - b),\ \dfrac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t}$

CIR: $dr_t = \kappa(b - r_t)dt + \sigma\sqrt{r_t}\, dW_t$ — no closed form

same mean $E[r_t] = b + e^{-\kappa t}(r_0 - b)$, but **not** normal

> **Use when:** Vasicek is OU shifted — substitute $X_t = r_t - b$ and it becomes I7.3 exactly. CIR keeps $r_t \geq 0$ (noise vanishes at 0), strictly positive if $2\kappa b \geq \sigma^2$ (Feller).

> **Trick:** Vasicek's flaw is that Gaussian rates can go negative, $P(r_t < 0) > 0$. That is precisely the motivation for CIR — a standard short-note answer.
> **Note:** the two share a mean because they share a drift (I5.2: the diffusion never enters the mean).

▼ **Proof**

> Vasicek: $X = r - b$ satisfies $dX = -\kappa X\, dt + \sigma\, dW$, an OU process (§I7.3); add b back.
> CIR mean: the drift is linear, so $m' = \kappa(b - m)$ (§I5.2), giving $m(t) = b + e^{-\kappa t}(r_0 - b)$. Not normal: the noise $\sigma\sqrt{r_t}$ depends on r_t (§I5.5), and $r_t \geq 0$, which a normal variable cannot be.

I7.6 GEOMETRIC BROWNIAN MOTION

$dN_t = rN_t\, dt + \alpha N_t\, dW_t,\ \ N_0 > 0$

$N_t = N_0 \exp\left\{\left(r - \frac{\alpha^2}{2}\right)t + \alpha W_t\right\}$ — **log-normal**

$E[N_t] = N_0 e^{rt},\ \ \text{median} = N_0 e^{(r - \alpha^2/2)t}$

> **Use when:** multiplicative noise. Solve by Itô on $\log N$ — the **only** place taking a logarithm is the right move, and it works because $N_t > 0$.

> **Cross-ref:** all numerical GBM questions are §11.2, including the two-convention trap of §11.1. This card is where that $\frac{\alpha^2}{2}$ comes from.
> **Rigour note:** $\log N_t$ presumes $N_t > 0$; close the gap by verifying the candidate with I6.3 instead.

▼ **Proof**

> Solution: §I4.6 reversed (or Itô on $\ln N$, §D5.3). Mean: $E[e^{\alpha W_t}] = e^{\alpha^2 t/2}$ cancels the $-\alpha^2/2$, giving $N_0 e^{rt}$. Median: the exponent is normal with median $(r - \frac{\alpha^2}{2})t$, and e^x is increasing.

I7.7 SUMMARY TABLE

$\mu\, dt + \sigma\, dW$ → Normal, variance $\sigma^2 t$

$-rX\, dt + \sigma\, dW$ → Normal, variance bounded (OU)

$rN\, dt + \alpha N\, dW$ → log-Normal (GBM)

$\kappa(b - r)dt + \sigma\sqrt{r}\, dW$ → no closed form (CIR)

> **Use when:** identify which row you are in before doing anything. **Noise depends on X?** If no → Gaussian, integrating factor works. If yes → either GBM (log) or nothing.

> **Trick:** the mean always comes from the drift alone (I5.2), so you can answer "find $E[X_t]$" for *every* row — even CIR, where no solution exists.

I8. Worked Past-Paper Walkthroughs

I8.1 ⭐ MIDSEM'25 Q4(A)

$dX_t = dt + 2\sqrt{X_t}\,dW_t$, $X_0 = 0$. Show $X_t = W_t^2$; find $E(X_t)$, $\mathrm{Var}(X_t)$.

Show: Itô with $f(x) = x^2$ on $X{=}W$ gives $d(W_t^2) = 2W_t\,dW_t + dt$.
With $X_t = W_t^2$ we have $\sqrt{X_t} = |W_t|$, so $2\sqrt{X_t}\,dW_t = 2W_t\,dW_t$ ✓, and $X_0 = 0$ ✓

Mean (I5.2): $a \equiv 1$, so $m' = 1$, $m(0) = 0 \Rightarrow E[X_t] = \mathbf{t}$. (Check: $E[W_t^2] = t$ ✓)

Variance (I5.3):
$\frac{d}{dt}E[X^2] = E[2X_t a_t + b_t^2] = E[2X_t + 4X_t] =$

$\Rightarrow E[X_t^2] = 3t^2 \Rightarrow \mathrm{Var}(X_t) = 3t^2 - t^2 = \mathbf{2t^2}$

Check:
$\mathrm{Var}(W_t^2) = E[W^4] - t^2 = 3t^2 - t^2 = 2t^2$ ✓

Note: Var is the part they circled on the paper. Do it both ways — the cross-check costs one line.

I8.2 ⭐ MIDSEM'25 Q4(B)

$X_t := e^{t/2}\cos(W_t)$. (i) find dX_t. (ii) evaluate $E[\cos W_t]$.

(i) $f(t,x) = e^{t/2}\cos x$: $\partial_t f = \frac{1}{2}f$,
$\partial_x f = -e^{t/2}\sin x$, $\partial_{xx} f = -f$.
$dX_t = \frac{1}{2}X_t\,dt - e^{t/2}\sin W_t\,dW_t - \frac{1}{2}X_t\,dt =$

(ii) Drift is zero ⇒ X_t is a martingale ⇒
$E[X_t] = X_0 = e^0\cos 0 = 1$.
$e^{t/2}E[\cos W_t] = 1 \;\Rightarrow\; E[\cos W_t] = \mathbf{e^{-t/2}}$
Check: $E[e^{iW_t}] = e^{-t/2}$; real part gives the same ✓ ($E[\sin W_t] = 0$ by symmetry.)

Trick: "using part (i) or otherwise" is the examiner telling you the martingale *is* the method. This is I5.4.

I8.3 ⭐ MIDSEM'24 Q4(A)

$dX_t = -X_t\,dt + dW_t$, $X_0 = x_0$. Express as a stochastic integral. OU? Find $E(X_t)$, $\mathrm{Var}(X_t)$.

Solve (I7.1):
$d(e^t X_t) = e^t X_t\,dt + e^t dX_t = e^t dW_t$ — the X cancels.
Integrate: $e^t X_t = x_0 + \int_0^t e^s dW_s$, so

$$X_t = \mathbf{x_0 e^{-t}} + \int_0^t \mathbf{e^{-(t-s)}\,dW_s}$$

OU? Yes — drift $-rX$ with $r = 1$, constant noise $\sigma = 1$: mean-reverting to 0.

Mean: the Itô integral has mean 0, so $E[X_t] = \mathbf{x_0 e^{-t}}$.
Variance (isometry):
$\int_0^t e^{-2(t-s)}ds = \frac{1}{2}\left(\mathbf{1 - e^{-2t}}\right)$
Distribution: noise is deterministic ⇒ $X_t \sim N\left(x_0 e^{-t},\ \frac{1}{2}(1 - e^{-2t})\right)$.

Trick: as $t \to \infty$, $X_t \Rightarrow N(0, \frac{1}{2})$ — add that line, it is the point of OU.

I8.4 MIDSEM'24 Q4(B) & ENDSEM'25 Q1(B,C)

$X_t = e^{W_t - t/2}$: MG? | $d(e^t W_t) = ?$ | is $e^t W_t$ a MG?

$X_t = e^{W_t - t/2}$: $f = e^{x - t/2}$, so $\partial_t f = -\frac{1}{2}f$, $\partial_x f = \partial_{xx} f = f$.
$dX = -\frac{1}{2}X\,dt + X\,dW + \frac{1}{2}X\,dt = \mathbf{X_t\,dW_t}$
⇒ zero drift ⇒ **Yes, a martingale**.

$d(e^t W_t)$: $\partial_{xx} f = 0$, so no correction (I4.5):
$d(e^t W_t) = \mathbf{e^t W_t\,dt + e^t\,dW_t}$

Is $e^t W_t$ a MG? Drift $= e^t W_t \neq 0 \Rightarrow$ **No**.

Trick: both questions are one Itô application each. Compute the differential, then read the dt coefficient — that single number answers the Yes/No.

I8.5 THE SHORT DIFFERENTIALS (1-MARK ANSWERS)

$d \sin W_t = \cos W_t \, dW_t - \frac{1}{2} \sin W_t \, dt$

[CT-2'25 Q8]

$d \cos W_t = -\sin W_t \, dW_t - \frac{1}{2} \cos W_t \, dt$

[Midsem'24 Q1(e)]

$\mathrm{Var}\left(\int_0^1 t^4 dW_t \right) = \frac{1}{9}$ [Endsem'25 Q1(a)]

$\int_0^1 t^2 dW_t \sim N(0, \frac{1}{5})$ [CT-2'25 Q7]

> **Use when:** these appear as single-mark table entries. Write the answer, show nothing. Each takes ten seconds with I4.3 and I1.5.

> **Trick:** for any $f(W_t)$ the drift is always $\frac{1}{2} f''(W_t)$ and the diffusion always $f'(W_t)$. **Two derivatives and you are done** — that covers every entry above.

Part IV — Option Pricing & Arbitrage

Sections A1–A2. One-period binomial, the arbitrage theorem, contingent claims. Builds on §M3–M4.

A1. Arbitrage & the Risk-Neutral Measure

A1.1 ARBITRAGE — DEFINITION

A scheme with cost ≤ 0 at $t = 0$ and payoff ≥ 0 in *every* outcome, > 0 in some.

Risk-free profit. "Every outcome" means with probability 1 — not on average. No probability appears in the definition.

> **Use when:** any question saying "sure win", "risk-free profit", "for which price is there no arbitrage". The answer is always: build a portfolio whose time-1 value is the *same number* in all states, then compare with its cost.

> **Trick:** if a certain gain is positive for one scale x, it is unbounded — scale x up. That is why an arbitrage price cannot survive in a market.

A1.2 ☆ NO-ARBITRAGE PRICE BY HEDGING (ONE PERIOD, TWO STATES)

Buy x stock, y options. Force V_1 equal in both states $\Rightarrow$ y in terms of x, then

$$\text{gain} = V_1 - (1+r)\,\text{cost} = 0 \implies c^*$$

> **Use when:** two possible time-1 prices and an option quoted at c. Works with *no probabilities at all* — only the two possible values matter.

> **Paper:** $S_0 = 100 \to \{200, 50\}$, $K = 150$, $r = 0$. $200x + 50y = 50x \Rightarrow y = -3x$; $V_1 = 50x$, cost $= 100x - 3xc$, gain $= x(3c - 50)$. So $c^* = \mathbf{50/3}$; $c = 20$ (buy stock, write 3 calls) or $c = 15$ (short stock, buy 3 calls) both print money.

▼ **Proof**

> The equalised portfolio pays the same amount in both states, so it is riskless. A riskless position with a non-zero net gain can be scaled up without limit: an arbitrage. So the only price without arbitrage is the one making that gain 0.

A1.3 ☆ RISK-NEUTRAL MEASURE

$$S_0 = \frac{E^*[S_1]}{1+r}, \qquad p^* = \frac{(1+r)S_0 - S_d}{S_u - S_d}$$

Equivalently: every wager has zero expected return under p^*; the discounted price is a martingale (§M3.4).

> **Use when:** any binomial pricing question. Need $d < 1 + r < u$, else $p^* \notin (0,1)$ and the market itself is an arbitrage.

> **Paper:** $100 \to \{130, 80\}$, $r = 0.05$: $105 = 80 + 50p^* \Rightarrow p^* = \mathbf{0.5}$ — unrelated to the real $p = 0.4$.

▼ **Proof**

> Solve $S_0(1+r) = p^*S_u + (1-p^*)S_d$ for p^*. With $S_u = uS_0$, $S_d = dS_0$, divide by S_0: $1 + r = p^*u + (1-p^*)d$. It is in $(0,1)$ iff $d < 1 + r < u$ (§M3.4).

A1.4 ☆ PRICING FORMULA FOR ANY CLAIM

$$C_0 = \frac{E^*\left[g(S_1)\right]}{1+r}, \qquad \text{call: } g(s) = (s -$$

> **Use when:** "price the call / put / claim". Two steps, never more: (1) p^* from the *stock*, (2) expectation of the *payoff*. Never use the real-world p.

> **Paper:** $100 \to \{130, 80\}$, $K = 110$, $r = 0.05$, $p^* = 0.5$: $C_0 = \frac{20(0.5)+0(0.5)}{1.05} = \mathbf{9.52}$.

▼ **Proof**

> The replicating portfolio (§A1.5) costs $\frac{p^*C_u + (1-p^*)C_d}{1+r} = \frac{E^*[g(S_1)]}{1+r}$ (§M4.4). Any other option price: buy the cheaper of {option, portfolio}, sell the dearer; the payoffs cancel and you keep the gap.

A1.5 ☆ REPLICATING PORTFOLIO (Δ, B)

$$\Delta = \frac{C_u - C_d}{S_u - S_d}, \qquad B = \frac{C_u - \Delta S_u}{1+r},$$

Δ = shares held, B = cash lent ($B < 0$ means borrowing). C_0 from replication always equals C_0 from §A1.4 — two routes, one price.

> **Use when:** "construct the hedge", "show the claim is attainable", or when you must *exhibit* the arbitrage rather than merely assert it.

> **Paper:** $\Delta = \frac{20-0}{130-80} = 0.4$, $B = \frac{20-52}{1.05} = -30.48$; $C_0 = 40 - 30.48 = \mathbf{9.52}$. Down state check: $0.4(80) - 32 = 0$ ✓.

▼ **Proof**

> Solve two equations (up and down state) in two unknowns: $\Delta S_u + B(1+r) = C_u$, $\Delta S_d + B(1+r) = ($. Subtract: $\Delta(S_u - S_d) = C_u - C_d$. Back-substitute for B. Its cost today is $\Delta S_0 + B$; any other option price lets you buy the cheaper of {option, portfolio} and sell the other: arbitrage.

A1.6 ⚠ THE ACTUARIAL TRAP

$\frac{E_p[\text{payoff}]}{1+r} \neq C_0$ when p is the *real-world* probability.

> **Why it fails:** the payoff is *replicable*. Two portfolios with identical cash flows in every state must have identical prices, whatever anyone believes about likelihoods. Insurance risk is diversifiable but not hedgeable, which is why the actuarial method is right there and wrong here.

> **Paper:** $p = 0.4$ gives $\frac{20(0.4)}{1.05} = 7.62$, but $C_0 = 9.52$. At 7.62: buy the call, short 0.4 shares $(+40)$, lend $30.48 \Rightarrow$ net $+1.90$ today, payoffs cancel at T. **Certain 1.90 per contract.**

▼ **Proof**

> The price must equal the replication cost (§A1.5), which uses p^* only. $E_p[\text{payoff}]/(1+r)$ equals it only when $p = p^*$; otherwise the gap is an arbitrage, as in the example.

A2. Contingent Claims, the Arbitrage Theorem & Incompleteness

A2.1 PAYOFF DICTIONARY

Forward (long): $S_T - K$ — an **obligation**, can be negative

European call: $(S_T - K)^+$ | European put: $(K - S_T)^+$

A *derivative security / contingent claim* is any $X = g(S_T)$ paid at T. $x^+ := \max(x, 0)$.

> **Use when:** writing down the payoff before anything else. Getting $(\cdot)^+$ vs no $(\cdot)^+$ right is the whole difference between an option and a forward.

> **Trick:** $K = 150$, $S_T \in \{200, 50\}$ → forward $+50/-100$, call $50/0$, put $0/100$. The call is the forward with the loss leg deleted — which is exactly what the premium buys. Futures = forwards, but exchange-traded and marked to market daily.

A2.2 AMERICAN VS EUROPEAN

$\text{price} = \sup_{\tau \le T} E^*\left[(1+r)^{-\tau}(S_\tau - K)^+\right]$, τ a **stopping time**

$(S_\tau - K)^+$ is the *payoff*, not the price. $\{\tau \le t\} \in \mathcal{F}_t$ is compulsory: "the time S peaks" is not admissible, since at t you cannot know the future (§M5).

Use when: "short note: American vs European". No dividends and $r \ge 0 \Rightarrow$ early exercise of a **call** is never optimal, so the two prices are **equal**; for a **put** early exercise can be optimal and the American is strictly dearer.

Trick: one-line reason for the call: exercising early throws away the interest on K and the remaining insurance value. See §M7.6.

▼ **Proof**

The price formula: the holder may pick any stopping time, so the fair price is the best discounted risk-neutral value over them. Call = European: proof in §M7.6.

A2.3 ⭐ ARBITRAGE THEOREM (DICHOTOMY)

Wagers $i = 1..n$, outcomes $j = 1..m$, return $r_i(j)$ per unit staked. **Exactly one** holds:

(i) $\exists\, \underline{p}$ probability vector with $\sum_j p_j r_i(j) = 0 \ \forall i$

(ii) $\exists\, \underline{x}$ with $\sum_i x_i r_i(j) > 0 \ \forall j$

No arbitrage $\iff$ a risk-neutral measure exists. Return = payoff − cost; $x_i < 0$ allowed (short).

Use when: asked to state or prove it. The "not both" half is three lines: $\sum_j p_j R(j) = \sum_i x_i \sum_j p_j r_i(j) = 0$, yet $R(j) > 0\,\forall j$ and $\sum_j p_j = 1$ force it > 0. Contradiction.

Trick: the "at least one" half is a separating-hyperplane (Farkas) argument — say so and move on; it is never asked in full.

▼ **Proof**

Not both. Suppose p and x both exist. Total return $R(j) = \sum_i x_i r_i(j) > 0$ for every j. Then $\sum_j p_j R(j) = \sum_i x_i \sum_j p_j r_i(j) = 0$; but a p-average of positive numbers is positive. Contradiction.

At least one. If no p makes all expected returns 0, the point 0 lies outside the convex hull of the return vectors; a separating hyperplane gives the stakes x with $\sum_i x_i r_i(j) > 0$ for all j (same geometry as §D3.3, Step 3).

A2.4 THREE STATES ⇒ INCOMPLETE MARKET

m states, n assets with $n < m$: p^* is not unique ⇒ an **interval** of arbitrage-free prices.

> **Use when:** the tree has three or more branches from one node. Do not hunt for a single price — there is none, and the claim cannot be replicated.

> **Paper:** $100 \to \{300, 200, 50\}$, $K = 150$, $r = 0$. Stock: $250p_1 + 150p_2 = 50 \Rightarrow p_2 = \frac{1-5p_1}{3}$; option: $c = 150p_1 + 50p_2 = \frac{200p_1+50}{3}$. Positivity gives $p_1 \in (0, \frac{1}{5})$, hence $c \in \left(\frac{\mathbf{50}}{\mathbf{3}}, \mathbf{30}\right)$.

▼ **Proof**

> EMM: the stock equation plus $\sum p = 1$ is 2 equations in 3 unknowns, a line of solutions. Replication: 3 states give 3 equations in 2 unknowns (Δ, b), generally no solution, so the claim is not attainable. $E^*[X]$ then moves along the line, and the open range allowed by all $p_k > 0$ is the interval of prices (§D3.4).

Part V — Discrete Markets & Black–Scholes

Sections D1–D5. Lectures 16–26: forwards, the general discrete model, finite single-period markets (FTAP, pricing, completeness), the multi-period binomial model and its limit, the Black–Scholes framework. Builds on §A1–A2 and §I7.6.

D1. Discounting & Forwards

D1.1 PRESENT VALUE

v paid at time t is worth $v(1+r)^{-t}$ (discrete) $= ve^{-rt}$ (continuous) today

> **Use when:** every price on this sheet is a *discounted* expected payoff. Binomial models use $(1+r)^{-T}$; Black–Scholes uses e^{-rT}. Never mix them in one answer.

> **Trick:** the two agree when $1+r_n = e^{r\delta_n}$ per step, which is exactly the choice in §D5.1.

▼ **Proof**

> One rupee in the bank grows to $(1+r)^t$ (or e^{rt}). So to have v at time t you need $v(1+r)^{-t}$ now. Any other price for a sure v at t: borrow or lend the difference: arbitrage.

D1.2 ⭐ FORWARD PRICE

long forward pays $S_T - K$ (an obligation, §A2.1)

fair delivery price: $K = S_0(1+r)^T$ (discrete) $= S_0e^{rT}$ (continuous)

value today of a forward with price K: $S_0 - K(1+r)^{-T}$

> **Use when:** "find K so the contract costs nothing". No probabilities and no p^* needed: only the stock and the bond.

> **Paper-style:** $S_0 = 100$, $r = 5\%$, $T = 1$: $K = \mathbf{105}$. If someone quotes $K = 110$: borrow 100, buy the stock, sell forward at 110; at T deliver, repay 105, keep $\mathbf{5}$ for sure.

▼ **Proof**

> Buy 1 share with borrowed money (S_0) and sell a forward at K. At T: deliver the share for K and repay $S_0(1+r)^T$. Profit $K - S_0(1+r)^T$ is certain, so it must be 0 (if negative, run the trade backwards: short the share, lend, buy forward).
> Same via E^*: the contract costs 0, so $E^*[S_T - K] = 0$, i.e. $K = E^*[S_T] = S_0(1+r)^T$.

D2. General Discrete Market Model

D2.1 MARKET & TRADING STRATEGY

$S_n = (S_n^0, S_n^1, \ldots, S_n^d)$, bond $S_n^0 = (1+r)^n$

strategy $\Phi_n = (\Phi_n^0, \ldots, \Phi_n^d)$ = number of units of each asset held

$$V_n(\Phi) = \Phi_n \cdot S_n = \sum_{i=0}^{d} \Phi_n^i S_n^i$$

> **Use when:** setting up any multi-asset question. Write the value as a dot product first; everything else (self-financing, arbitrage) is a statement about V_n.

> **Trick:** Φ_n may only use information up to time $n-1$ (you choose the portfolio *before* the price moves). That is what "predictable" means here, as in §M7.2.

D2.2 ⭐ SELF-FINANCING

$\Phi_n \cdot S_n = \Phi_{n+1} \cdot S_n$ (rebalance at today's prices, no cash in or out)

$\Rightarrow \Delta V_n = V_{n+1} - V_n = \Phi_{n+1} \cdot \Delta S_n$

> **Use when:** "verify the strategy is self-financing": value the *old* holdings at the *new* prices, then check it equals the cost of the *new* holdings. Gains then come only from price moves.

> **Trick:** the binomial version is $\Delta_t S_t + b_t(1+r)^t = \Delta_{t+1} S_t + b_{t+1}(1+r)^t$ (§D4.1). §M4.7 is the same idea with cash B instead of bond units b.

▼ **Proof**

> $V_{n+1} - V_n = \Phi_{n+1} \cdot S_{n+1} - \Phi_n \cdot S_n = \Phi_{n+1} \cdot$, using self-financing to swap $\Phi_n \cdot S_n$ for $\Phi_{n+1} \cdot S_n$.

D2.3 NO-ARBITRAGE PRINCIPLE

there is **no** self-financing Φ with

$$V_0(\Phi) = 0, \quad V_n(\Phi) \geq 0, \quad P\big(V_n(\Phi) >$$

Use when: "define arbitrage / the NA condition formally". Three parts: costs nothing, never loses, sometimes wins.

Trick: to *check* NA never search all strategies; find an EMM instead (§D3.3). To *show* arbitrage, exhibit one Φ meeting all three parts.

D3. Finite Single-Period Market (no-arbitrage, pricing, completeness)

D3.1 ☆ MODEL & EMM

$\Omega = \{\omega_1, \ldots, \omega_N\}$, n stocks
$S_t = (S_{t,1}, \ldots, S_{t,n})$, bond
$B_t = (1+r)^t$, $t = 0, 1$
EMM P^*: $p_k^* = P^*(\omega_k) > 0$ for every k
and $E^*\left[\dfrac{S_1}{1+r}\right] = S_0$

Use when: "find all EMMs": unknowns $p_1^*, \ldots, p_N^*$; one equation per stock, plus $\sum_k p_k^* = 1$; then keep only solutions with *every* $p_k^* > 0$.

Paper: binomial ($n = 1$, $N = 2$): NA $\iff S_0 \in \left(\dfrac{S_1(d)}{1+r}, \dfrac{S_1(u)}{1+r}\right) \iff d < 1 + r$; unique $p^* = \frac{1+r-d}{u-d}$. Three states, one stock: a whole line of EMMs (§A2.4).

▼ **Proof**

One stock, two states: by §D3.3 NA iff $S_0 \in D^0$, and D^0 is the open interval between $\frac{S_1(d)}{1+r}$ and $\frac{S_1(u)}{1+r}$. With $S_1(u) = uS_0$, $S_1(d) = dS_0$, divide by S_0: $\frac{d}{1+r} < 1 < \frac{u}{1+r}$, i.e. $d < 1 + r < u$.

D3.2 CONVEX HULL & SEPARATION

$\operatorname{conv}\{y_1 .. y_N\} = \left\{ \sum_j \alpha_j y_j : \ \alpha_j \geq 0, \ \right.$; relative interior D^0: all $\alpha_j > 0$

Separation: D convex, $x_0 \notin D^0 \Rightarrow$ $\exists\, a \neq 0$: $a \cdot (x - x_0) > 0 \ \ \forall x \in D^0$

Use when: only inside the proof of §D3.3. Picture: the hull is the polygon spanned by the discounted payoff vectors; D^0 is "strictly inside".

Trick: one stock ⇒ the hull is the interval $\left[\min_k \frac{S_1(\omega_k)}{1+r}, \max_k \frac{S_1(\omega_k)}{1+r}\right]$ and D^0 the open interval. Separation then just says "a point outside is to one side".

▼ **Proof**

One dimension: D^0 is an open interval (α, β); a point $x_0 \notin D^0$ has $x_0 \leq \alpha$ (take $a = +1$) or $x_0 \geq \beta$ (take $a = -1$).
General idea when x_0 is outside the closed hull: let y be the nearest hull point and $a = y - x_0$. Convexity gives $a \cdot (x - y) \geq 0$ for hull points x, so $a \cdot (x - x_0) \geq |a|^2 > 0$. (The boundary case is quoted.)

D3.3 ☆ FIRST FUNDAMENTAL THEOREM (THEOREM 2)

$$\text{NA} \iff \exists\,\text{EMM}\ P^* \iff S_0 \in D^0,$$

> **Use when:** "show the market is arbitrage-free" ⇒ exhibit one EMM with all $p_k^* > 0$. "Show there is arbitrage" ⇒ show the EMM equations have no positive solution, then build Δ.

> **Trick:** "discounted stock price is a martingale under P^*" is the same sentence as $E^*[S_1/(1+r)] = S_0$.

▼ **Proof**

> **Step 1 (EMM ⇔ $S_0 \in D^0$).** $S_0 = \sum_k p_k^* \frac{S_1(\omega_k)}{1+r}$ with $p_k^* > 0$, $\sum p_k^* = 1$ is word-for-word the definition of D^0 with $\alpha_k = p_k^*$.
> **Step 2 (EMM ⇒ NA).** Suppose (Δ, b) is an arbitrage: $V_0 = \Delta \cdot S_0 + b = 0$, so $b = -\Delta \cdot S_0$ and $\frac{V_1}{1+r} = \Delta \cdot \left(\frac{S_1}{1+r} - S_0\right) \geq 0$ always, > 0 in some state. Take E^*: $\Delta \cdot \left(E^*[\frac{S_1}{1+r}] - S_0\right) = 0$. But a variable ≥ 0 that is > 0 on a state with $p_k^* > 0$ has positive mean. Contradiction.
> **Step 3 (NA ⇒ $S_0 \in D^0$).** If $S_0 \notin D^0$, separation (§D3.2) gives Δ with $\Delta \cdot \left(\frac{S_1(\omega)}{1+r} - S_0\right) \geq 0$ for every ω, > 0 for some. Then $(\Delta,\ b = -\Delta \cdot S_0)$ costs 0 and pays $(1+r)\Delta \cdot \left(\frac{S_1}{1+r} - S_0\right) \geq 0$, sometimes > 0: an arbitrage. Contradiction.

D3.4 ☆ PRICING ATTAINABLE CLAIMS (THEOREM 3)

any strategy (Δ, b): $E^*\left[\dfrac{V_1}{1+r}\right] = V_0$

X attainable ($\exists(\Delta, b)$ with $V_1 = X$ in every state) $\Rightarrow$ $X^* = V_0 = E^*\left[\dfrac{X}{1+r}\right]$, the **same for every EMM**

> **Use when:** "price the claim" when there are several EMMs. If $E^*[X/(1+r)]$ changes as you move along the EMM line, X is *not* attainable and you get an interval (§A2.4).

> **Trick:** to test attainability, compute $E^*[X/(1+r)]$ with the free parameter left in. Parameter cancels ⇒ attainable, one price.

▼ **Proof**

> $\frac{V_1}{1+r} = \Delta \cdot \frac{S_1}{1+r} + b$. Take E^* and use $E^*[\frac{S_1}{1+r}] = S_0$: $E^*[\frac{V_1}{1+r}] = \Delta \cdot S_0 + b = V_0$. If X is attainable, $V_1 = X$ in every state, so $E^*[\frac{X}{1+r}] = V_0$ for *every* EMM, and V_0 does not depend on which P^* you used. Any other price: sell the dearer of {claim, replicating portfolio}, buy the cheaper, pocket the gap.

D3.5 ☆ COMPLETENESS (THEOREM 4)

complete (every claim attainable) $\iff$ the EMM is **unique**

counting rule: unique EMM needs **states** $=$ **assets**, $N = n + 1$, and a non-singular matrix A

> **Use when:** "is the market complete?" Count N states vs $n + 1$ assets (stocks + bond), then solve the EMM system: one solution ⇒ complete; a line of solutions ⇒ incomplete.

> **Paper:** binomial: $N = 2$, $n + 1 = 2$ ⇒ complete. One stock, three prices (§A2.4): $N = 3 > 2$ ⇒ incomplete, price interval $(\frac{50}{3}, 30)$.

▼ **Proof**

> (⇐) Unique EMM ⇒ the EMM system is square with $\det A \neq 0$. The replication system $\Delta \cdot S_1(\omega_k) + b(1 + r) = X(\omega_k)$, $k = 1..N$, has matrix A^{T}, also non-singular, so it has a solution for *every* X: complete.
>
> (⇒) Complete, and suppose two EMMs $P^*, \widetilde{P}^*$. Take the claim X_i that pays 1 in state ω_i and 0 elsewhere. It is attainable, so by Theorem 3 $E^*[X_i] = \widetilde{E}^*[X_i]$, i.e. $p_i^* = \widetilde{p}_i^*$. True for every i, so $P^* = \widetilde{P}^*$.

D4. Multi-Period Binomial Market

D4.1 SETUP & SELF-FINANCING

$S_t = uS_{t-1}$ or dS_{t-1}, $B_t = (1 + r)^t$, (Δ_t, b_t) held over $(t - 1, t]$, chosen from $S_0 .. S_{t-1}$

self-financing: $\Delta_t S_t + b_t(1 + r)^t = \Delta_{t+1} S_t + b_{t+1}(1 +$, $V_0 = \Delta_1 S_0 + b_1$

> **Use when:** "verify self-financing" at a node: old portfolio at the new price $=$ cost of the new portfolio. b_t counts *bonds*; their value is $b_t(1 + r)^t$.

> **Trick:** at $t = 1$: $V_1 = \Delta_1 S_1 + b_1(1 + r) = \Delta_2 S_1 + b_2(1 + r)$ — you only reshuffle (Δ_1, b_1) into (Δ_2, b_2).

D4.2 ☆ BACKWARD INDUCTION (PRICE + HEDGE AT EVERY NODE)

at node $S_{t-1} = s$:

$$\Delta_t(s) = \frac{V_t[us] - V_t[ds]}{us - ds}, \qquad b_t(s) = \frac{u}{}$$

$$V_{t-1}[s] = \frac{p^* V_t[us] + (1-p^*)V_t[ds]}{1+r},$$

> **Use when:** any tree question: "price / replicating strategy at each node". Start at the leaves (payoffs), step back one period at a time.

> **Worked:** $S_0 = 100$, $u = 2$, $d = \frac{1}{2}$, $r = \frac{1}{4} \Rightarrow p^* = \frac{1}{2}$. Call $K = 100$, $T = 2$. Leaves $400, 100, 25$ → payoffs $300, 0, 0$. $V_1[200] = \frac{150}{1.25} = 120$, $V_1[50] = 0$, $V_0 = \frac{60}{1.25} = \mathbf{48}$. Hedge: $\Delta_1 = \frac{120}{150} = 0.8$, $b_1 = \frac{-60}{1.875} = -32$ (check $80 - 32 = 48$ ✓); at 200: $\Delta_2 = 1$, $b_2 = -64$ (check $200 - 80 = 120$ ✓).

▼ **Proof**

> At a node it is the one-period problem (§A1.5) with bond worth $(1+r)^t$: solve $\Delta us + b(1+r)^t = V_u$, $\Delta ds + b(1+r)^t = V_d$. Subtract for Δ, back-substitute for b.
> Cost at $t-1$: $\Delta s + b(1+r)^{t-1} = \frac{V_u - V_d}{u-d} + \frac{uV_d - dV_u}{(1+r)(u-d)} = \frac{(1+r}{}$.

D4.3 ☆ GENERAL VALUE FORMULA & MARTINGALE

$$V_t[s] = \frac{1}{(1+r)^{T-t}} \sum_{j=0}^{T-t} \binom{T-t}{j} (p^*)^j (1$$

$$V_0 = E^*\left[\frac{g(S_0 e^{Z_T})}{(1+r)^T}\right],$$

$Z_T = Y_1 + \cdots + Y_T$, $Y_k = \ln u$ (prob p^*) or $\ln d$

$S_t^* = S_t(1+r)^{-t}$ is a MG under p^*, and p^* is the **only** such measure ⇒ arbitrage-free and complete

> **Use when:** pricing $g(S_T)$ straight from $t = 0$ without drawing the tree. Only the *number* of up-moves j matters, not their order, hence $\binom{T}{j}$.

> **Check:** worked example of §D4.2: $\frac{1}{1.25^2}\left[\frac{1}{4} \cdot 300\right] = \frac{75}{1.5625} = \mathbf{48}$ ✓.

▼ **Proof**

> Apply §D4.2 repeatedly. Each step back divides by $(1+r)$ and averages the two children with weights $p^*, 1-p^*$. After $T-t$ steps you are averaging over $T-t$ independent up/down moves; the number of ups is $\mathrm{Bin}(T-t, p^*)$, and j ups land at $u^j d^{T-t-j} s$.

D4.4 ⭐ T-PERIOD CALL PRICE (CRR)

$$C = \frac{1}{(1+r)^T} \sum_{j=m}^{T} \binom{T}{j} (p^*)^j (1-p^*)^{T-j}$$

$$m = \left\lfloor \frac{\ln K - \ln S_0 - T \ln d}{\ln u - \ln d} \right\rfloor + 1, \quad U^*$$

Use when: "price a European call over T periods". Find m first (fewest ups to finish in the money); all terms with $j < m$ are 0.

Worked (§D4.2 data): $m = \lfloor \frac{0+2\ln 2}{2\ln 2} \rfloor + 1 = 2$; $\hat{p} = \frac{2 \cdot 0.5}{1.25} = 0.8$.
$C = 100(0.8)^2 - \frac{100}{1.5625}(0.5)^2 = 64 - 16 = \mathbf{48}$ ✓.

▼ **Proof**

$(u^j d^{T-j} S_0 - K)^+ > 0 \iff j \ln u + (T-j) \ln$. Split the sum into the S_0 part and the K part. In the S_0 part, $\frac{u^j d^{T-j} (p^*)^j (1-p^*)^{T-j}}{(1+r)^T} = \hat{p}^{\,j} (1-\hat{p})^{T-j}$ with $1 - \hat{p} = \frac{d(1-p^*)}{1+r}$; these add to 1 because $up^* + d(1-p^*) = 1 + r$. So both parts are binomial tail probabilities.

D5. Black–Scholes

D5.1 BINOMIAL → BLACK–SCHOLES (THE LIMIT)

n steps of length $\delta_n = T/n$:

$u_n = e^{\sigma\sqrt{\delta_n}},\ d_n = e^{-\sigma\sqrt{\delta_n}},\ 1 + r_n = e^{r\delta_n}$

$$p_n^* = \frac{e^{r\delta_n} - e^{-\sigma\sqrt{\delta_n}}}{e^{\sigma\sqrt{\delta_n}} - e^{-\sigma\sqrt{\delta_n}}} \approx \frac{1}{2} + \frac{r - \sigma^2/2}{2\sigma}$$

$S_T = S_0 e^{Z_n}$, Z_n = sum of n i.i.d. jumps $\pm\sigma\sqrt{T/n}$; CLT on the two tails of §D4.4 gives §D5.2

> **Use when:** "derive the Black–Scholes formula from the binomial model". Write the three choices (u_n, d_n, r_n), then p_n^*, then apply the CLT to $P(U^* \geq m)$ and $P(\hat{U} \geq m)$.

> **Note:** the lecture writes $p_n^* \approx \frac{1}{2} + \frac{r}{2\sigma}\sqrt{\delta_n}$ (first order). Keeping the $\frac{\sigma^2}{2}\delta_n$ term from $e^{-\sigma\sqrt{\delta_n}}$ is what produces the $-\frac{\sigma^2}{2}$ in $h - \sigma\sqrt{T}$; quote it if asked to show the limit in full.

▼ **Proof**

$P(U^* \geq m) \approx \Phi\left(\frac{np_n^* - m}{\sqrt{np_n^*(1-p_n^*)}}\right)$ by the CLT.

With $\ln d_n = -\sigma\sqrt{\delta_n}$: $m \approx \frac{\ln(K/S_0)}{2\sigma\sqrt{\delta_n}} + \frac{n}{2}$ and $np_n^* \approx \frac{n}{2} + \frac{n(r-\sigma^2/2)\sqrt{\delta_n}}{2\sigma}$. So $np_n^* - m \approx \frac{\ln(S_0/K) + (r-\sigma^2/2)T}{2\sigma\sqrt{\delta_n}}$, and dividing by $\sqrt{n}/2$ gives $\frac{\ln(S_0/K) + (r-\sigma^2/2)T}{\sigma\sqrt{T}} = h - \sigma\sqrt{T}$.

The same steps with $\hat{p}_n$ (one extra $+\sigma^2$ from u_n) give h. Finally $(1 + r_n)^{-n} = e^{-rT}$.

D5.2 ⭐ BLACK–SCHOLES FORMULA

$$C = S_0\,\Phi(d_1) - Ke^{-rT}\,\Phi(d_2)$$

$$d_1 = h = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}},$$

put (by parity, §M4.6 with e^{-rT}):

$$P = Ke^{-rT}\Phi(-d_2) - S_0\Phi(-d_1)$$

> **Use when:** numeric option price in continuous time. The real drift μ **does not appear** — only r and σ.

> **Worked:** $S_0 = K = 100$, $r = 0.05$, $\sigma = 0.2$, $T = 1$: $d_1 = \frac{0.07}{0.2} = 0.35$, $d_2 = 0.15$; $\Phi(0.35) = 0.6368$, $\Phi(0.15) = 0.5596$. $C = 63.68 - 95.12(0.5596) = \mathbf{10.45}$; $P = 10.45 - 100 + 95.12 = \mathbf{5.57}$.

▼ **Proof**

Under P^* (§D5.4) $S_T = S_0 e^{(r-\sigma^2/2)T + \sigma\sqrt{T}Z}$, $Z \sim N(0,1)$, and $C = e^{-rT}E^*[(S_T - K)^+]$. $S_T > K \iff Z > -d_2$. So $e^{-rT}E^*[K\mathbf{1}_{\{Z>-d_2\}}] = Ke^{-rT}\Phi(d_2)$. $e^{-rT}E^*[S_T\mathbf{1}_{\{Z>-d_2\}}] = S_0\int_{-d_2}^{\infty} e^{-\sigma^2T/2 + \sigma\sqrt{T}z}\varphi$ (complete the square, §1.2). Subtract.

D5.3 BLACK–SCHOLES MARKET

bond $B_t = e^{rt}$ ($dB_t = rB_t\,dt$); stock $dS_t = \mu S_t\,dt + \sigma S_t\,dW_t$

$$S_t = S_0 \exp\left\{\left(\mu - \tfrac{\sigma^2}{2}\right)t + \sigma W_t\right\}$$

(GBM, §I7.6)

> **Use when:** "solve the Black–Scholes SDE" or as the starting line of any BS question.

> **Trick:** the $-\frac{\sigma^2}{2}$ is the Itô correction; under P^* just replace μ by r (§D5.4).

▼ **Proof**

Itô on $\ln S$: $d\ln S_t = \frac{dS_t}{S_t} - \frac{(dS_t)^2}{2S_t^2} = \mu\,dt + \sigma\,dW_t - \frac{\sigma^2}{2}dt$. Integrate: $\ln S_t - \ln S_0 = (\mu - \frac{\sigma^2}{2})t + \sigma W_t$; exponentiate.

D5.4 ☆ EMM IN THE BLACK–SCHOLES MARKET

$Z_t = S_0 \exp\left\{\int_0^t a_s\,ds + \sigma W_t\right\}$:

$e^{-rt}Z_t$ is a MG $\iff$ $a_t = r - \dfrac{\sigma^2}{2}$

so under P^*:

$S_t = S_0 \exp\left\{(r - \frac{\sigma^2}{2})t + \sigma W_t^*\right\}$ ($\mu \to r$)

arbitrage-free **and** complete;

$P_t(X) = E^*\left[e^{-r(T-t)}X \mid \mathcal{F}_t\right]$,

$P_0(X) = E^*\left[e^{-rT}X\right]$

> **Use when:** "is the BS market arbitrage-free? complete? find the EMM". Answer: yes, yes, the drift becomes r. Then price any claim as a discounted E^*.

> **Trick:** "MG ⇔ zero dt coefficient" (§I5.1) is the whole proof; the lecture writes it as a product of two Itô processes.

▼ **Proof**

> Write $e^{-rt}Z_t = S_0 X_t Y_t$ with $X_t = e^{\int_0^t a_s\,ds}$, $Y_t = e^{-rt+\sigma W_t}$. Then $dX_t = a_t X_t\,dt$ and $dY_t = Y_t(-r + \frac{\sigma^2}{2})\,dt + \sigma Y_t\,dW_t$.
> Product rule ($dX\,dY = 0$):
> $d(XY) = X_tY_t\big(a_t - r + \frac{\sigma^2}{2}\big)dt + \sigma X_tY_t\,dW_t$. Since $X_tY_t > 0$, the dt coefficient is 0 iff $a_t = r - \frac{\sigma^2}{2}$. Such a measure exists and is unique, so by the three facts (§D5.5) the market is arbitrage-free and complete.

D5.5 MARTINGALES & CLAIM PRICING: THE THREE FACTS

(1) an EMM P^* exists with $S_t^* = (1+r)^{-t}S_t$ / $e^{-rt}S_t$ a MG ⇒ **arbitrage-free**

(2) attainable X: $P_t(X) = E^*\left[(1+r)^{t-T}X \mid \mathcal{F}_t\right]$ / $E^*\left[e^{-r(T-t)}X \mid \mathcal{F}_t\right]$

(3) P^* unique ⇒ **complete**: every claim attainable, so every claim has one price

> **Use when:** short note on "martingales and claim pricing", or as the conclusion line after finding the EMM in any model.

> **Trick:** these are §D3.3, §D3.4, §D3.5 restated for any time grid. Discrete: discount with $(1+r)$; continuous: with e^r.

▼ **Proof**

> (1) is Step 2 of §D3.3 (continuous time: §D5.4). (2) is §D3.4 applied one period at a time, with the tower property joining the periods (§D4.3). (3) is §D3.5.

FinMath Formula Sheet — Foundations, Expectation, Martingales, Random Walks, Brownian Motion, Stochastic Calculus, Option Pricing & Black–Scholes. Markov decision processes are out of syllabus. Cases and worked instances cross-checked against MA60225 Midsem 2024, Midsem 2025, Endsem 2025, CT-1 2026 (Sets 1–4) and CT-2 2025.